ASPECTS OF CONFORMAL AND SUPERCONFORMAL REPRESENTATION THEORY IN FOUR DIMENSIONS

Thesis

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Abstract

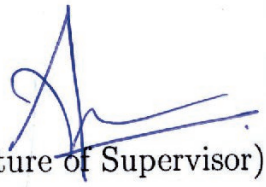
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This thesis was taken in partial fulfillment of the requirements for the integrated M.Sc. (Hons) degree in physics from BITS Pilani. The thesis was performed off-campus during my term as a visiting student at Tata Institute of Fundamental Research (TIFR) under the supervision of Prof. Abhijit Gadde of the Department of Theoretical Physics.

The aim of this thesis was to review various topics in conformal and superconformal representation theory. In the recent years conformal field theories have taken the central stage in theoretical high energy physics research. Conformal field theories (or their supersymmetric extensions) seem to appear everywhere - from string theory to phase transitions in statistical systems. CFTs also lie at the very core of our understanding of quantum field theories and their behaviour at different energy scales. On the other hand, superconformal field theories also incorporate the maximal symmetry group allowed by nature and hence are vital in our understanding of fundamental physics at very high energy scales. The thesis was divided into three parts which reflect as three chapters into this report. The aim of the first part was to study conformal field theories in higher dimensions (d>2). The representations of the conformal group were constructed and topics like the state operator correspondence and operator product expansions were discussed. Finally, the unitarity bounds on physical theories were derived. The aim of the second part was to get acquainted with the basic ideas of supersymmetry. We discussed the supersymmetric extension of the more familiar case of Poincare group and derived the relevant representations. The emphasis was on understanding the algebra itself and building its representations rather than the field theoretic aspects. Therefore the second part acted as a mediator from conformal theories to superconformal theories, which are conformal theories with the addition of supersymmetry. The aim of the final part was to get familiar with the superconformal algebra, which is the supersymmetric extension of the higher dimensional (>2) conformal group. Eventually, the short and semi-short representations of the superconformal group in four dimensions were reviewed for the N=1,2,4 supersymmetry and the multiplet recombinations were derived using character analysis. Finally, the analysis of N=3 supersymmetry was worked out including possible shortening/semi-shortening conditions and recombination relations.

CERTIFICATE

This is to certify that the thesis entitled, Aspects of conformal and super-conformal representation theory in four dimensions is submitted by Avi Wadhwa, ID No. 2014B5TS0897G, in partial fullfillment of the requirements of BITS421T. Thesis embodies the work done by him under my supervision.



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Notations and Conventions

- Throughout this report we work in natural units in which $\hbar = c = 1$. In these units every quantity has units of mass to some power. For example length $L = \hbar c M^{-1} \rightarrow M^{-1}(in\ natural\ units)$. Similarly, time $T = \hbar M^{-1} \rightarrow M^{-1}$ and energy $E = c^2 M \rightarrow M$. Therefore when we say that some quantity has dimension N we mean that N is the exponent of the mass. If we let [q] denote the dimension of some quantity q then we have

$$[m] = [E] = +1$$

$$[x^\mu] = -1$$

$$\Rightarrow [\partial_\mu] = +1$$

  Since an action S must be dimensionless we also have for the lagrangian $\mathcal{L}$

$$[\mathcal{L}] = +d$$

- Throughout the report we use a flat metric which we denote by η. Different signatures of the metric are used in different parts but the same symbol η is used to denote all of them. The signature of metric used is mentioned in each context. In the first chapter we mostly use the signature (-,+,+,+,..+) or (+,+,+,...+). Sometimes the signature (-,++...,-) or (+,+,...+,-) are also used. The choice is mentioned in each instance. Throughout the second chapter we use the signature (+,-,-,...-). Again in the third chapter we mostly use (-,+,+..+) or (-,+,+...,-). **Note that in this report $\eta_{\mu\nu}$ denotes a flat metric in any arbitrary coordinates. Hence, in this report it does not just mean the matrix with diagonal entries of 1 and -1.**

Chapter 1

Conformal Field Theory in More Than Two Dimensions

1.1 Background

Conformal field theories are quantum field theories which are invariant under conformal transformations. A conformal transformation is basically a transformation which preserves angles between vectors in the space. Conformal field theories have a very wide range of applications. For example, conformal field theories are used to describe critical behavior of statistical systems. A statistical system usually has a characteristic correlation length which describes the scale at which the correlations between individual elements become relevant. A statistical system, when near a critical point, has infinite correlation length and hence behaves like a scale invariant field theory. Now under some technical assumptions, scale invariance often implies invariance under the larger conformal group. Hence a scale invariant theory is likely to be conformally invariant as well and therefore will have properties of a CFT.

Another application of CFTs is in studying quantum field theories. As we know, a QFT comes with an ultraviolet cutoff Λ which describes the energy scale (or equivalently the length scale) upto which it is valid. At higher energies one needs more degrees of freedom to describe the physics. One is often interested in studying how the different degrees of freedom or couplings change with energy scale. This change is described as a **Renormalisation group (RG) flow** via the β functions. A point at which $\beta = 0$ is called a **fixed point**. Near a fixed point, a coupling is called **relevant** if it takes you away from it, while it is called **irrelevant** if it takes you towards it. At a fixed point, all couplings are zero and the theory becomes scale invariant and hence conformally invariant. Therefore, despite starting from general Langrangians with general couplings, only a few small couplings are important at low energies. Therefore, any QFT can be though of as a perturbation to a CFT by relevant operators. In this sense, CFTs lie at the heart of understanding quantum field theories. Therefore studying CFTs helps us understand the emergence of possible fixed points of RG flows and thus understand the space of QFTs.

Another important use for CFTs is that they can allow us to define quantum field theories without any reference to a Lagrangian, since in principle one can solve a CFT without ever mentioning a

Lagrangian. The method of doing so is called the **conformal bootstrap**. CFTs also provide one of the best understandings of quantum gravity via the **AdS/CFT correspondence**. Finally, conformal field theories also have applications in string theory. In string theory, the fundamental entities are one dimensional objects called **strings** (instead of zero dimensional particles). These strings evolve in time giving two dimensional world-sheets. The string dynamics are described by a world-sheet theory which is actually a 2d CFT. Although we will not discuss 2d CFTs here, some of the results in higher dimensional CFTs also hold in 2d and hence are useful in this context.

1.2 Conformal Transformations - Definition

Let M be a d dimensional Lorentzian manifold with a metric g. A **conformal transformation** (more formally a conformal isometry) is a diffeomorphism $\Psi : M \to M$ such that

$$\boldsymbol{(\Psi^* g)_{ab} = \Omega^2 g_{ab}} \tag{1.1}$$

where Ψ^* 'push forward' map with respect to the diffeomorphism Ψ and $\Omega^2(M)$ is a strictly positive function. In terms of local coordinates this becomes

$$g_{\rho\sigma}(x') \frac{\partial x'^{\rho}}{\partial x^{\mu}} \frac{\partial x'^{\sigma}}{\partial x^{\nu}} = \Omega^2(x) g_{\mu\nu}(x) \tag{1.2}$$

In the rest of the report, we specialize to a flat metric, $g_{\mu\nu} = \eta_{\mu\nu}$. The exact signature of $\eta_{\mu\nu}$ will be specified when relevant.

1.3 Infinitesimal Conformal Transformations

Consider an infinitesimal coordinate transformation $\Psi : x \to x'$ given by

$$x'^{\mu} = x^{\mu} + \epsilon^{\mu}(x) + O(\epsilon^2) \tag{1.3}$$

Such a transformation is conformal if it satisfies (1.1). Therefore, substituting (1.3) in (1.1) we have

$$\partial_{\mu}\epsilon_{\nu} + \partial_{\nu}\epsilon_{\mu} = c(x)\eta_{\mu\nu}$$

where $\Omega^2(x) = 1 + c(x)$. The function c(x) can be found out by taking the trace on both sides with $\eta^{\mu\nu}$. This gives $c(x) = \frac{2}{d}(\partial.\epsilon) + O(\epsilon^2)$. Therefore

$$\Omega(x) = \left(1 + \frac{2}{d}(\partial.\epsilon) + O(\epsilon^2)\right)^{1/2} = 1 + \frac{1}{d}(\partial.\epsilon) + O(\epsilon^2) \tag{1.4}$$

and

$$\boxed{\boldsymbol{\partial_{\mu}\epsilon_{\nu} + \partial_{\nu}\epsilon_{\mu} = \frac{2}{d}(\partial.\epsilon)\,\eta_{\mu\nu}}} \tag{1.5}$$

This is called the **Conformal Killing equation.** To find solutions to (1.5), we apply $\partial^\mu \partial^\nu$ on both sides. This gives us

$$\frac{2}{d}(d-1)\,\partial^2(\partial.\epsilon) = 0$$

If $d \neq 1$, this implies $\partial^2(\partial.\epsilon) = 0$. Then taking $\partial^\nu \partial_\rho$ in (1.5) gives

$$\left(1 - \frac{2}{d}\right)\partial_\rho \partial_\mu(\partial.\epsilon) + \partial_\rho \partial^2 \epsilon_\mu = 0$$

Symmetrising this equation and using (1.5) again leads to

$$\left(1 - \frac{2}{d}\right)\partial_\rho \partial_\mu(\partial.\epsilon) + \frac{2}{d}\eta_{\rho\mu}\partial^2(\partial.\epsilon) = 0$$

Then if $d \neq 1$ we get

$$\left(1 - \frac{2}{d}\right)\partial_\rho \partial_\mu(\partial.\epsilon) = 0 \tag{1.6}$$

If $d \neq 2$, we have $\partial_\rho \partial_\mu(\partial.\epsilon) = 0$. Taking two derivatives of (1.5) and using (1.6) we find

$$\partial_\alpha(\partial_\beta \partial_\gamma \epsilon_\delta + \partial_\beta \partial_\delta \epsilon_\gamma) = 0$$

$$\partial_\alpha(\partial_\gamma \partial_\beta \epsilon_\delta + \partial_\gamma \partial_\delta \epsilon_\beta) = 0$$

$$\partial_\alpha(\partial_\delta \partial_\gamma \epsilon_\beta + \partial_\delta \partial_\beta \epsilon_\gamma) = 0$$

These equations together imply $\partial_\alpha \partial_\beta \partial_\gamma \epsilon_\delta = 0$, and hence ϵ_μ is at-most quadratic in x. Then with a little bit of analysis [6], one can show that the most general solution can be written as -

$$\boldsymbol{\epsilon_\mu(x) = a_\mu + m_{\mu\nu}x^\nu + \alpha x_\mu + 2(x.b)x_\mu - x^2 b_\mu} \tag{1.7}$$

where $m_{\mu\nu}$ is anti-symmetric and $a_\mu, m_{\mu\nu}, \alpha, b_\mu \ll 1$. Note that this is only when $d \neq 1$ and $d \neq 2$. One can easily recognize that the first two terms correspond to translations and rotations respectively. The third term corresponds to a scale transformation (dilatation) with $(1+\alpha)$ as the scaling factor. The last two terms correspond to what are known as **special conformal transformations** (SCT). Their physical interpretation will be discussed in the next section.

1.4 Finite Conformal Transformations

We will only discuss continuous finite transformations that can be built from identity by repeated action of infinitesimal transformations. Consider an infinitesimal conformal transformation as in (1.3). Then we can write

$$\frac{\partial x'^\mu}{\partial x^\nu} = \delta^\mu_\nu + \partial_\nu \epsilon^\mu = \left(1 + \frac{1}{d}(\partial.\epsilon)\right)\left(\delta^\mu_\nu + \frac{1}{2}\left(\partial_\nu \epsilon^\mu - \partial^\mu \epsilon_\nu\right)\right) = \Omega(x)\left(\delta^\mu_\nu + m^\mu_\nu\right)$$

Therefore a conformal transformation can in general be thought of as a rotation followed by a local re-scaling. Hence, any finite conformal transformation would look like

$$\boldsymbol{\frac{\partial x'^{\mu}}{\partial x^{\nu}} = \Omega(x)\mathcal{R}^{\mu}{}_{\nu}} \tag{1.8}$$

where $\mathcal{R}^{\mu}{}_{\nu}$ is a Lorentz rotation satisfying $\mathcal{R}^{\mu}{}_{\rho}\mathcal{R}^{\nu}{}_{\sigma}\eta^{\rho\sigma} = \eta^{\mu\nu}$. It is easy to see that (1.8) leads to (1.2) naturally. These define the **conformal group** which consists of combinations of translations, rotations, dilatations and special conformal transformations. The finite versions of the first three are known. The form of finite special conformal transformations can be obtained by integrating the following equation

$$\delta x^{\mu} = 2(b.x)x^{\mu} - x^2 b^{\mu}$$

This immediately gives

$$\delta\left(\frac{x^{\mu}}{x^2}\right) = \frac{\delta x^{\mu}}{x^2} - 2x^{\mu}\left(\frac{x.\delta x}{x^4}\right) = -b^{\mu}$$

Therefore for a finite transformation with parameter b^{μ}

$$\frac{x'^{\mu}}{x'^2} - \frac{x^{\mu}}{x^2} = -b^{\mu}$$

Dividing this equation by the square of itself gives us

$$x'^{\mu} = \frac{x^{\mu} - x^2 b^{\mu}}{1 - 2\,(b.x) - b^2 x^2} \tag{1.9}$$

Here the conformal factor is $\Omega^2(x) = \left(1 - 2b.x + b^2x^2\right)^2$. Now we can describe the physical meaning of SCT. For this we define an inversion map as $I : x^{\mu} \to x'^{\mu} = \frac{x^{\mu}}{x^2}$. This is a discrete conformal transformation. An SCT can then be identified as an inversion followed by a translation again followed by an inversion.

$$x^{\mu} \underrightarrow{(inv.)} \frac{x^{\mu}}{x^2} \underrightarrow{(trans.)} \left(\frac{x^{\mu}}{x^2} - b^{\mu}\right) \underrightarrow{(inv.)} \frac{\left(\frac{x^{\mu}}{x^2} - b^{\mu}\right)}{\left(\frac{x^{\mu}}{x^2} - b^{\mu}\right)^2} = \left(\frac{x^{\mu} - x^2 b^{\mu}}{1 - 2\,(b.x) - b^2 x^2}\right)$$

1.5 Symmetries - Noether's Theorem and Ward Identities

1.5.1 Noether's Theorem

Let us briefly review the Noether's theorem for classical fields and state the important results. Consider a coordinate transformation

$$x'^{\mu} = x^{\mu} + \epsilon^{\mu} \tag{1.10}$$

Consider a field configuration $\left\{\Phi^A(x)\right\}$ (belonging to some symmetry representation) defined on the space. Then a coordinate transformation like (1.3) would lead to a transformation of the fields as well. This transformation comes from two parts - a change in the coordinates from $x \to x'$ as well as (if the field has non zero spin) a change in the shape of the fields from $\Phi^A \to \Phi'^A$. Hence we can define the

variation in field as

$$\delta_\epsilon \Phi^A(x) = \Phi'^A(x') - \Phi^A(x) \tag{1.11}$$

We can also define a modified variation in the field by taking the difference at the same spatial point.

$$\bar{\delta}_\epsilon \Phi^A(x) = \Phi'^A(x) - \Phi^A(x) \tag{1.12}$$

The two variations are related as [17]

$$\bar{\delta}_\epsilon \Phi^A(x) = -\epsilon_\mu \partial^\mu \Phi^A(x) + \delta_\epsilon \Phi^A(x) \tag{1.13}$$

The first term in RHS of (1.13) is the change in field due to change in the coordinates and the second term as the change due to change in shape. The corresponding change in the action of the field is given by

$$\delta S = \int dx' \mathcal{L}\left(\Phi', \partial\Phi'\right) - \int dx \mathcal{L}\left(\Phi, \partial\Phi\right) = \int dx \epsilon_\mu \mathcal{J}_\epsilon^\mu(x) \tag{1.14}$$

where the current $\mathcal{J}_\epsilon^\mu$ is given by

$$\mathcal{J}_\epsilon^\mu = \frac{\partial \mathcal{L}}{\partial\left(\partial_\mu \Phi^A\right)} \bar{\delta}\Phi^A - \mathcal{L}\epsilon^\mu = \frac{\partial \mathcal{L}}{\partial\left(\partial_\mu \Phi^A\right)} \delta\Phi^A - \mathcal{T}_\mathcal{C}^{\mu\nu} \epsilon_\nu \tag{1.15}$$

Where $\mathcal{T}_\mathcal{C}^{\mu\nu} = \frac{\partial \mathcal{L}}{\partial(\partial^\mu \Phi^A)} \partial_\nu \Phi^A - \eta_{\mu\nu}\mathcal{L}$, is the **canonical stress energy tensor**. If the transformation is a symmetry, $\delta S = 0$ and therefore the current is conserved, $\partial_\mu \mathcal{J}_\epsilon^\mu = 0$.

1.5.2 Ward Identities

In general classical invariance doesn't necessarily imply quantum invariance. Apart from invariance of action, quantum invariance has the additional condition that it leaves the path integral measure $\mathfrak{D}\Phi$ invariant, i.e.

$$\int \mathfrak{D}\Phi' ... = \int \mathfrak{D}\Phi ...$$

Just like a symmetry for a classical field leads to a conservation law, a symmetry for a quantum field leads to a Ward Identity. These are given by [16]

$$\boldsymbol{\partial_\mu \left\langle \mathcal{J}_\epsilon^\mu(x) \Phi^1(x_1) \Phi^2(x_2) ... \Phi^n(x_n) \right\rangle = -i\delta(x - x_1) \left\langle \left(\bar{\delta}_\epsilon \Phi^1(x_1)\right) \Phi^2(x_2) ... \Phi^n(x_n) \right\rangle +}$$

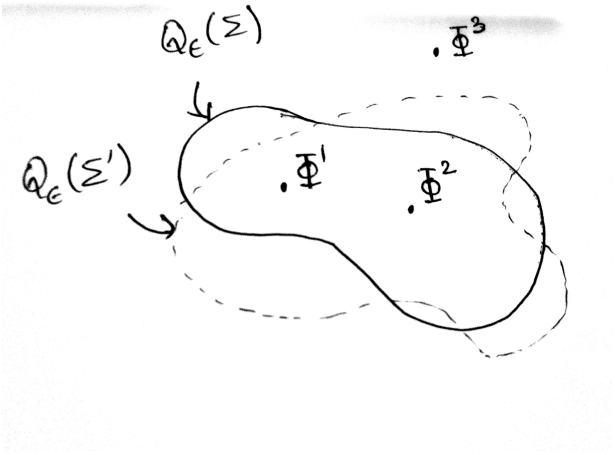
$$\boldsymbol{-i\delta(x - x_2) \left\langle \Phi^1(x_1) \left(\bar{\delta}_\epsilon \Phi(x_2)\right) ... \Phi^n(x_n) \right\rangle ... - i\delta(x - x_n) \left\langle \Phi^1(x_1) \Phi^2(x_2) ... \left(\bar{\delta}_\epsilon \Phi^n(x_n)\right) \right\rangle} \tag{1.16}$$

where $\langle .. \rangle$ is shorthand for the correlation function of the fields and $\mathcal{J}_\epsilon^\mu$ is the associated current as defined in (1.15).
Then we can define a topological surface charge operator $Q_\epsilon(\Sigma)$ as

$$Q_\epsilon(\Sigma) = \int\limits_\Sigma dS_\mu \mathcal{J}_\epsilon^\mu(x) \tag{1.17}$$

where dS_μ is orthogonal to the hypersurface Σ. Then using the Ward identity (1.16) we can see that

the correlator of $Q_\epsilon(\Sigma)$ with any number of fields $\Phi^i(x_i)$ is unchanged as we reshape Σ, as long as it does not cross any of the x_i in the process.



Let Σ be the boundary of a ball containing only one point x. Then using (1.16) and (1.17)

$$\langle Q_\epsilon(\Sigma)\Phi(x)\rangle = -i\left\langle \bar{\delta}_\epsilon \Phi(x)\right\rangle \tag{1.18}$$

Therefore the action of surrounding a field operator with Q_ϵ can be thought of as acting on it by $\bar{\delta}_\epsilon$. But, from the study of quantum field theory [16, 17], we already know that given a particular quantisation of the theory, the RHS is the commutator of the surface charge operator with the field operator

$$\boldsymbol{\left[Q_\epsilon, \Phi^A\right] = -i\left(\bar{\delta}_\epsilon \Phi^A\right)} \tag{1.19}$$

Therefore surrounding with a surface charge has the same effect as commuting with it in any particular quantisation.

$$\boldsymbol{\langle Q_\epsilon(\Sigma)\Phi(x)\rangle = \left\langle \left[Q_\epsilon, \Phi^A\right]\right\rangle} \tag{1.20}$$

Equation (1.20) represents the equivalence of the two ways of thinking about the action of a surface charge operator on the field. We can either think of it as a path integral insertion or as a commutator of operators in a give quantisation.

1.6 The Conformal Group and its Field Representations

Consider a symmetry transformation $x \to x' = g(x)$ belonging to a symmetry group G. The field configuration $\Phi^A(x)$ then belongs to a representation of the symmetry group,

$$\Phi'^A(x') = D(g)^A_B \Phi^B(x) \tag{1.21}$$

Hence, the matrices $D(g)$ form a finite dimensional representation on the space of fields. The corresponding symmetry generators are represented by $\delta\Phi^A$. This representation is called the **spin representation** on fields. We get a more interesting possibility if we consider the variation $\bar{\delta}\Phi^A$ as a representation for the symmetry generators instead. Then we get an infinite dimensional representa-

tion of the symmetry group g. This is because in the first case we are comparing fields at the same space time point with different labels x and x', and hence the base space is $\left\{\Phi^A(x)\right\}$, which is finite dimensional. On the other hand in the second case we are comparing fields at different space-time points and hence the base space is the space of all possible functions $\Phi^A(x)$ and hence is infinite dimensional. Now we know that the two representations are related by, $\bar{\delta}\Phi^A = -\epsilon^\mu \partial_\mu \Phi^A + \delta\Phi^A$, where the first term is the orbital part and the second term is the spin part. To derive the Lie Algebra of the conformal group we first find the generators $\bar{\delta}_\epsilon$ and then compute their commutation relations by hand. Let us compute these for a scalar field. In this case, the advantage is that in equation (1.13) the spin part becomes trivial, $\delta\Phi = 0$. Hence we simply have $\bar{\delta}_\epsilon \Phi = -\epsilon^\mu \partial_\mu \Phi$. Therefore

$$\boldsymbol{\left[\bar{\delta}_{\epsilon_1}, \bar{\delta}_{\epsilon_2}\right] = \left[\epsilon_1^\mu \left(\partial_\mu \epsilon_2^\nu\right) - \epsilon_2^\mu \left(\partial_\mu \epsilon_1^\nu\right)\right] \partial_\nu = \left[\epsilon_1, \epsilon_2\right]^\nu \partial_\nu = \bar{\delta}_{-[\epsilon_1, \epsilon_2]}} \tag{1.22}$$

This is the Lie algebra satisfied by the generators. Also, using (1.13) we can find the generators for all four types of conformal transformations individually -

i]Translations For translations, $\epsilon^\mu = a^\mu$, $Q_P = P^\mu a_\mu$, and therefore $\bar{\delta}_P = -a_\mu \partial^\mu \Longrightarrow [P_\mu, \Phi] = i\partial_\mu \Phi$

ii]Rotations For rotations, $\epsilon^\mu = m_{\mu\nu} x^\nu$, $Q_M = \frac{1}{2} m_{\mu\nu} M^{\mu\nu}$, and therefore $\bar{\delta}_M = m^{\mu\nu}\left(x_\mu \partial_\nu - x_\nu \partial_\mu\right) \Longrightarrow \left[M_{\mu\nu}, \Phi^A\right] = -i\left(x_\mu \partial_\nu - x_\nu \partial_\mu\right)\Phi$

iii]Scaling For scaling, $\epsilon^\mu = \alpha x^\mu$, $Q_D = \alpha D$, $\bar{\delta}_D = -\alpha x_\mu \partial^\mu \Longrightarrow [D, \Phi] = i x^\mu \partial_\mu \Phi$, where D is called the **dilatation operator**.

iv]SCT For SCT, $\epsilon^\mu = 2(x.b)x^\mu - x^2 b^\mu$, $Q_K = b^\mu K_\mu$, $\bar{\delta}_K = -b^\mu\left(2x_\mu (x.\partial) - x^2 \partial_\mu\right) \Longrightarrow [K_\mu, \Phi] = i\left(2x_\mu (x.\partial) - x^2 \partial_\mu\right)\Phi$, where K^μ is called the **special conformal charge operator**.

(Note that in most standard references one usually defines the charge and current after taking out the parameter part (for ex. one usually defines $\mathcal{J}_\epsilon^\mu \sim \mathcal{T}^{\mu\nu}$ and $Q_P \sim P_\mu$, without the parameter a_μ). We instead, keep the parameter (which will anyways cancel in (1.19) and (1.35) from both sides giving the same algebra) so that the charge remains dimensionless and index free.)
Now using $\left|\frac{\partial x'}{\partial x}\right| = \Omega(x)^d \, det(\mathcal{R}) = \Omega(x)^d$, let us define a **primary field** as any field $\Phi^A(x)$ which under a conformal transformation transforms as

$$\boldsymbol{\Phi'^A(x') = \Omega(x)^{-\Delta} D_B^A(\mathcal{R}) \Phi^B(x) = \left|\frac{\partial x'}{\partial x}\right|^{-\Delta/d} D_B^A(\mathcal{R}) \Phi^B(x)} \tag{1.23}$$

where Δ is the scaling dimension of the field, defined by $\Phi^A(\lambda x) = \lambda^{-\Delta}\Phi^A(x)$, and $D_B^A(\mathcal{R})$ is an irreducible representation of the Lorentz subgroup SO(d-1,1) on the fields (the irreducibility condition is important as we'll see later), i.e for example if the field is

i]Scalar we have $D(\mathcal{R}) = \delta_B^A$

ii]Vector we have $D^\mu{}_\nu(\mathcal{R}) = \mathcal{R}^\mu{}_\nu$

iii]Weyl we have $D_\beta^\alpha(\mathcal{R}) = \left(e^{\frac{i}{2} m_{\mu\nu}\sigma^{\mu\nu}}\right)_\beta^\alpha$(see chapter 2 section 2.2.1)

We could've gone more formally about it and derived the most general form of field transformations via the method of induced representations which will be introduced in the next chapter. But for now we would refrain from doing so until chapter 3. For now we take (1.23) as the definition of a primary field.

For a continuous Lorentz rotation we can write the representation matrices as the exponent of generators

$$D_B^A(\mathcal{R}) = \left(e^{\frac{i}{2}m^{\mu\nu}S_{\mu\nu}}\right)_B^A \tag{1.24}$$

where $S_{\mu\nu}$ is the corresponding spin generator in the representation. For an infinitesimal rotation then

$$D_B^A(\mathcal{R}) = \delta_B^A + \frac{i}{2}m^{\mu\nu}\left(S_{\mu\nu}\right)_B^A = \delta_B^A + \frac{i}{4}\left(\partial^\nu\epsilon^\mu - \partial^\mu\epsilon^\nu\right)\left(S_{\mu\nu}\right)_B^A = \delta_B^A - \frac{i}{2}\left(\partial^\mu\epsilon^\nu\right)\left(S_{\mu\nu}\right)_B^A \tag{1.25}$$

where in the last equation we used the fact that $S_{\mu\nu}$ is anti-symmetric (because $m_{\mu\nu}$ is anti-symmetric and therefore only the anti-symmetric part of $S_{\mu\nu}$ would contribute to the product $m^{\mu\nu}S_{\mu\nu}$). Therefore we can derive the infinitesimal change in fields using (1.23).

$$\begin{aligned}
\Phi'^A(x') &= \left(1 + \frac{2}{d}\left(\partial.\epsilon\right)\right)^{-\Delta/2}\left(\delta_B^A - \frac{i}{2}\left(\partial^\mu\epsilon^\nu\right)\left(S_{\mu\nu}\right)_B^A\right)\Phi^B(x) \\
&= \left(1 - \frac{\Delta}{d}\left(\partial.\epsilon\right) + O(\epsilon^2)\right)\left(\delta_B^A - \frac{i}{2}\left(\partial^\mu\epsilon^\nu\right)\left(S_{\mu\nu}\right)_B^A\right)\Phi^B(x) \\
&= \Phi^A(x) - \frac{\Delta}{d}\left(\partial.\epsilon\right)\Phi^A(x) - \frac{i}{2}\left(\partial^\mu\epsilon^\nu\right)\left(S_{\mu\nu}\right)_B^A\Phi^B(x)
\end{aligned}$$

Therefore

$$\delta_\epsilon\Phi^A(x) = -\frac{\Delta}{d}\left(\partial.\epsilon\right)\Phi^A(x) - \frac{i}{2}\left(\partial^\mu\epsilon^\nu\right)\left(S_{\mu\nu}\right)_B^A\Phi^B(x) \tag{1.26}$$

and using (1.13)

$$\bar{\delta}_\epsilon\Phi^A(x) = -\epsilon_\mu\partial^\mu\Phi^A(x) - \frac{\Delta}{d}\left(\partial.\epsilon\right)\Phi^A(x) - \frac{i}{2}\left(\partial^\mu\epsilon^\nu\right)\left(S_{\mu\nu}\right)_B^A\Phi^B(x) \tag{1.27}$$

Therefore using (1.19) the full commutator $[Q_\epsilon, \Phi]$ is

$$\boldsymbol{\left[Q_\epsilon, \Phi(x)\right] = i\left[\epsilon_\mu\partial^\mu + \frac{\Delta}{d}\left(\partial.\epsilon\right)\right]\Phi(x) - \frac{1}{2}\left(\partial^\mu\epsilon^\nu\right)S_{\mu\nu}\Phi(x)} \tag{1.28}$$

where we've suppressed the SO(d-1,1) index for clarity. Using (1.28) we can derive the individual commutators as follows

i]Translations For translations, $\epsilon^\mu = a^\mu$, $Q_P = P^\mu a_\mu$, and therefore

$$\left[P_\mu, \Phi(x)\right] = i\partial_\mu\Phi(x) \tag{1.29}$$

ii]Rotations For rotations, $\epsilon^\mu = m^{\mu\nu}x_\nu$, $Q_L = \frac{1}{2}m^{\mu\nu}M_{\mu\nu}$, and therefore

$$\left[M_{\mu\nu}, \Phi(x)\right] = -i\left(x_\mu\partial_\nu - x_\nu\partial_\mu\right)\Phi(x) + S_{\mu\nu}\Phi(x) \tag{1.30}$$

iii]Scaling For scaling, $\epsilon^\mu = \alpha x^\mu$, $Q_D = \alpha D$, and therefore

$$[D, \Phi] = ix^\mu \partial_\mu \Phi(x) + i\Delta\Phi(x) \tag{1.31}$$

iv]SCT For SCT, $\epsilon^\mu = 2(x.b)x^\mu - x^2 b^\mu$, $Q_K = b^\mu K_\mu$, and therefore

$$[K_\mu, \Phi(x)] = i\left(2x_\mu (x.\partial) - x^2\partial_\mu\right)\Phi(x) + 2i\left(\Delta x_\mu - x^\nu S_{\mu\nu}\right)\Phi(x) \tag{1.32}$$

1.7 Scale vs. Conformal Invariance

The canonical stress energy tensor $\mathcal{T}^{\mu\nu}_{\mathcal{C}}$ is defined as the conserved current corresponding to translations. By this definition, this tensor need not necessarily be symmetric. We can get around this by adding a divergence term to the tensor which will make it symmetric.

$$\mathcal{T}^{\mu\nu} = \mathcal{T}^{\mu\nu}_C + \partial_\sigma \chi^{\sigma\mu\nu} \tag{1.33}$$

where $\chi^{\sigma\mu\nu} = -\chi^{\mu\sigma\nu}$., as required by the conservation law $\partial_\mu \mathcal{T}^{\mu\nu} = 0$ for the new tensor. The charges are not affected by this change. The symmetrisation procedure requires rotational invariance and gives us the Belifante stress tensor [17]. Therefore it is possible to symmetrise the stress tensor in any theory with rotational invariance. There is an alternate way of defining the stress tensor in which it is manifestly symmetric. In this definition the stress tensor is defined as the response to a metric perturbation.

$$\boldsymbol{\mathcal{T}^{\mu\nu} = 2\frac{\delta S}{\delta \eta_{\mu\nu}}} \tag{1.34}$$

For dimension d>2, it can be shown that in any theory with scale invariance it is also possible to make the stress energy tensor traceless [6] The tracelessness of the stress energy tensor then implies full conformal invariance. This can be shown because for a change in coordinates like in equation (1.3), the change in action under a transformation is given by equation (1.13)

$$\delta S = \int dx \partial_\mu \mathcal{J}^\mu = -\int dx \mathcal{T}^{\mu\nu}\partial_\mu \epsilon_\nu = -\frac{1}{2}\int dx \mathcal{T}^{\mu\nu}\left(\partial_\mu\epsilon_\nu + \partial_\nu \epsilon_\mu\right)$$

Then for a conformal transformation, using (1.5) this becomes

$$\delta S = -\int dx. \left(\frac{\partial.\epsilon}{d}\right)\mathcal{T}^\mu{}_\mu$$

Therefore, if the stress tensor is traceless then $\delta S = 0$ and the theory is conformally invariant. Hence we've proved that in d>2 dimensions, scale invariance implies conformal invariance, at least classically.

1.8 Representation of the Conformal Algebra on a Hilbert Space

From (1.19) we expect the charges to satisfy the same algebra as the generators

$$\boldsymbol{[Q_{\epsilon_1}, Q_{\epsilon_2}] = iQ_{-[\epsilon_1,\epsilon_2]}} \tag{1.35}$$

This is not always the case though. To confirm, let us try to prove this. For this we first we need to find the commutation relation $[Q_\epsilon, \mathcal{T}^{\mu\nu}]$. This commutator can only contain combinations of ϵ, ∂, and $\mathcal{T}$. This commutator can be completely fixed by dimensional analysis, the conformal killing equation, conservation and tracelessness of $\mathcal{T}$. Also, we only want terms that are first order in ϵ. Then using dimensional analysis we know that the commutator has dimension d (since Q_ϵ is dimensionless in our convention and $\mathcal{T}$ has dimensions d). We also know that ϵ and ∂ have dimensions -1 and 1 respectively. Therefore if $d > 2$, each term should have equal number of ϵ's and ∂'s and only one $\mathcal{T}$. But since we need only first order terms we must have only one of each. Then the only possible terms are

$$[Q_\epsilon, \mathcal{T}^{\mu\nu}] = i\epsilon.\partial\mathcal{T}^{\mu\nu} \pm i\mathcal{T}^{\mu\nu}\partial.\epsilon \pm i\mathcal{T}^{\rho(\nu}\partial_\rho\epsilon^{\mu)} \pm i\mathcal{T}^{\rho(\mu}\partial^{\nu)}\epsilon_\rho$$

where we have kept a factor of i, by convention. The first term has a positive sign since we must have $[P_\rho, \mathcal{T}^{\mu\nu}] = i\partial_\rho\mathcal{T}^{\mu\nu}$. Also since the stress tensor has dimension d, we must have $[D, \mathcal{T}^{\mu\nu}] = i\left(x.\partial + d\right)\mathcal{T}^{\mu\nu}$. Hence the second term must also have a positive sign and and the third and fourth terms must have opposite signs so that they cancel for this case. Therefore

$$[Q_\epsilon, \mathcal{T}^{\mu\nu}] = i\epsilon.\partial\mathcal{T}^{\mu\nu} + i\mathcal{T}^{\mu\nu}\partial.\epsilon - i\mathcal{T}^{\rho(\nu}\partial_\rho\epsilon^{\mu)} + i\mathcal{T}^{\rho(\mu}\partial^{\nu)}\epsilon_\rho$$

Now that the signs are fixed we can remove the symmetrisation in third and fourth terms by realizing that the combination is already symmetric. Therefore we finally have the commutator as

$$\boldsymbol{[Q_\epsilon, \mathcal{T}^{\mu\nu}] = i\epsilon.\partial\mathcal{T}^{\mu\nu} + i\mathcal{T}^{\mu\nu}\partial.\epsilon - i\mathcal{T}^{\rho\nu}\partial_\rho\epsilon^{\mu} + i\mathcal{T}^{\rho\mu}\partial^{\nu}\epsilon_\rho} \tag{1.36}$$

Notice that for d=2, by dimensional analysis we can also have an additional term with two ∂s, one ϵ, and no $\mathcal{T}$. The only consistent term of this type is $\partial^\mu\partial^\nu\left(\partial.\epsilon\right)$, which by the way also becomes zero in higher dimensions (see equation (1,6)). Now using (1.36) and (1.17) we can derive the algebra (1.35) as follows

$$[Q_{\epsilon_1}, Q_{\epsilon_2}] = -\int dS_\mu\epsilon_{1\nu}\left[\mathcal{T}^{\mu\nu}, Q_{\epsilon_2}\right]$$

$$= i\int dS_\mu\mathcal{T}^{\rho\mu}\left(\epsilon_{1\nu}\partial^\nu\epsilon_{2\rho} - \epsilon_{2\nu}\partial^\nu\epsilon_{1\rho}\right) = i\int dS_\mu\mathcal{T}^{\rho\mu}\left[\epsilon_1, \epsilon_2\right]_\rho = iQ_{-[\epsilon_1,\epsilon_2]}$$

Now let us see what this algebra looks like explicitly. We can treat each of the four types of conformal transformations individually.

Then using (1.35) we get the Lie algebra for conformal group as -

$$\begin{aligned}
[\boldsymbol{D}, \boldsymbol{P_\mu}] =& \boldsymbol{iP_\mu .....(i)} \\
[\boldsymbol{D}, \boldsymbol{K_\mu}] =& \boldsymbol{-iK_\mu .....(ii)} \\
[\boldsymbol{K_\mu}, \boldsymbol{P_\nu}] =& \boldsymbol{-2i\left(\eta_{\mu\nu}D - M_{\mu\nu}\right).....(iii)} \\
[\boldsymbol{K_\rho}, \boldsymbol{M_{\mu\nu}}] =& \boldsymbol{i\left(\eta_{\rho\mu}K_\nu - \eta_{\rho\nu}K_\mu\right).....(iv)} \\
[\boldsymbol{P_\rho}, \boldsymbol{M_{\mu\nu}}] =& \boldsymbol{i\left(\eta_{\rho\nu}P_\mu - \eta_{\rho\mu}P_\nu\right).....(v)} \\
[\boldsymbol{M_{\mu\nu}}, \boldsymbol{M_{\rho\sigma}}] =& \boldsymbol{i\left(\eta_{\mu\rho}M_{\nu\sigma} - \eta_{\nu\rho}M_{\mu\sigma} - \eta_{\mu\sigma}M_{\nu\rho} + \eta_{\nu\sigma}M_{\mu\rho}\right).....(vi)}
\end{aligned} \tag{1.37}$$

The number of generators are = 1 dilatation + d translations + d SCT + $\frac{\mathrm{d(d-1)}}{2}$ rotations = $\frac{(d+2)(d+1)}{2}$. One can define alternate generators $J_{mn} = -J_{nm}$ with $m, n = 0, 1, 2..., d+1$.

$$\boldsymbol{J_{\mu\nu} \equiv M_{\mu\nu}} \tag{1.38}$$

$$\boldsymbol{J_{d,\mu} \equiv \frac{1}{2}\left(P_\mu - K_\mu\right)} \tag{1.39}$$

$$\boldsymbol{J_{d+1,\mu} \equiv \frac{1}{2}\left(P_\mu + K_\mu\right)} \tag{1.40}$$

$$\boldsymbol{J_{d,d+1} \equiv D} \tag{1.41}$$

Then the conformal algebra can be written as

$$\boldsymbol{[J_{mn}, J_{pq}] = i\left(\eta_{mp}J_{nq} - \eta_{mq}J_{np} - \eta_{np}J_{mq} + \eta_{nq}J_{mp}\right)} \tag{1.42}$$

where the d+2 dimensional metric η_{mn} used here has the signature (-,+,+,..-). The algebra in equation (1.42) is simply the algebra of a rotation group in d+2 dimensions. The exact group SO(p,q) depends on the metric used. If $\eta_{\mu\nu}$ is Lorentzian (-,++...+) then η_{mn} is (-,+,+...+,-) as before and the algebra is of SO(d,2). On the other hand, if we take $\eta_{\mu\nu}$ to be Euclidean (++...++) then η_{mn} becomes Lorentzian (+,+...+,-) and the group is SO(d+1,1). More generally, the conformal group of a p+q dimensional space is SO(p+1,q+1). Therefore the conformal group is locally isomorphic to SO(d,2) or SO(d+1,1) depending on the metric used.. This is the core idea behind the embedding space formalism to be discussed in 1.10.

1.9 Primaries and Descendants

In the language of conformal field theory, the term local operator (represented by $\mathcal{O}(x)$) is used to refer to any local field in the theory. These include any functions of the primary fields $\Phi^A(x)$, their derivatives $\partial_\mu\Phi^A(x)$, $\partial_\mu\partial_\nu\Phi^A(x)$, the stress tensor $\mathcal{T}^{\mu\nu}(x)$ etc. In a CFT, an operator can either be a **primary** or a **descendant**. Any operator $\mathcal{O}(x)$ for which

$$[K_\mu, \mathcal{O}(0)] = 0 \Rrightarrow primary \tag{1.43}$$

is a primary. From (1.32) we see that the primary fields $\Phi^A(x)$ are primary operators. Also, from (1.36) it is easy to see that $\mathcal{T}^{\mu\nu}(x)$ are also a primary operator. Any operator that is not a primary, is a descendant. A descendant operator can be obtained from a primary by commuting with P_μ, i.e. taking derivatives. Therefore, $\partial_\mu\Phi^A$, $\partial_\mu\partial_\nu\Phi^A\ldots$ are all descendants. In general, if $\mathcal{O}(x)$ is a primary then $\partial_\mu\mathcal{O}(x)$ is a descendant. To gain more insight into this, note that K_μ and P_μ are lowering and raising operators for dimension respectively. That is if $|\Delta\rangle$ are the eigenstates of D,

$$D\,|\Delta\rangle = i\Delta\,|\Delta\rangle$$

then

$$D\,(P_\mu\,|\Delta\rangle) = ([D,P_\mu] + P_\mu D)\,|\Delta\rangle = (iP_\mu + P_\mu D)\,|\Delta\rangle = i\,(\Delta+1)\,(P_\mu\,|\Delta\rangle)$$

Therefore

$$\boldsymbol{P_\mu\,|\Delta\rangle \propto |\Delta+1\rangle} \tag{1.44}$$

Similarly

$$\boldsymbol{K_\mu\,|\Delta\rangle \propto |\Delta-1\rangle} \tag{1.45}$$

This is also apparent from the commutation relations (37)-(i) and 37-(ii) which look exactly like those of ladder operators. The same happens for local operators (any local operator will be an eigenstate of D in the sense of the following equation)

$$\boldsymbol{[D,\mathcal{O}_\Delta(0)] = -i\Delta\mathcal{O}_\Delta(0)}$$

$$[D,[P_\mu,\mathcal{O}_\Delta(0)]] = [D,i\partial_\mu\mathcal{O}_\Delta(0)] = -\,[P_\mu,[\mathcal{O}_\Delta(0),D]] - [\mathcal{O}_\Delta(0),[D,P_\mu]]$$

$$= [P_\mu,i\Delta\mathcal{O}_\Delta(0)] + [iP_\mu,\mathcal{O}.(0)] = i\,(\Delta+1)\,[P_\mu,\mathcal{O}_\Delta(0)] = -\,(\Delta+1)\,\partial_\mu\mathcal{O}(0)$$

Therefore

$$\boldsymbol{[P_\mu,\mathcal{O}_\Delta(0)] = i\partial_\mu\mathcal{O}_\Delta(0) \propto \mathcal{O}_{\Delta+1}(0)} \tag{1.46}$$

Similarly

$$\boldsymbol{[K_\mu,\mathcal{O}_\Delta(0)] \propto \mathcal{O}_{\Delta-1}(0)} \tag{1.47}$$

Therefore, acting by K_μ and P_μ enables us to move between lower and higher dimensional operators. Therefore primary operators have the lowest dimension in a multiplet. Any physical theory is expected to have a lower bound on the dimension of an operator. This lower bound comes from the condition of unitarity which any physical theory is expected to follow. The implications of unitarity will be discussed in section 1.12 . Therefore, in any physical theory there must exist an operator which when commuted with K_μ gives zero. Therefore, any physical theory must have at least one primary operator. The higher dimensional descendant operators can then be constructed by acting with P_μ, or taking derivatives. Note that a descendant operator does not transform under conformal transformation like in (1.23) since (1.23) leads to (1.43).

1.10 Constraints on Correlators - Embedding Space Formalism

Now that we've seen how fields transform under conformal transformations, our next aim is to analyze the conditions imposed by conformal invariance on the functional form of correlators. The most straightforward way to do this would be to simply impose the relevant symmetry conditions. For example imposing rotational invariance fixes a correlator $\left\langle \mathcal{O}^A(x)\mathcal{O}^B(y)\right\rangle$ to be of the form $f\left(|x-y|\right)$. For imposing conformal invariance, this approach requires a lot of work and is even more involved if we consider operators with non-zero spin. Therefore, we will follow a much easier approach which involves working in a higher dimensional embedding space. From equation (1.42) we know that the algebra of the conformal group in d dimensions is the same as that of rotations in d+2 dimensions. Therefore, we can assume our d dimensional space-time to be embedded in a d+2 dimensional space of appropriate signature. Then rotational invariance in this embedding space is equivalent to conformal invariance in the physical d dimensional space. The constraints defining the embedding must be invariant under these transformations. Let this embedding space have coordinates $\left\{X^A\right\} \equiv \left\{X^0, X^1, X^2 ... X^{d-1}, X^d, X^{d+1}\right\}$. Let's start by defining the light cone coordinates $X^{\pm}$ as

$$X^{\pm} = X^d \pm X^{d+1} \tag{1.48}$$

Now, we need to find a d dimensional subspace embedded in our d+2 dimensional embedding space which can be mapped to the physical d dimensional space in which the theory was constructed. To restrict ourselves to a d dimensional surface we impose the condition that the points on our physical space lie on the null cone

$$X^2 = 0 \Rightarrow X^\mu X_\mu + X^+ X^- = 0 \tag{1.49}$$

This gets rid of one dimension. To remove one more, we impose the constraint that the overall scale of X^A is arbitrary, i.e. all points related by the scaling are equivalent and describe the same point on the physical space

$$X^A \sim \lambda X^A \tag{1.50}$$

Therefore, the light cone is divided into equivalence classes of points with (1.50) as the equivalence relation. These classes then form the required d dimensional space which can be mapped to our physical d dimensional space with coordinates $\{x^\mu\}$. To make a connection we need to choose unique representations for each of these classes by fixing λ. To do this we take the intersection of the light cone with $X^+ = 1$ and map it to our physical d dimensional space.This means we make the following choice for representing each class for $X^+ \neq 0$

$$x^\mu = \frac{X^\mu}{X^+} \Leftrightarrow X^\mu = X^+ x^\mu \tag{1.51}$$

Note that this choice is invariant under (1.50) and hence forms a representation. The inverse map then is

$$X^A = \left\{X^\mu, X^+, X^-\right\} = X^+ \left\{x^\mu, 1, -x^2\right\} \tag{1.52}$$

where X^- is mapped to $-x^2$ in accordance with (1.49). The class with $X^+ = 0$ is simply mapped to infinity. Using these relations

$$\eta_{AB}dX^A dX^B = (X^+)^2\eta_{\mu\nu}dx^\mu dx^\nu + 2\eta_{\mu\nu}X^\mu dX^\nu \frac{dX^+}{X^+} + \eta_{\mu\nu}X^\mu X^\nu \left(\frac{dX^+}{X^+}\right)^2 + dX^+dX^-$$

$$\Rightarrow \eta_{AB}dX^A dX^B = (X^+)^2\eta_{\mu\nu}dx^\mu dx^\nu \tag{1.53}$$

where we've used the equation (1.49) to get rid of the other terms. Hence for any rotations $X^A \to X'^A$, the RHS of (1.53) is invariant and hence

$$(X^+)^2\eta_{\mu\nu}dx^\mu dx^\nu = (X'^+)^2\eta_{\mu\nu}dx'^\mu dx'^\nu$$

$$\eta_{\mu\nu}dx^\mu dx^\nu = \left(\frac{X'^+}{X^+}\right)^2 \eta_{\mu\nu}dx'^\mu dx'^\nu \tag{1.54}$$

Hence the corresponding transformation in our d dimensional space is conformal, as expected. Now to derive constraints on correlators, first we must extend the corresponding operators to the null cone. To keep things simple, we will only consider scalar operators here. Therefore we define an operator $\Theta(X)$ (corresponding to an operator $\mathcal{O}(x)$ on the physical space) on the light cone as

$$\Theta(X) = \left(X^+\right)^{-\Delta} \mathcal{O}\left(X^\mu/X^+\right) \tag{1.55}$$

The inverse map is then

$$\mathcal{O}(x) = \left(X^+\right)^{\Delta} \Theta(X) \tag{1.56}$$

Then by (1.55) we see that $\Theta(X)$ is homogeneous in X

$$\Theta(\lambda X) = \lambda^{-\Delta}\Theta(X) \tag{1.57}$$

This is consistent with (1.50) since the RHS in (1.56) doesn't change under a re-scaling of X and hence we get the same operator on LHS for all equivalent points.

Now we are in a position to derive constraints on correlators of operators due to conformal invariance. We start with the two point function $\langle\mathcal{O}(x)\mathcal{O}(y)\rangle$. Consider the corresponding function $\langle\Theta(X)\Theta(Y)\rangle$ in the larger space. It must have the form

$$\langle\Theta(X)\Theta(Y)\rangle = \frac{c}{(X.Y)^\Delta} \tag{1.58}$$

This is the most general Lorentz invariant function consistent with scaling rule (1.57). Note that due to (1.49) we can't have X^Δ or Y^Δ in the denominator. For the corresponding function $\langle\mathcal{O}(x)\mathcal{O}(y)\rangle$ we have

$$X.Y = (x-y)^2 X^+Y^+$$

and hence

$$\langle\mathcal{O}(x)\mathcal{O}(y)\rangle = \frac{c}{|x-y|^{2\Delta}} \tag{1.59}$$

If we have two different operators $\mathcal{O}_1$ and $\mathcal{O}_2$ with scaling dimensions Δ_1 and Δ_2, then we see that there is no possible Lorentz invariant function since we cannot have $(X_1)^{\Delta_1}$ or $(X_2)^{\Delta_2}$ in the denominator as previously explained. Therefore, $\langle \Theta_1(X_1)\Theta_2(X_2)\rangle = 0$ and hence $\langle \mathcal{O}_1(x_1)\mathcal{O}_2(x_2)\rangle = 0$. Therefore

$$\boldsymbol{\langle \mathcal{O}_1(x_1)\mathcal{O}_2(x_2)\rangle = \begin{cases} \frac{c}{|x_1-x_2|^{2\Delta}} & \Delta_1 = \Delta_2 = \Delta \\ 0 & \Delta_1 \neq \Delta_2 \end{cases}} \tag{1.60}$$

For three point functions, we have to take dot products of X_1, X_2 and X_3 with each other.

$$\langle \Theta_1(X_1)\Theta_2(X_2)\Theta_3(X_3)\rangle = \frac{d}{(X_1.X_2)^a (X_2.X_3)^b (X_3.X_1)^c} \tag{1.61}$$

Then with respect to (1.57), the transformation $X_i \to \lambda X_i$ gives us the constraint

$$2a + 2b + 2c = \Delta_1 + \Delta_2 + \Delta_3 = \Delta \tag{1.62}$$

Scaling X_1, X_2 and X_3 separately in accordance with (1.57)

$$a + c = \Delta_1 \tag{1.63}$$

$$a + b = \Delta_2 \tag{1.64}$$

$$b + c = \Delta_3 \tag{1.65}$$

Solving for a,b,c we get

$$a = \Delta - 2\Delta_3 \tag{1.66}$$

$$b = \Delta - 2\Delta_1 \tag{1.67}$$

$$c = \Delta - 2\Delta_2 \tag{1.68}$$

Therefore finally we have

$$\boldsymbol{\langle \mathcal{O}_1(x_1)\mathcal{O}_2(x_2)\mathcal{O}_3(x_3)\rangle = \frac{d}{|x_1 - x_2|^{\Delta-2\Delta_3} |x_2 - x_3|^{\Delta-2\Delta_1} |x_3 - x_1|^{\Delta-2\Delta_2}}} \tag{1.69}$$

Let's consider taking this to four point functions. We'll have inner products of X_1, X_2, X_3 and X_4 with powers of a,b,c,d,e and f. These powers cannot be fully determined since scaling will give us only five equations for six variables. Hence conformal invariance doesn't fix the higher order correlators completely.

1.11 Radial Quantisation and State-Operator Correspondence

For this and subsequent sections it is more convenient to work in the Euclidean space $\mathbb{R}^d$ instead of the Minkowski space $\mathbb{R}^{d-1,1}$. Therefore $\eta_{\mu\nu} = \delta_{\mu\nu}$. This transition can be made by a trivial Wick rotation and doesn't change any of the results derived above. This change in the metric though will affect the Hermitian properties of the generators. The implication of this transition on the Hermiticity of the

generators will be discussed in section 1.12.

The discussion so far has been independent of a particular choice of quantisation of spacetime. In a general QFT invariant under time translations, the usual choice of quantisation is the one in which we foliate the spacetime in slices on which time is constant. For a Lorentzian theory, we have a preferred time direction. But for a Euclidean theory we can choose any direction and call it time. The foliation can then be done in any arbitrary direction. We then define our Hilbert space on these spatial slices. The evolution of these slices is carried by the time translation generator, i.e. the Hamiltonian operator $\mathcal{H}$.

In a CFT, due to scale invariance it is more convenient to quantise along the radial direction, i.e. to foliate spacetime with spheres $\mathbb{S}^{d-1}$ around the origin and consider evolving states from smaller spheres to larger spheres using the dilatation operator D. This particular choice of quantisation is called **radial quantisation**. Radial quantisation is convenient only in a scale invariant theory because in a scale invariant theory the Hilbert spaces defined on spheres of different sizes are related by a symmetry transformation (dilatation), just like Hilbert spaces defined on constant time slices are related by time translation symmetry. In radial quantisation, a correlator naturally becomes the vacuum expectation value of a radially ordered product

$$\langle \mathcal{O}_1(x_1)\,\mathcal{O}_2(x_2)\,...\mathcal{O}_n(x_n)\rangle = \langle 0\,|\mathcal{R}\mathcal{O}_1(x_1)\,\mathcal{O}_2(x_2)\,...\mathcal{O}_n(x_n)|\,0\rangle \tag{1.70}$$

The Hilbert space is defined on the sphere $\mathbb{S}^{d-1}$. Therefore it will have all the symmetries that leave the sphere $\mathbb{S}^{d-1}$ invariant. Therefore, the Hilbert space must be invariant under the SO(d) group. Therefore the states can be labeled by there scaling dimension Δ and their SO(d) spin label

$$D\,|\Delta,s\rangle^A = i\Delta\,|\Delta,s\rangle^A \tag{1.71}$$

$$M_{\mu\nu}\,|\Delta,s\rangle^A = (S_{\mu\nu})^A_B\,|\Delta,s\rangle^B \tag{1.72}$$

where A,B are the SO(d) labels

Another way of looking at radial quantisation is by noting that the metric of the flat space in radial coordinates is related to that of a cylinder by a Weyl Transformation

$$ds^2_{\mathbb{R}^d} = dr^2 + r^2 ds^2_{\mathbb{S}^{d-1}} = r^2\left(\frac{dr^2}{r^2} + ds^2_{\mathbb{S}^{d-1}}\right)$$

If we now define $\tau = log(r)$, then

$$ds^2_{\mathbb{R}^d} = e^{2\tau}\left(d\tau^2 + ds^2_{\mathbb{S}^{d-1}}\right) = e^{2\tau} ds^2_{\mathbb{R}\times\mathbb{S}^{d-1}} \tag{1.73}$$

Therefore dilatations $r \to \lambda r$ become time translations of radial time $\tau \to \tau + log\lambda$ on the cylinder geometry. This mapping is possible only in conformally invariant theories where the Weyl factor (here $e^{2\tau}$) doesn't matter. More explicitly, if the theory is conformal, then for any metric equivalent to the flat one through a Weyl transformation, the correlation functions are related by simple re-scaling

factors.

$$\langle \mathcal{O}_1(x_1)\mathcal{O}_2(x_2)...\mathcal{O}_n(x_n)\rangle_g = \left(\prod_i \left(e^{2\omega(x_i)}\right)^{\Delta_i}\right)\langle \mathcal{O}_1(x_1)\mathcal{O}_2(x_2)...\mathcal{O}_n(x_n)\rangle_{e^{2\omega}g} \tag{1.74}$$

Thus, given an operator $\mathcal{O}(x)$ in $\mathbb{R}^d$, we can define an operator on the cylinder as

$$\mathcal{O}_{cyl}(\tau, n) = e^{\Delta\tau}\mathcal{O}_{flat}(x = e^{\tau}n) \tag{1.75}$$

The corresponding states on the cylinder live on surfaces with constant τ. On any such surface we can define 'in' states by acting by an operator inserted in the infinite past ($\tau \to -\infty$) and 'out' states by acting by an operator inserted in the infinite future ($\tau \to \infty$) on the vacuum state. The region $\tau \to -\infty$ on the cylinder is mapped back to a single point r=0 in the flat space. Hence for any operator inserted at the origin, there is a unique state defined on the sphere. This correspondence is not found in ordinary QFT because due to lack of conformal invariance the region at $-\infty$ cannot be mapped to a point and hence corresponding to any operator insertion at $-\infty$, there can be many states on the surface. Therefore in a CFT for any operator $\mathcal{O}(0)$ inserted at the origin there exists a unique state $\mathcal{O}(0)|0\rangle$ on the sphere. The vacuum state corresponds to inserting a unit operator. This vacuum state is defined to be conformally invariant, hence

$$D|0\rangle = 0 \tag{1.76}$$

$$M_{\mu\nu}|0\rangle = 0 \tag{1.77}$$

Let us define this map more explicitly. Let $|\Phi_b, r_b\rangle$ be a complete set of field eigenstates on the sphere of radius r_b.

$$\Phi(r, n)|\Phi_b, r_b\rangle = \Phi_b(n)|\Phi_b, r_b\rangle$$

Then the state $\mathcal{O}(0)|0\rangle$ is defined as an insertion in the path integral at the origin

$$\boldsymbol{\langle \Phi_b, r_b|\,\mathcal{O}(0)\big|0\big\rangle = \int_{\Phi(r_b,n)=\Phi_b(n);r\leq r_b} \mathcal{D}\Phi(r,n)e^{iS[\Phi]}\mathcal{O}(0)} \tag{1.78}$$

Therefore the vacuum state is defined as

$$\langle \Phi_b, r_b|\,0\big\rangle = \int_{\Phi(r_b,n)=\Phi_b(n);r\leq r_b} \mathcal{D}\Phi(r,n)e^{iS[\Phi]} \tag{1.79}$$

There also exists an inverse map between operators and states. For any state $|\mathcal{O}\rangle$ on the surface, the corresponding operator $\mathcal{O}(0)$ is defined such that

$$\boldsymbol{\langle \mathcal{O}_1(x_1)\mathcal{O}_2(x_2)...\mathcal{O}(0)\rangle = \big\langle 0\big|\mathcal{R}\mathcal{O}_1(x_1)\mathcal{O}_2(x_2)...\big|\mathcal{O}\big\rangle} \tag{1.80}$$

The correspondence $\mathcal{O}(0) \Leftrightarrow |\mathcal{O}\rangle = \mathcal{O}(0)|0\rangle$ as described above, is called the **state operator correspondence.** Note that according to this definition, the state $\mathcal{O}_\Delta\big|0\big\rangle$ is an eigenstate of the dilatation

operator

$$D\left(\mathcal{O}_\Delta|0\rangle\right) = \left([D,\mathcal{O}_\Delta] + \mathcal{O}_\Delta D\right)\Big|0\Big\rangle = i\Delta\left(\mathcal{O}_\Delta|0\rangle\right)$$

where we've used (1.76) since the vacuum in conformally invariant. Therefore,

$$|\Delta\rangle = \mathcal{O}_\Delta|0\rangle \tag{1.81}$$

Therefore we can interpret the action of a conformal transformation on an operator by taking a commutator. Equivalently, we can interpret it as the direct action on the corresponding state. i.e$[P_\mu, \mathcal{O}(0)] \Leftrightarrow P_\mu|\mathcal{O}\rangle$, $[L_{\mu\nu}, \mathcal{O}(0)] \Leftrightarrow L_{\mu\nu}|\mathcal{O}\rangle$, $[K_\mu, \mathcal{O}(0)] \Leftrightarrow K_\mu|\mathcal{O}\rangle$, $[D, \mathcal{O}(0)] \Leftrightarrow D|\mathcal{O}\rangle$. Then in this sense, any local operator is always a dilatation eigenstate.

1.12 Unitarity Bounds

As mentioned before, one expects a physical theory to be unitary. This unitarity condition leads to lower bounds on the dimension of any operator in a CFT. Formally, the condition of unitarity states that the time evolution operator must be unitary, i.e. they should preserve inner products between states as they evolve in time. This implies that the conserved charges (including the Hamiltonian) or the generators must become Hermitian operators under quantisation. Now we want our theory to be unitary in Lorentzian signature. But since we need Euclidean signature for radial quantisation and state operator correspondence we prefer working in the Wick-rotated Euclidean theory. Therefore we wish to derive conditions imposed on the Euclidean theory by the Unitarity condition in Lorentzian. Unitarity in Lorentzian signature translates to what is called '**Reflection positivity**' in the Euclidean signature. Let $\mathcal{O}_L(t,\bar{x})$ be an operator in Lorentzian signature and $\mathcal{O}_E(t_E,\bar{x})$ be an equivalent operator in Euclidean signature. Then

$$\mathcal{O}_L(t,x) = e^{iHt - i\bar{x}.\bar{P}_L}\mathcal{O}_L(0,0)\,e^{-iHt+i\bar{x}.\bar{P}_L} \tag{1.82}$$

If $\mathcal{O}_L(0,0)$ is Hermitian, then it follows from (1.82) that $\mathcal{O}_L(t,x)$ is also Hermitian. This follows because the Hamiltonian and the momentum are Hermitian and hence the exponents are unitary. Let us Wick rotate to Euclidean signature using the identification $t = -it_E$,

$$\mathcal{O}_E(t_E,x) = e^{Ht_E - i\bar{x}.\bar{P}_L}\mathcal{O}_L(0,0)\,e^{-Ht_E+i\bar{x}.\bar{P}_L} \tag{1.83}$$

Then it is easy to see that the Euclidean operator satisfies

$$\mathcal{O}_E(t_E,\bar{x})^\dagger = \mathcal{O}_E(-t_E,\bar{x}) \tag{1.84}$$

To Wick rotate a tensor operator, we also need to multiply the time components by a -i factor. Therefore for such operators we also need to put a minus sign for each time component. Therefore (1.84) generalizes to

$$\mathcal{O}^{\mu_1\mu_2\ldots\mu_l}(t_E,\bar{x})^\dagger = \theta^{\mu_1}_{\nu_1}\theta^{\mu_2}_{\nu_2}\theta^{\mu_3}_{\nu_3}\ldots\theta^{\mu_l}_{\nu l}\mathcal{O}^{\nu_1\nu_2\ldots\nu_l}(-t_E,x) \tag{1.85}$$

where $\theta^\mu_\nu = \delta^\mu_\nu - 2\delta^\mu_0\delta^0_\nu$. An operator which is Hermitian in Lorentzian signature is called '**real**' in Euclidean signature, i.e. a Euclidean operator is real if it satisfies (1.85). Then by reflection positivity in a Euclidean theory we mean that conserved charges must be real operators in the sense of equation (1.85). Therefore condition of Hermiticity of an operator depends on the choice of quantisation, i.e. which direction we call time. Hence, an operator might be Hermitian in one quantisation but not in the other. We can also think of the opposite question. Given a Euclidean theory that satisfies reflection positivity, can we construct a unitary Lorentzian theory by Wick rotation? The **Osterwalder-Schrader reconstruction theorem** [19] assures that under some technical assumptions this can be done. Hence unitarity and reflection positivity are equivalent properties.

Now we want to impose reflection positivity on the CFT on a cylinder $\mathbb{R} \times S^{d-1}$ as defined in radial quantisation. Therefore, the scalar operators must satisfy

$$\mathcal{O}_{cyl}\left(\tau, n\right)^\dagger = \mathcal{O}(-\tau, n) \tag{1.86}$$

Therefore using (1.75)

$$\mathcal{O}_{cyl}\left(\tau, n\right)^\dagger = e^{\Delta\tau}\mathcal{O}_{flat}(x)^\dagger$$

$$\mathcal{O}_{cyl}\left(-\tau, n\right) = e^{-\Delta\tau}\mathcal{O}_{flat}(x)$$

Therefore using the identification $x^2 = r^2 = e^{2\tau}$ and $x = e^\tau n$, (1.86) translates to

$$\mathcal{O}_{flat}(x)^\dagger = x^{-2\Delta}\mathcal{O}_{flat}\left(\frac{x^\mu}{x^2}\right) \tag{1.87}$$

Therefore the Hermitian conjugation of real operators in radial quantisation must be equivalent to inversion apart from a factor of $x^{-2\Delta}$. For tensor operators, this generalizes to

$$\mathcal{O}^{\mu_1\mu_2\ldots\mu_l}\left(x\right)^\dagger = \mathcal{I}^{\mu_1}_{\nu_1}\left(x\right)\mathcal{I}^{\mu_2}_{\nu_2}\left(x\right)\ldots\mathcal{I}^{\mu_l}_{\nu_l}\left(x\right)x^{-2\Delta}\mathcal{O}^{\nu_1\nu_2..\nu_l}\left(\frac{x^\mu}{x^2}\right) \tag{1.88}$$

where $\mathcal{I}^\mu_\nu\left(x\right) = \delta^\mu_\nu - \frac{2x^\mu x_\nu}{x^2}$ is the **inversion tensor**. Applying (1.88) to the stress tensor we find the condition for the charge operators as

$$Q^\dagger_\epsilon = -Q_{I\epsilon I} \tag{1.89}$$

In particular

$$M^\dagger_{\mu\nu} = M_{\mu\nu} \tag{1.90}$$

$$D^\dagger = -D \tag{1.91}$$

$$P^\dagger_\mu = K_\mu \tag{1.92}$$

$$K^\dagger_\mu = P_\mu \tag{1.93}$$

We can now use these relations to find lower bounds on the dimension of an operator. Let $|\mathcal{O}\rangle$ be a primary state corresponding to a scalar operator $\mathcal{O}(x)$ of dimension Δ. Then by definition, any inner product must be positive. Therefore

$$\left\langle \mathcal{O}\middle| P^\dagger_\nu P_\mu \middle|\mathcal{O}\right\rangle \geq 0$$

$$\Rightarrow \Big\langle \mathcal{O}\Big| K_\nu P_\mu \Big|\mathcal{O}\Big\rangle = \Big\langle \mathcal{O}\Big| \left[K_\nu, P_\mu\right] \Big|\mathcal{O}\Big\rangle = -\Big\langle \mathcal{O}\Big| 2i\left(\eta_{\mu\nu} D - M_{\mu\nu}\right) \Big|\mathcal{O}\Big\rangle$$
$$= 2\Delta\eta_{\mu\nu}\Big\langle \mathcal{O}\Big|\mathcal{O}\Big\rangle \geq 0$$
$$\Rightarrow \Delta \geq 0$$

This gives us the first lower bound. Similarly evaluating

$$\Big\langle \mathcal{O}\Big| P_\nu^\dagger P^{\nu\dagger} P^\mu P_\mu \Big|\mathcal{O}\Big\rangle \geq 0$$

gives a stronger condition: $\Delta = 0$ or $\Delta \geq \frac{d-2}{2}$. If we instead consider operators with spin

$${}_A\Big\langle \mathcal{O}, s\Big| P_\nu^\dagger P_\mu \Big|\mathcal{O}, s\Big\rangle^B \geq 0 \Rightarrow -{}_A\Big\langle \mathcal{O}, s\Big| 2i\left(\eta_{\mu\nu} D - M_{\mu\nu}\right) \Big|\mathcal{O}, s\Big\rangle^B \geq 0 \Rightarrow 2\Delta\eta_{\mu\nu}\delta_A^B - 2i\left(S_{\mu\nu}\right)_A^B \geq 0$$

$$\Rightarrow \Delta \geq 2i \times max.\ eigenvalue\left(S_{\mu\nu}\right) = s + d - 2$$

where we've assumed $s > 0$. The max eigenvalue of $S_{\mu\nu}$ was found out as follows - we write the matrix $(S_{\mu\nu})_A^B$ as

$$(S_{\mu\nu})_A^B = \frac{i}{2}\left(L^{\alpha\beta}\right)_{\mu\nu}(S_{\alpha\beta})_A^B$$

where $\left(L^{\alpha\beta}\right)_{\mu\nu}$ is defined as

$$\left(L^{\alpha\beta}\right)_{\mu\nu} = -i\left(\delta_\mu^\alpha\delta_\nu^\beta - \delta_\nu^\alpha\delta_\mu^\beta\right)$$

This is just the defining representation (vector representation) of SO(d). Therefore we use a familiar trick from quantum mechanics. In ordinary quantum mechanics, while studying spin-orbit interaction one comes across a similar problem where one is required to diagonalize $L^i S_i$. One does that by writing

$$L^i S_i = \frac{1}{2}\left((L+S)^2 - L^2 - S^2\right)$$

where L^2 and S^2 are Casimirs and have eigenvalues $l(l+1)/2$ and $s(s+1)/2$, whereas $(L+S)^2$ is a Casimir of the tensor product $\boldsymbol{l} \otimes \boldsymbol{s}$ and has eigenvalue $j(j+1)/2$ where $|l-s| < j < |l+s|$. Similarly here we have

$$\frac{i}{2}\left(L^{\alpha\beta}\right)_{\mu\nu}(S_{\alpha\beta})_A^B = \frac{i}{4}\left(-Cas\left(R_1\right) - Cas\left(R_s\right) + Cas\left(R_1 \otimes R_s\right)\right)$$

where R_1 is the vector representation (spin-1 traceless symmetric tensor representation) of SO(d) and R_s is a spin-s traceless symmetric tensor representation of SO(d). Using the eigenvalue s(s+d-2) of the Casimir of R_s and the fact that for s>0, $\Rightarrow R_s \otimes R_1 = R_{s-1} \oplus .....$, we have

$$\Delta \geq s + d - 2$$

where we've taken the minimum value of $Cas\left(R_1 \otimes R_s\right)$ which is $Cas\left(R_{s-1}\right)$. Therefore overall we get

the final unitarity bounds as

$$\Delta \Rightarrow \begin{cases} = 0 & \textit{unit operator} \\ \geq \frac{d-2}{2} & s = 0 \\ \geq s + d - 2 & s > 0 \end{cases} \tag{1.94}$$

One may think that we may get stronger bounds if we take inner products of even higher descendant states. It turns out though that the above bounds are sufficient and we don't get anything new by taking inner products of higher descendants.

1.13 Operator Product Expansion

The operator product expansion is a general concept in QFT. The idea is that the product of any two local operators $\mathcal{O}_1(x_1)$ and $\mathcal{O}_2(x_2)$ may be expressed in terms of a series of other local operators at a point x, times coefficient functions which depend on x_1 and x_2. The set of operators in the expansion depend on the particular quantum field theory. For example, for a free scalar field the existence of OPE can be easily derived using the Wick's Theorem.

In CFTs , the OPE acquires some additional properties due to the state-operator correspondence. Let us derive the OPE for a CFT. Consider two operators $\mathcal{O}_1(x)$ and $\mathcal{O}_2(0)$ inserted inside a sphere. These will give a state $\mathcal{O}_1(x)\mathcal{O}_2(0)\left|0\right\rangle$. We can expand this state in terms of dilatation eigenstates. Then by state-operator correspondence, these dilatation eigenstates will be in one to one correspondence with local operators which are either primaries or their derivatives (descendants). Therefore, we can write

$$\boldsymbol{\mathcal{O}_1(x)\mathcal{O}_2(0)\left|0\right\rangle = \sum_{\mathcal{O}\ primaries} \mathcal{C}_{\mathcal{O}}(x,\partial_y)\mathcal{O}(y)\Big|_{y=0}\left|0\right\rangle} \tag{1.95}$$

where $\mathcal{C}_{\mathcal{O}}(x,\partial_y)$ is a power series in ∂_y. We can constrain the coefficients by checking for consistency with the conformal algebra. Let us take a particular term in the series and write it in the form

$$\mathcal{O}_1(x)\mathcal{O}_2(0)\left|0\right\rangle = \frac{constant}{|x|^k}\left[\mathcal{O}(0) + (derivatives\ of\ \mathcal{O}(0))\right]\left|0\right\rangle + (contributions\ from\ other\ primaries)$$

The power k can be fixed by scaling. Applying the dilatation operator

$$D\mathcal{O}_1(x)\mathcal{O}_2(0)\left|0\right\rangle = -i(\Delta_1 + x_\mu\partial^\mu)\mathcal{O}_1(x)\mathcal{O}_2(0)|0\rangle - i\Delta_2\mathcal{O}_1(x)\mathcal{O}_2(0)|0\rangle$$

$$\Rightarrow -\frac{constant}{|x|^k}\left[i\Delta_{\mathcal{O}}O(0) + ...\right]|0\rangle = -i(\Delta_1 + \Delta_2 - k)\frac{constant}{|x|^k}[O(0) + ..]\left|0\right\rangle + ...$$

By comparing the first term on both sides we get $k = \Delta_1 + \Delta_2 - \Delta_{\mathcal{O}}$. Consider now the first descendant term

$$\mathcal{O}_1(x)\mathcal{O}_2(0)|0\rangle = \frac{constant}{|x|^k}[O(0) + cx^\mu\partial_\mu O(0) + ...]|0\rangle + (contributions\ from\ other\ primaries)$$

The coefficient c can be found out by acting on both sides by K_μ to get

$$LHS \Rightarrow K_\mu \mathcal{O}_1(x)\mathcal{O}_2(0)|0\rangle = (\sim x_\mu + \sim x^2\partial_\mu)\mathcal{O}_1(x)\mathcal{O}_2(0)|0\rangle = \frac{constant \times x_\mu}{|x|^k}\left[\mathcal{O}(0) + ...\right]|0\rangle$$

$$RHS \Rightarrow K_\mu \frac{constant}{|x|^k}[O(0) + cx^\nu \partial_\nu O(0) + ...]|0\rangle = \frac{constant}{|x|^k}[K_\mu O(0) + cK_\mu x^\nu \partial_\nu O(0) + ...]|0\rangle$$

$$\sim \frac{constant}{|x|^k}[c\Delta x_\mu O(0) + ...]|0\rangle$$

Matching the coefficients, we see that we can determine c. Analogously, the coefficients of the higher descendants can also be determined recursively. Conformal invariance thus fixes completely the function $\mathcal{C}_\mathcal{O}(x, \partial_y)$, up to a numerical pre-factor $\lambda_\mathcal{O}$. Practically, the above procedure is difficult to carry out. A much easier way to get the coefficients is to just take the correlation function of both sides of (1.95) with a third operator $\mathcal{O}_3(z)$.

$$\langle \mathcal{O}_1(x)\mathcal{O}_2(0)\mathcal{O}_3(z)\rangle = \sum_\mathcal{O} \lambda_\mathcal{O}\mathcal{C}_\mathcal{O}(x, \partial_y)\left\langle \mathcal{O}(y)\Big|_{y=0}\mathcal{O}_3(z)\right\rangle$$

The three-point function on the left-hand side is fixed by conformal invariance. The inner product on the RHS is just a two point function and by (1.60) we know that will be non zero only if $\mathcal{O} = \mathcal{O}_3$. The sum then collapses to a single term.

$$\langle \mathcal{O}_1(x)\mathcal{O}_2(0)\mathcal{O}_3(z)\rangle = \lambda_{\mathcal{O}_3}\mathcal{C}_{\mathcal{O}_3}(x, \partial_y)\left\langle \mathcal{O}_3(y)\Big|_{y=0}\mathcal{O}_3(z)\right\rangle$$

Then by (1.60) and (1.69) we know the functional form of the two point and the three point functions. Substituting it in the above equation and expanding around x=0 we can find the coefficient $\mathcal{C}_{\mathcal{O}_3}$. For example, for the special case $\Delta_1 = \Delta_2 = \Delta$ we get

$$\mathcal{C}_{\mathcal{O}_3}(x, \partial_y) = \frac{1}{|x|^{2\Delta - \Delta_3}}\left[1 + \frac{1}{2}x^\mu\partial_\mu + \alpha x^\mu x^\nu \partial_\mu\partial_\nu + \beta x^2\partial^2 + ...\right] \tag{1.96}$$

where $\alpha = \frac{\Delta_3+2}{8(\Delta_3+1)}$ and $\beta = \frac{\Delta_3}{16\left(\Delta_3 - \frac{\Delta}{2}+1\right)(\Delta_3+1)}$.
In usual QFT an OPE only exists for the short distance limit. But as we saw above, for a CFT it is possible to interpret the OPE as an expansion in the Hilbert space and hence it exists everywhere.

Chapter 2

Super-Poincare Algebra and Its Representations

2.1 Introduction

The story of supersymmetry started with the **Coleman-Mandula theorem**. In 1967, Coleman and Mandula showed that under some technical assumptions, the most general possible symmetry group of the S matrix is the direct product of Poincare group and some internal symmetry group. Therefore this "no-go theorem" ruled out any possibility of a non-trivial intermixing of space-time Poincare symmetries with internal symmetries like gauge symmetries or any global symmetries. It also ruled out the possibility of enlarging the symmetry group of nature to include more symmetries. However, the Coleman-Mandula theorem had a loophole. Coleman and Mandula only considered bosonic symmetry generators. In the early 1970s people started realising this and started extending the Poincare algebra to include fermionic generators. In 1975, **Haag**, **Lopuskanski** and **Sohnius** extended the Coleman-Mandula theorem to include the case of fermionic generators, which relate particles of different spins. They showed that the most general symmetry of the S-matrix is a direct product of super-Poincare and internal symmetries. The super-Poincare algebra is a graded extension of the Poincare group to include symmetry transformations that turn bosons into fermions and fermions into boson. Such transformations are generated by fermionic generators and are called **supersymmetry transformations**. Therefore, there was only one non-trivial way to extend the spacetime symmetries, and that is to incorporate supersymmetry. Since, supersymmetry relates a particle of integer spin to one with half-integer spin, it predicts the existence of particle-superparticle pairs which have the same mass but differ by half a spin. These fermionic generators, in general, can also have non-trivial commutators with internal symmetries leading to what are known as **R-symmetries.**

In this chapter we will start by formally constructing irreducible representations of the Lorentz and Poincare groups. We will discuss representations on fields as well as on the Hilbert space. We will see that the states in these irreducible representations are what one refers to as particles. The results obtained will help understand the super Poincare algebra and its representations, which will be the next topic. We will formally construct the representations of super-Poincare algebra and state some general results. In the case of super Poincare, the states of an irreducible representation are made

up of many particles of different spins. Thus, one can say that a 'particle' in an irreducible super-Poincare representation is made up of many particles of Poincare representation. To avoid confusion, one reserves the word particle for states in the Poincare representation and call the representation states in a super-Poincare irreducible representation as **supermultiplets**.

2.2 Representations of Lorentz and Poincare Groups

2.2.1 Representations of the Lorentz Group

A Lorentz transformation is a homogeneous space-time transformation defined as

$$\boldsymbol{x'^{\mu} = \mathcal{R}^{\mu}_{\nu} x^{\nu}} \tag{2.1}$$

such that

$$\boldsymbol{\mathcal{R}^{\mu}_{\rho} \mathcal{R}^{\nu}_{\sigma} \eta_{\mu\nu} = \eta_{\rho\sigma}} \tag{2.2}$$

This group can be identified as O(3,1). If one specializes to proper orthochronous Lorentz transformations (i.e. those which can be constructed from the identity by their infinitesimal counterparts) one gets the SO(3,1) group.
Consider an infinitesimal Lorentz transformation $\mathcal{R}^{\mu}_{\nu} = \delta^{\mu}_{\nu} + m^{\mu}_{\nu}$. Then any element of the Lorentz group can be written via the exponential map

$$lorentz\ group = \left\{ e^{\frac{i}{2} m_{\mu\nu} M^{\mu\nu}} \right\}$$

where $M_{\mu\nu}$ are the generators of the Lorentz group. The generators are anti-symmetric and hence have six independent components. This connects the group to the Lorentz algebra so(3,1) satisfied by the generators and given by -

$$\boldsymbol{[M_{\mu\nu}, M_{\rho\sigma}] = i\left(\eta_{\mu\rho} M_{\nu\sigma} - \eta_{\nu\rho} M_{\mu\sigma} - \eta_{\mu\sigma} M_{\nu\rho} + \eta_{\nu\sigma} M_{\mu\rho}\right)} \tag{2.3}$$

More precisely, in any representation over a set of objects $\{\phi^a\}$, the transformation looks like

$$\phi'^{a} = \left[e^{\frac{i}{2} m_{\mu\nu} M_R^{\mu\nu}} \right]^{a}_{b} \phi^{b}$$

where $\left(M_R^{\mu\nu}\right)^{a}_{b}$ is a representation of generators on $\{\phi^a\}$ and satisfy (2.3).
The six generators can also be written in terms of two spatial vectors corresponding to spatial rotations and boosts respectively

$$J^i = \frac{1}{2} \epsilon^{ijk} M^{jk} \Rightarrow spatial\ rotations \tag{2.4}$$

$$K^i = M^{i0} \Rightarrow boosts \tag{2.5}$$

Then the Lorentz algebra can then be rewritten as

$$\left[J^i, J^j\right] = i\epsilon^{ijk} J^k \tag{2.6}$$

$$\left[K^i, K^j\right] = -i\epsilon^{ijk}J^k \tag{2.7}$$

$$\left[J^i, K^j\right] = i\epsilon^{ijk}K^k \tag{2.8}$$

As you can see from (2.6) that the generators J^i of spatial rotations generate a SO(3) subgroup of the Lorentz group.

Let us now construct the representations of SO(3,1). The defining representation is the vector representation where the group elements are simply $\mathcal{R}^\mu_\nu$. Taking the infinitesimal form it is easy to show that the generators in the vector representation are given by

$$\left(M^{\mu\nu}_R\right)^\rho_\sigma = \eta^{\nu\rho}\delta^\mu_\sigma - \eta^{\mu\rho}\delta^\nu_\sigma \tag{2.9}$$

Using the vector representation we can easily construct higher dimensional tensor representations by taking direct products. In general, a representation on second rank tensors is reducible to the subspaces of symmetric traceless tensors, trace and anti-symmetric tensors. In the notation of representation theory we denote each representation by their dimension. Therefore a vector representation is denoted by **4** and a scalar by **1** and so on. Hence we have

$$\mathbf{4} \otimes \mathbf{4} = \mathbf{9} \oplus \mathbf{1} \oplus \mathbf{6}$$

The spin of each of these representations is given by their transformation properties under the SO(3) subgroup. For example, a four vector representation (which we denoted by **4**) is irreducible under SO(3,1) but reducible under the SO(3) subgroup since the SO(3) subgroup leaves the time component invariant. Therefore the vector representation is reducible under SO(3) and splits up into a one dimensional scalar representation with spin 0 and a spatial vector representation with spin 1. If we denote the SO(3) representations with their spin **j** instead of their dimension 2j+1, we can write this as

$$\textit{vector rep.} = \mathbf{0} \oplus \mathbf{1}$$

Note that all the above tensor represntations correspond to integer spin. To construct representations which also include spinors we need to consider the representations of the universal covering group SL(2,C) of the Lorentz group. First note that if we rewrite the generators as

$$A^i = \frac{J^i - iK^i}{2} \tag{2.10}$$

$$B^i = \frac{J^i + iK^i}{2} \tag{2.11}$$

The algebra can then be rewritten as

$$\left[A^i, A^j\right] = i\epsilon^{ijk}A^k \tag{2.12}$$

$$\left[B^i, B^j\right] = i\epsilon^{ijk}B^k \tag{2.13}$$

$$\left[A^i, B^j\right] = 0 \tag{2.14}$$

Therefore we see that the SO(3,1) group is locally isomorphic to a direct product of two SU(2) groups

$$SO(3,1) \cong SU(2) \times SU(2)$$

More precisely, one may realise that $SU(2) \times SU(2)$ is the covering group of SO(4) which is the compact version of SO(3,1). On the other hand, the covering group SO(3,1) is SL(2,C). This means that SL(2,C) is homomorphic to SO(3,1), as can be explicitly shown. For this, let us define a map from the Minkowski space M_4 to a space of 2×2 Hermitian complex matrices. This map takes a vector V^μ to a Hermitian matrix $V_{\alpha\dot{\alpha}}$ given by

$$V_{\alpha\dot{\alpha}} = (\sigma_\mu)_{\alpha\dot{\alpha}} V^\mu = \begin{pmatrix} V_0 + V_3 & V_1 - iV_3 \\ V_1 + iV_2 & V_0 - V_3 \end{pmatrix} \tag{2.15}$$

which has $det\,(V) = V^\mu V_\mu$. The inverse map is given by

$$V^\mu = \frac{1}{2} Tr\,[V\bar{\sigma}^\mu] \tag{2.16}$$

Here we've defined $\sigma^\mu = \left(1, \sigma^i\right)$ and $\bar{\sigma}^\mu = \left(1, -\sigma^i\right)$. Now just like elements of the Lorentz group act on vectors V^μ, the elements of SL(2,C) act on the Hermitian matrices $V_{\alpha\dot{\alpha}}$. The action of SL(2,C) matrices N^α_β on $V_{\alpha\dot{\alpha}}$ is defined as

$$V_{\alpha\dot{\alpha}} \longrightarrow N^\alpha_\beta V_{\beta\dot{\gamma}} N^{*\dot{\gamma}}_{\dot{\alpha}} \Leftrightarrow V \longrightarrow NVN^\dagger \tag{2.17}$$

This representation of SL(2,C) on the $V_{\alpha\dot{\alpha}}$ matrices is called the **adjoint representation**. Now since det(N) =1, we see that $det(V) = V_\mu V^\mu$ is invariant under the action of SL(2,C) just like it is under a Lorentz transformation on V_μ. Therefore using (2.16) and (2.17) we can explicitly find the map between SL(2,C) matrices N^α_β and Lorentz rotations $\mathcal{R}^\mu_\nu$. This relation is given by

$$\mathcal{R}^\mu_\nu(N) = \frac{1}{2} Tr\left[\bar{\sigma}^\mu N \sigma_\nu N^\dagger\right] \tag{2.18}$$

On the other hand the inverse relationship is given by [10]

$$N(\mathcal{R}) = \pm \frac{1}{\left\{det\left[\mathcal{R}^\mu_\nu \sigma_\mu \bar{\sigma}^\nu\right]\right\}^{1/2}} \mathcal{R}^\mu_\nu \sigma_\mu \bar{\sigma}^\nu \tag{2.19}$$

Note that from (115) it is clear that this map is a homeomorphism and not an isomorphism.

The irreducible representations of SL(2,C) are labeled by charges (a,b) of the two SU(2) algebras. These irreducible representations are (2a+1)(2b+1) dimensional, where a and b can take half integer and integer values. Now, $J^i = A^i + B^i$. Therefore, in an (a,b) representation, the spin j has values between $|a - b|$ and (a+b). The (0,0) representation is the one dimensional scalar representation with spin 0. The next two irreducible representations are $\left(0, \frac{1}{2}\right)$ and $\left(\frac{1}{2}, 0\right)$. These are both two dimensional and have spin 1/2. The spinors in $\left(\frac{1}{2}, 0\right)$ are denoted by ψ_α and those in $\left(0, \frac{1}{2}\right)$ are denoted by $\bar{\chi}_{\dot{\alpha}}$. These are the left handed and right handed Weyl Spinors respectively. In the $\left(\frac{1}{2}, 0\right)$ representation,

$A^i = \frac{1}{2}\sigma^i$ and $B^i = 0$, therefore

$$J^i = \frac{\sigma^i}{2}$$

$$K^i = \frac{i}{2}\sigma^i$$

and hence

$$M^{\mu\nu} = \sigma^{\mu\nu}$$

where we've defined

$$\sigma^{\mu\nu} = \frac{i}{4}\left(\sigma^\mu\bar{\sigma}^\nu - \sigma^\nu\bar{\sigma}^\mu\right) \tag{2.20}$$

$$\bar{\sigma}^{\mu\nu} = \frac{i}{4}\left(\bar{\sigma}^\mu\sigma^\nu - \bar{\sigma}^\nu\sigma^\mu\right) \tag{2.21}$$

Similarly, in $\left(0, \frac{1}{2}\right)$ representation $A^i = 0$ and $B^i = \frac{1}{2}\sigma^i$

$$J^i = \frac{\sigma^i}{2}$$

$$K^i = -\frac{i}{2}\sigma^i$$

and

$$M^{\mu\nu} = \bar{\sigma}^{\mu\nu}$$

Note that K^i are not Hermitian. This is due to the fact that the Lorentz group is non compact and hence doesn't have finite dimensional unitary irreducible representations with Hermitian generators. The left handed spinors form the fundamental (defining) representation of SL(2,C) while the right handed spinors form its conjugate representation. Therefore in the fundamental representation, the spinors transform as

$$\psi^\alpha = \left(e^{\frac{i}{2}m_{\mu\nu}\sigma^{\mu\nu}}\right)^\alpha_\beta \psi^\beta = N^\alpha_\beta \psi^\beta \tag{2.22}$$

and in the conjugate representation

$$\bar{\chi}^{\dot{\alpha}} = \bar{\chi}^{\dot{\beta}}\left(e^{\frac{i}{2}m_{\mu\nu}\bar{\sigma}^{\mu\nu}}\right)^{\dot{\alpha}}_{\dot{\beta}} = \bar{\chi}^{\dot{\beta}} N^{*\dot{\alpha}}_{\dot{\beta}} \tag{2.23}$$

The Weyl representations discussed above can be used to build all the representations of the Lorentz group.

In the space of spinor representations discussed above, $\epsilon^{\alpha\beta} = -\epsilon_{\alpha\beta}$ is an invariant tensor under SL(2,C)

$$\epsilon'_{\alpha\beta} = N^\beta_\alpha N^\gamma_\beta \epsilon_{\beta\gamma} = \epsilon_{\alpha\beta}.det(N) = \epsilon_{\alpha\beta}$$

since det(N)=1.Therefore we can use it to raise or lower indices as

$$\psi^\alpha = \epsilon^{\alpha\beta}\psi_\beta; \bar{\chi}^{\dot{\alpha}} = \epsilon^{\dot{\alpha}\dot{\beta}}\bar{\chi}_{\dot{\beta}}$$

Next we have the $\left(\frac{1}{2}, \frac{1}{2}\right) = \left(\frac{1}{2}, 0\right) \otimes \left(0, \frac{1}{2}\right)$ representation. The elements in this representation are denoted by $V_{\alpha\dot{\alpha}}$ and correspond to vectors of SO(3,1) via the homeomorphism (2.16). The fact that the

vector representation has spin j=0 or 1 is apparent here. Further we can construct higher dimensional representations (a,b) by taking direct products of Weyl representations.

$$\left(\frac{1}{2},0\right)^{\otimes 2a} \otimes \left(0,\frac{1}{2}\right)^{\otimes 2b} = (a,b) \oplus ...$$

These may be denoted by 2a $'\alpha'$ indices and 2b $'\dot{\alpha}'$ indices like $\Theta_{\alpha_1\alpha_2...\alpha_{2a},\dot{\alpha}_1\dot{\alpha}_2...\dot{\alpha}_{2b}}$. We can obtain the other irreducible terms by the elementary SU(2) rule for $j \neq 0 \Rightarrow \left(\frac{j}{2}\right) \otimes \left(\frac{1}{2}\right) = \left(\frac{j-1}{2}\right) \oplus \left(\frac{j+1}{2}\right)$. For example $\left(\frac{1}{2},0\right) \otimes \left(\frac{1}{2},0\right) = (1,0) \oplus (0,0)$ where (1,0) denotes the symmetric part $\Theta_{\alpha\beta} = \Theta_{\beta\alpha}$ and (0,0) the trace $\epsilon^{\beta\alpha}\Theta_{\alpha\beta}$. In general, in all such cases the (a,b) irreducible representation is given by the symmetrising all the α and $\dot{\alpha}$ indices. Therefore an object in the (a,b) representation looks like

$$\Theta_{\alpha_1\alpha_2...\alpha_{2a},\dot{\alpha}_1\dot{\alpha}_2...\dot{\alpha}_{2b}} = \Theta_{(\alpha_1\alpha_2...\alpha_{2a}),(\dot{\alpha}_1\dot{\alpha}_2...\dot{\alpha}_{2b})}$$

where (.) denotes symmetrisation of the components inside it.
Under a parity transformation, $K^i \to -K^i$ (vector) and $J^i \to J^i$ (pseudo-vector). Therefore under parity transformation, $A^i \longleftrightarrow B^i$. Hence, any object invariant under parity must belong to a representation that is symmetric under (a,b). A Dirac field is defined as a spin 1/2 field that is invariant under Lorentz and parity and hence must belong to the reducible spinor representation $\left(\frac{1}{2},0\right) \oplus \left(0,\frac{1}{2}\right)$. Therefore a four component Dirac spinor has the form

$$\begin{pmatrix} \psi \\ \bar{\chi} \end{pmatrix}$$

Under a Lorentz transformation this transforms as

$$\begin{pmatrix} \psi' \\ \bar{\chi}' \end{pmatrix} = \begin{pmatrix} N & 0 \\ 0 & N^* \end{pmatrix} \begin{pmatrix} \psi \\ \bar{\chi} \end{pmatrix}$$

It is useful to define gamma matrices as

$$\gamma^\mu = \begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix}$$

which satisfy the Clifford Algebra $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$. Then the generators $J^{\mu\nu}$ are represented as

$$\Sigma^{\mu\nu} = \frac{i}{4}\gamma^{\mu\nu} = \begin{pmatrix} \sigma^{\mu\nu} & 0 \\ 0 & \bar{\sigma}^{\mu\nu} \end{pmatrix}$$

Now using the above results we can also construct infinite dimensional representation on fields as in chapter 1. A scalar field is defined as a field which transforms as a Lorentz scalar - $\phi'(x') = \phi(x)$. Therefore the spin representation for it is trivial and it has only the orbital part which as we know is given by

$$\bar{\delta}\phi(x) = -\delta x^\mu \partial_\mu \phi(x) = -im_{\mu\nu}\left(x^\mu\partial^\nu - x^\nu\partial^\mu\right)\phi(x)$$

Similarly a (left handed) Weyl Field is defined as a two-component field $\psi^\alpha(x)$ that transforms as a

(left handed) Weyl spinor under a Lorentz transformation -

$$\psi'^{\alpha}(x') = N^{\alpha}_{\beta}\psi^{\beta}(x) = \left(e^{\frac{i}{2}m_{\mu\nu}\sigma^{\mu\nu}}\right)^{\alpha}_{\beta}\psi^{\beta}$$

Then again, the generators in the field representation are given by

$$\bar{\delta}\psi^{\alpha}(x) = m^{\mu\nu}\left(x_{\mu}\partial_{\nu} - x_{\nu}\partial_{\mu}\right)\psi^{\alpha}(x) + \frac{i}{2}m^{\mu\nu}\left(\sigma_{\mu\nu}\right)^{\alpha}_{\beta}\psi^{\beta}(x)$$

We can define the right handed Weyl field in the same way.

A Dirac Field is a four component field $\Psi^{A}(x)$, which transforms as a Dirac spinor under a Lorentz transformation

$$\Psi'^{A}(x') = \left(e^{\frac{i}{2}m_{\mu\nu}\Sigma^{\mu\nu}}\right)^{A}_{B}\Psi^{B}(x)$$

The generators in the field representation (corresponding to the modified variation) then have the infinitesimal form

$$\bar{\delta}\Psi^{A}(x) = m^{\mu\nu}\left(x_{\mu}\partial_{\nu} - x_{\mu}\partial_{\nu}\right)\Psi^{A}(x) + \frac{i}{2}m^{\mu\nu}\left(\Sigma_{\mu\nu}\right)^{A}_{B}\Psi^{B}(x)$$

2.2.2 Representations of the Poincare Group

The Poincare group is the semi-direct product of the proper orthochronous Lorentz group and the group of space-time translations. It describes the most general symmetry of a Minkowski space-time

$$Poincare = SO(3,1) \rtimes T_4$$

where T_4 is the translation group in four dimensions. The Poincare Group consists of Lorentz rotations generated by $M_{\mu\nu}$ as well as space-time translations which are generated by four momenta P_{μ}. Apart from (2.3), the Poincare algebra is given by

$$\boldsymbol{[P^{\mu}, M^{\rho\sigma}] = i\left(\eta^{\mu\rho}P^{\sigma} - \eta^{\mu\sigma}P^{\rho}\right)} \tag{2.24}$$

In addition, the translation generators P_{μ} commute with each other. We already know the field representations of the Lorentz generators, all we need to do is find the representations for P^{μ}. We require that all fields, irrespective of their transformation property under the Lorentz group, behave as scalars under space-time translation. That is under an infinitesimal translation $x'^{\mu} = x^{\mu} + a^{\mu}$, we have

$$\phi'(x') = \phi(x)$$

and therefore

$$\bar{\delta}\phi(x) = -a^{\mu}\partial_{\mu}\phi(x)$$

Therefore in field representation of the Poincare group, the fields transform as

$$\bar{\delta}\phi^{a} = -a^{\mu}\partial_{\mu}\phi^{a}(x) + m^{\mu\nu}\left(x_{\mu}\partial_{\nu} - x_{\mu}\partial_{\nu}\right)\phi^{a}(x) + \frac{i}{2}m^{\mu\nu}\left(S_{\mu\nu}\right)^{a}_{b}\phi^{b}(x)$$

where $S_{\mu\nu}$ is the spin generator in the corresponding representation.
We are also interested in constructing infinite dimensional representations on the Hilbert space. The Poincare group is non-compact. Therefore there can be no finite dimensional unitary irreducible representations. Since we are constructing infinite dimensional irreducible representations, we can make them unitary. The irreducible unitary representations of the Poincare group on the Hilbert space are what we identify as particles. The approach to be followed here is a special case of the method of induced representations called **Wigner's Little Group Method**. We start by identifying the quadratic Casimirs of the algebra. There are two Casimirs. One is obviously $P_\mu P^\mu$, since it commutes with all the generators. The other one is given by $W_\mu W^\mu$, where W^μ is the **Pauli-Lubanski pseudo-vector** given by

$$W^\mu = -\frac{1}{2}\epsilon^{\mu\nu\rho\sigma} M_{\nu\rho} P_\sigma \tag{2.25}$$

Since $W_\mu W^\mu$ is a scalar, it obviously commutes with $M_{\mu\nu}$. From the explicit expression (2.25) it can be checked that it commutes with P_μ as well. Hence the representation is spanned by one particle states that are labeled by the eigenvalue of P^2, which is just $p^2 = m^2$, the squared mass of the one particle state, and that of W^2. The operator W^μ is just the generalization of spin to four vectors. For example for a particle with non zero mass, in the rest frame $P^\mu = (m, 0, 0, 0)$, we have $W^0 = 0$ and

$$W^i = mJ^i$$

Note that this interpretation cannot be carried to massless particles. Now to construct the representation we need to identify an appropriate little group. Any subgroup of the symmetry group which leaves a particular choice of momentum p^μ invariant is called a '**little group**'. If G_p is a little group that keep p^μ fixed and $p'^\mu = m^\mu_\nu p^\nu$, then it is straightforward to show that

$$G_p \cong G_{p'}$$

Therefore we can classify Poincare representations by the equivalence classes of four vectors p^μ related by a Lorentz transformation, and a representation of G_p. We have the following equivalence classes for p^μ-
i]$p^2 = m^2 > 0$, $p^0 > 0 \Longrightarrow (massive\ particle)$
ii]$p^2 = m^2 > 0$, $p^0 < 0 \Longrightarrow (massive\ antiparticle)$
iii]$p^2 = 0$, $p^0 > 0 \Longrightarrow (massless\ particle)$
iv]$p^2 = 0, p^0 < 0 \Longrightarrow (massless\ antiparticle)$
v]$p^\mu = 0 \Longrightarrow (vaccuum)$
vi]$p^2 < 0 \Longrightarrow (tachyons)$
Out of all the six possibilities, only vi] is non physical. Since the representations of antiparticles are similar to those of particles, we will only discuss cases i] and iii] here.

Massive Representations

For the equivalence class i] we can choose a representation $p^\mu = (m, 0, 0, 0)$, i.e. the rest frame of the particle. Then the little group is the subgroup which keeps this choice of p^μ fixed. This subgroup

is obviously the group of three dimensional rotations. Therefore, the little group can be identified as $SU(2)$. The generators of this subgroup are obviously J^1, J^2, J^3. Also in this case the Casimir is - $W^2 = m^2 J^2$ and hence the massive particles are characterized by the mass m and the spin j which can take half integer and integer values. Therefore $W^2 = m^2 j(j+1)$. This gives us a massive representation. The states in this space can be further labeled by additional quantum numbers corresponding to commuting operators, like momentum and Helicity. Therefore, a state in this representation can be identified as $|m, j, p^\mu, h\rangle$.

Massless Representations

For the equivalence class iii], we don't have a rest frame. We can instead represent the equivalence class with a vector, $p^\mu = (\omega, 0, 0, \omega)$. Then in this frame, $W_0 = -W_3 = -\omega J_3$ and $W_1 = \omega\,(J_1 - K_2)$, $W_2 = \omega\,(J_2 - K_1)$ and hence W^2 is given by

$$W^2 = -\omega^2 \left[(K_2 - J_1)^2 + (K_1 + J_2)^2\right]$$

The little group is then the subgroup which leaves the above choice of p^μ invariant. One can check that this group turns out to be the Euclidean group E(2) (group of translations and rotations in two dimensions) generated by J_3, A, and B, where $A = K_1 + J_2$and $B = K_2 - J_1$. The algebra satisfied by the little group is

$$[J_3, A] = -iB$$

$$[J_3, B] = iA$$

$$[A, B] = 0$$

which is that of E(2) as expected. The Casimir W^2 becomes

$$W^2 = -\omega^2 \left[A^2 + B^2\right]$$

Therefore we can label the states in the massless representation by the eigenvalues a,b and J_3. It turns out that the representations with $a, b \neq 0$ are not physical. Therefore we only consider representations in which $a = b = 0$ and therefore $W^2 = 0$. Then on these states the little group simply becomes SO(2) since we've removed the translations. Therefore such states can be represented simply by the eigenvalue h of J_3, which is just the Helicity (since $\bar{p}$ is in z direction in our choice of frame). Therefore massless particles are only characterized by their Helicity $h = \hat{p}.\bar{J}$. The Helicity h can take half integer and integer values and correspond to states of different particles. Since momentum commutes with Helicity, a one particle state in the massless representation can be identified as $|p^\mu, h\rangle$.

2.3 Supersymmetry Algebra

2.3.1 Graded Lie Algebra

To include fermionic generators in our algebra we need to consider a generalization of Lie algebra called a **graded Lie algebra**. Let us first define what we mean by a **graded algebra**. For our case we only

need to consider the $\mathcal{Z}_2$ grading. A **$\mathcal{Z}_2$ graded algebra** consists of a vector space L which is a direct sum of two vector spaces L_0 and L_1

$$L = L_0 \oplus L_1$$

and a product $\circ : L \times L \longrightarrow L$ such that

- If $u_1, u_2 \in L_0$, then $u_1 \circ u_2 \in L_0$
- If $u \in L_0, v \in L_1$, then $u \circ v \in L_1$
- If $v_1, v_2 \in L_1$, then $v_1 \circ v_2 \in L_0$

A **$\mathcal{Z}_2$ graded Lie algebra** (also called a **Lie superalgebra**) can be defined by modifying the above product as follows

- $u_1 \circ u_2 = -u_2 \circ u_1$ for $u_1, u_2 \in L_0$
- $u \circ v = -v \circ u$ for $u \in L_0, v \in L_1$
- $v_1 \circ v_2 = v_2 \circ v_1$ for $v_1, v_2 \in L_1$
- The product $\circ$ satisfies the generalized Jacobi identity, i.e if $x_a \in L_a$ (with i=0 or 1) then

$$x_a \circ (x_b \circ x_c)\,(-1)^{ac} + x_b \circ (x_c \circ x_a)\,(-1)^{ba} + x_c \circ (x_a \circ x_b)\,(-1)^{cb} = 0 \tag{2.26}$$

Note that this is not a Lie algebra since the product is not necessarily anti-symmetric. Although, the product $\circ$ does define a Lie algebra on the subspace L_0. The subspace L_0 is called the **bosonic sector** and the subspace L_1 is called the **fermionic sector**.

2.3.2 Super Poincare Algebra

To bypass the restrictions of the Coleman-Mandula theorem, we wish to extend the Poincare algebra to include fermionic generators. We do this by defining a Lie superalgebra as defined above with the bosonic sector L_0 spanned by the Poincare generators and the fermionic sector L_1 spanned by new generators Q^I_α and $\bar{Q}_{\dot{\alpha}I}$. Here, the fermionic generators are realized as left and right handed Weyl spinors and hence α and $\dot{\alpha}$ are spinor indices in the left (fundamental) and right handed (anti-fundamental) spinor representation. The product $\circ$ is defined as a commutator or anti-commutator depending on the generators involved

$$x_a \circ x_b = x_a x_b - (-1)^{ab}\, x_b x_a \tag{2.27}$$

The algebra then has the structure

$$x_a \circ x_b = x_a x_b - (-1)^{ab}\, x_b x_a = i f_{abc} x_c \tag{2.28}$$

The indices $I = 1, 2, \ldots \mathcal{N}$ (writing index 'I' up or down is the same thing since there is no invariant tensor with this index). For the case $\mathcal{N} = 1$ the algebra is called a **simple supersymmetry algebra**

and for $\mathcal{N} > 1$ it is called **extended supersymmetry algebra**. The fermionic generators Q are called **supercharges** and on the Hilbert space have the effect of turning a bosonic state into a fermionic state and vice-versa. The commutators/anti-commutators of the generators can be derived by imposing the Jacobi identities [3]. Then we have the SuSy algebra (apart from the commutators of the Poincare algebra) as

$$\begin{aligned}
\left\{Q^I_\alpha, \bar{Q}_{\dot{\alpha}J}\right\} &= 2\left(\sigma^\mu\right)_{\alpha\dot{\alpha}} P_\mu \delta^I_J = 2P_{\alpha\dot{\alpha}}\delta^I_J \ldots\ldots\ldots (i) \\
\left\{Q^I_\alpha, Q^J_\beta\right\} &= \epsilon_{\alpha\beta} Z^{IJ} \ldots\ldots\ldots (ii) \\
\left\{\bar{Q}_{\dot{\alpha}I}, \bar{Q}_{\dot{\beta}J}\right\} &= \epsilon_{\dot{\alpha}\dot{\beta}} \left(Z^\dagger\right)^{IJ} \ldots\ldots\ldots (iii) \\
\left[Q^I_\alpha, M^{\mu\nu}\right] &= \left(\sigma^{\mu\nu}\right)^\beta_\alpha Q^I_\beta \Longrightarrow \left[Q^I_\alpha, M^\gamma_\beta\right] = \frac{1}{2}\delta^\gamma_\beta Q^I_\alpha - \delta^\gamma_\alpha Q^I_\beta \ldots\ldots\ldots (iv) \\
\left[\bar{Q}_{\dot{\alpha}I}, M^{\mu\nu}\right] &= \left(\bar{\sigma}^{\mu\nu}\right)^{\dot{\beta}}_{\dot{\alpha}} \bar{Q}_{\dot{\beta}I} \Longrightarrow \left[\bar{Q}_{\dot{\alpha}I}, \bar{M}^{\dot{\gamma}}_{\dot{\beta}}\right] = -\frac{1}{2}\delta^{\dot{\gamma}}_{\dot{\beta}}\bar{Q}_{\dot{\alpha}I} + \delta^{\dot{\gamma}}_{\dot{\alpha}}\bar{Q}_{\dot{\beta}I} \ldots\ldots\ldots (v) \\
\left[Q^I_\alpha, P^\mu\right] &= 0 = \left[Q^I_\alpha, P_{\beta\dot{\beta}}\right] \ldots\ldots\ldots (vi)
\end{aligned} \tag{2.29}$$

where $P_{\alpha\dot{\alpha}} = \left(\sigma^\mu P_\mu\right)_{\alpha\dot{\alpha}}, \tilde{P}^{\dot{\alpha}\alpha} = \left(\bar{\sigma}^\mu P_\mu\right)^{\dot{\alpha}\alpha}$ and $M^\beta_\alpha = \left(\sigma^{\mu\nu} M_{\mu\nu}\right)^\beta_\alpha, \bar{M}^{\dot{\beta}}_{\dot{\alpha}} = \left(\bar{\sigma}^{\mu\nu} M_{\mu\nu}\right)^{\dot{\beta}}_{\dot{\alpha}}$ and their conjugates defined similarly in accordance with the correspondence (2.15). The rest of the commutators/anti-commutators not mentioned above (apart from Poincare) are zero. The first commutator verifies the fact that SuSy is a space-time symmetry since we get a translation in RHS from an anti-commutator of fermionic charges. The quantity Z^{IJ} is required to be anti-symmetric (since the LHS is symmetric in exchanging $I \leftrightarrow J$ and $\alpha \leftrightarrow \beta$ simultaneously and $\epsilon_{\alpha\beta}$ is anti-symmteric) and is called the **central charge**. The central charge commutes with all the generators. Due to their antisymmetry, the central charges are zero for $\mathcal{N} = 1$. Apart from these there might be some internal symmetry generated by generators B_l satisfying

$$[B_l, B_m] = i f_{lmn} B_n$$

These generators by definition commute with the Poincare generators. For most cases, the internal symmetries also commute with the supersymmetry generators. For some cases though, we might have some internal symmetries which have a non-trivial commutator with the SuSy generators. Therefore if a subgroup $span\{R_l\}$ of the internal symmetry group $span\{B_l\}$ has a non trivial commutator with supercharges,

$$\left[Q^I_\alpha, R_l\right] = \left(\mathcal{R}_l\right)^I_J Q^J_\alpha \neq 0 \tag{2.30}$$

then such a subgroup is called an **R-symmetry.**

2.4 Representations of Super Poincare Algebra

2.4.1 Some General Results on the Representations of Super Poincare Algebra

We state some general results related to representations of SuSy algebra on the Hilbert space.

1. The energy of any state in the representation of SuSy algebra is always positive. This can be shown using the fact that $Q^\dagger = \bar{Q}$ and the first anticommutator in the algebra. Consider any

arbitrary state $|\psi\rangle$ then using the fact that $Tr\left(\sigma^{\mu}\right)=\sigma^{\mu}_{\alpha\alpha}=2\delta^{\mu 0}$ we have

$$2\sigma^{\mu}_{\alpha\alpha}\left\langle\psi\right|P_{\mu}\left|\psi\right\rangle=\left\langle\psi\right|\left\{Q^{I}_{\alpha},\bar{Q}_{\alpha I}\right\}\left|\psi\right\rangle=\left\langle\psi\right|\left\{Q^{I}_{\alpha},Q^{I\dagger}_{\alpha}\right\}\left|\psi\right\rangle=\left\|Q^{I}_{\alpha}|\psi\rangle\right\|^{2}+\left\|\bar{Q}_{\alpha I}|\psi\rangle\right\|^{2}\geq 0$$

$$\Rightarrow 4\delta^{\mu 0}\left\langle\psi\right|P_{\mu}\left|\psi\right\rangle=4\left\langle\psi\right|P_{0}\left|\psi\right\rangle\geq 0$$

2. There are equal number of bosonic and fermionic states in a supermultiplet. To prove this we define the fermion number operator $(-1)^{n_F}$ whose action on the Hilbert space is as follows

$$(-1)^{n_F}\left|B\right\rangle=\left|B\right\rangle$$

$$(-1)^{n_F}\left|F\right\rangle=-\left|F\right\rangle$$

where $|B\rangle$ and $|F\rangle$ denote bosonic and fermionic states respectively. Then

$$(-1)^{n_F}Q^{I}_{\alpha}\left|F\right\rangle=Q^{I}_{\alpha}\left|F\right\rangle=-Q^{I}_{\alpha}(-1)^{n_F}\left|F\right\rangle\Rightarrow\left\{(-1)^{n_F},Q^{I}_{\alpha}\right\}\left|F\right\rangle=0$$

and

$$(-1)^{n_F}Q^{I}_{\alpha}\left|B\right\rangle=-Q^{I}_{\alpha}\left|B\right\rangle=-Q^{I}_{\alpha}(-1)^{n_F}\left|B\right\rangle\Rightarrow\left\{(-1)^{n_F},Q^{I}_{\alpha}\right\}\left|B\right\rangle=0$$

therefore for all states

$$\left\{(-1)^{n_F},Q^{I}_{\alpha}\right\}=0 \tag{2.31}$$

Then

$$Tr\left((-1)^{n_F}\left\{Q^{I}_{\alpha},\bar{Q}_{\dot{\beta}J}\right\}\right)=Tr\left((-1)^{n_F}Q^{I}_{\alpha}\bar{Q}_{\dot{\beta}J}+(-1)^{n_F}\bar{Q}_{\dot{\beta}J}Q^{I}_{\alpha}\right)$$

$$=Tr\left(-Q^{I}_{\alpha}(-1)^{n_F}\bar{Q}_{\dot{\beta}J}+Q^{I}_{\alpha}(-1)^{n_F}\bar{Q}_{\dot{\beta}J}\right)=0$$

Also

$$Tr\left((-1)^{n_F}\left\{Q^{I}_{\alpha},\bar{Q}_{\dot{\beta}J}\right\}\right)=Tr\left((-1)^{n_F}2\left(\sigma^{\mu}\right)_{\alpha\dot{\alpha}}P_{\mu}\delta^{I}_{J}\right)=2\left(\sigma^{\mu}\right)_{\alpha\dot{\alpha}}p_{\mu}\delta^{I}_{J}Tr\left((-1)^{n_F}\right)$$

Now if we assume that the eigenvalue p_μ of P_μ is non-zero, then

$$Tr\left((-1)^{n_F}\right)=0$$

$$\Rightarrow\sum_{B}\left\langle B\right|(-1)^{n_F}\left|B\right\rangle+\sum_{F}\left\langle F\right|(-1)^{n_F}\left|F\right\rangle=\sum_{B}\left\langle B|B\right\rangle-\sum_{F}\left\langle F|F\right\rangle=n_B-n_F=0$$

Therefore $n_B=n_F$ for any multiplet. The result does not hold when $p_\mu=0$ like in the case of a QFT vaccuum. Then the quantity $Tr\left((-1)^{n_F}\right)$ is known as the **Witten Index.**

3. The Casimirs for Super Poincare group are not the same as those that we found for Poincare. $C_1=P_\mu P^\mu$ is still a Casimir but $W_\mu W^\mu$ is not since it doesn't commute with the supercharges. For $\mathcal{N}=1$, the new Casimir denoted as $C_2=C_{\mu\nu}C^{\mu\nu}$ is a bit complicated. We define

$$B_{\mu}=W_{\mu}-\frac{1}{4}\bar{Q}_{\dot{\alpha}}\left(\bar{\sigma}_{\mu}\right)^{\dot{\alpha}\dot{\beta}}Q_{\beta}$$

then

$$C_{\mu\nu} = B_\mu P_\nu - B_\nu P_\mu$$
$$C_2 = C_{\mu\nu} C^{\mu\nu} \tag{2.32}$$

$C_{\mu\nu}$ is called the **superspin.**

2.4.2 Massless Representations

Let us first build the massless representations. Just like in the Poincare case we can specialize to a frame with $P^\mu = (\omega, 0, 0, \omega)$. Again the Casimirs P^2 and $C^{\mu\nu}C_{\mu\nu}$ are zero and therefore each state is characterised by momentum and Helicity $|p^\mu, h\rangle$. Also

$$\left\{Q^I_\alpha, \bar{Q}_{\dot{\alpha}J}\right\} = 2\left(\sigma^\mu\right)_{\alpha\dot{\alpha}} P_\mu \delta^{IJ} = 4\omega\delta^{IJ}\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \tag{2.33}$$

implying Q^I_1 is zero in this representation. Therefore the central charges are automatically zero since $\mathcal{Z}^{IJ}\epsilon_{12} = \left\{Q^I_1, Q^J_2\right\} = 0$. We can define $\mathcal{N}$ sets of ladder operators as

$$a_I = \frac{Q^I_2}{\sqrt{4\omega}}; a^\dagger_I = \frac{\bar{Q}_{\dot{2}I}}{\sqrt{4\omega}} \tag{2.34}$$

Then using (2.33)

$$\left\{a_I, a^\dagger_J\right\} = \delta_{IJ}$$

Also from (2.29)-(iv)

$$\left[J_3, a^\dagger_I\right] = \frac{1}{2}a^\dagger_I; \left[J_3, a_I\right] = -\frac{1}{2}a_I \tag{2.35}$$

Therefore given a state $|p^\mu, h\rangle$ we have

$$a^\dagger_I |p^\mu, h\rangle \propto \left|p^\mu, h + \frac{1}{2}\right\rangle; a_I |p^\mu, h\rangle \propto \left|p^\mu, h - \frac{1}{2}\right\rangle \tag{2.36}$$

Applying $a^\dagger_I$ twice would give zero since $\left\{a^\dagger_I, a^\dagger_I\right\} = 0$. Therefore starting from a state $|\Omega\rangle$ annihilated by all a_Is (known as the **Clifford vacuum**), we can build the representation by acting by different $a^\dagger_I$s to get states with increasing Helicity.

States	Helicity	Number of States
$\|\Omega\rangle$	h_0	${}^{\mathcal{N}}C_0 = 1$
$a_I^\dagger \|\Omega\rangle$	$h_0 + \frac{1}{2}$	${}^{\mathcal{N}}C_1 = \mathcal{N}$
$a_J^\dagger a_I^\dagger \|\Omega\rangle$	$h_0 + 1$	${}^{\mathcal{N}}C_2 = \frac{\mathcal{N}(\mathcal{N}-1)}{2}$
$\vdots$	$\vdots$	$\vdots$
$a_{I_k}^\dagger ... a_{I_1}^\dagger \|\Omega\rangle$	$h_0 + \frac{k}{2}$	${}^{\mathcal{N}}C_k$
$\vdots$	$\vdots$	$\vdots$
$a_{I_{\mathcal{N}-1}}^\dagger ... a_{I_1}^\dagger \|\Omega\rangle$	$h_0 + \frac{\mathcal{N}-1}{2}$	${}^{\mathcal{N}}C_{\mathcal{N}} = \mathcal{N}$
$a_{I_{\mathcal{N}}}^\dagger a_{I_{\mathcal{N}-1}}^\dagger ... a_{I_1}^\dagger \|\Omega\rangle$	$h_0 + \frac{\mathcal{N}}{2}$	${}^{\mathcal{N}}C_{\mathcal{N}} = 1$

Therefore total number of states in the representation are $\sum_{k=0}^{k=\mathcal{N}} {}^{\mathcal{N}}C_k = 2^{\mathcal{N}}$. Note that in any multiplet $h_{max} - h_{min} = \frac{\mathcal{N}}{2}$. Also we know that for renormalizable theories $|h| \leq 1$ and hence $\mathcal{N} \leq 4$. Now CPT flips the sign of the Helicity. Therefore, in order to have a CPT-invariant theory one should add the CPT conjugates to the supermultiplet, i.e for each state with Helicity h we must add a state with Helicity $-h$. This is not required if the multiplet is already CPT invariant.

2.4.3 Massive Representations - BPS States

The massive case is a bit involved due to the presence of central charges. For simplicity let us first discuss the case where central charges $\mathcal{Z}^{IJ}$ are zero. This is always true for $\mathcal{N} = 1$. Here again like in the Poincare case we can specialize to a frame with $P_\mu = (m, 0, 0, 0)$. Then we have

$$\left\{Q_\alpha^I, \bar{Q}_{\dot{\alpha}J}\right\} = 2\left(\sigma^\mu\right)_{\alpha\dot{\alpha}} P_\mu \delta^{IJ} = 2m\delta^{IJ} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Therefore we can define $2\mathcal{N}$ sets of ladder operators

$$a_{\alpha I} = \frac{Q_\alpha^I}{\sqrt{2m}}; a_{\alpha I}^\dagger = \frac{\bar{Q}_{\dot{\alpha}I}}{\sqrt{2m}} \tag{2.37}$$

But now we have

$$\left[J_3, a_{2I}^\dagger\right] = \frac{1}{2} a_{2I}^\dagger; [J_3, a_{2I}] = -\frac{1}{2} a_{2I} \tag{2.38}$$

$$\left[J_3, a_{1I}^\dagger\right] = -\frac{1}{2} a_I^\dagger; [J_3, a_{1I}] = \frac{1}{2} a_{1I} \tag{2.39}$$

Therefore

$$a_{2I}^\dagger \left|j_3\right\rangle \propto \left|j_3 + \frac{1}{2}\right\rangle; a_{2I} \left|j_3\right\rangle \propto \left|j_3 - \frac{1}{2}\right\rangle \tag{2.40}$$

$$a_{1I}^\dagger \left|j_3\right\rangle \propto \left|j_3 - \frac{1}{2}\right\rangle; a_{2I} \left|j_3\right\rangle \propto \left|j_3 + \frac{1}{2}\right\rangle \tag{2.41}$$

Now any state is characterized by the eigenvalues of the Casimirs P^2 and $C_{\mu\nu}C^{\mu\nu}$. If we start with a state $|\Omega\rangle$ annihilated by all $a_{\alpha I}$, then for that particular state (with our choice of momentum)

$$B_\mu \left|\Omega\right\rangle = W_\mu \left|\Omega\right\rangle$$

$$C_{\mu\nu}C^{\mu\nu}\left|\Omega\right\rangle = \left(W_\mu P_\nu - W_\nu P_\mu\right)\left(W^\mu P^\nu - W^\nu P^\mu\right)\left|\Omega\right\rangle = -2m^4 J^2\left|\Omega\right\rangle$$

and therefore for this state the superspin and the spin are the same. The state $|\Omega\rangle$ can then be written as $|\Omega\rangle = |m, s = j_0, p_\mu, j_3\rangle \sim |m, j_0, p_\mu, j_3\rangle$, where we've used 's' to denote eigenvalues of Casimir $C_2 = C_{\mu\nu}C^{\mu\nu}$. The state $|\Omega\rangle$ has a degeneracy of $2j_0 + 1$ since j_3 can take those many different values. The whole multiplet of $2^{2\mathcal{N}}$ states can then be obtained by acting with different combinations of creation operators. Consider for example the case of $\mathcal{N} = 1$. If we start with $|\Omega\rangle = |m, j_0, p_\mu, j_3\rangle$ we get the following four states

$$\begin{aligned}
\left|\Omega\right\rangle &= \left|m, j_0, p_\mu, j_3\right\rangle \\
a_1^\dagger\left|\Omega\right\rangle &= \left|m, j_0, p_\mu, j_3 + \frac{1}{2}\right\rangle \\
a_2^\dagger\left|\Omega\right\rangle &= \left|m, j_0, p_\mu, j_3 - \frac{1}{2}\right\rangle \\
a_1^\dagger a_2^\dagger\left|\Omega\right\rangle &= \left|m, j_0, p_\mu, j_3\right\rangle = \left|\Omega'\right\rangle
\end{aligned} \tag{2.42}$$

These four states each have $2j_0 + 1$ degeneracy. The states in the middle two equations can be written as linear combinations of states with spin $j = j_0 + \frac{1}{2}$ and $j = j_0 - \frac{1}{2}$. The states $|\Omega\rangle$ and $|\Omega'\rangle$ have the same eigenvalues. These states are different though because $|\Omega'\rangle$ is not annihilated by a_αs. In fact, $a_\alpha^\dagger\left|\Omega'\right\rangle = 0$ instead.

Now let us consider the case where we have non-trivial central charges $\mathcal{Z}^{IJ}$. Then we cannot use the above definition of ladder operators since we won't have $\{a_{\alpha I}, a_{\beta J}\} = \left\{a_{\alpha I}^\dagger, a_{\beta J}^\dagger\right\} = 0$ like before. To get past this we first do a unitary transformation on $\mathcal{Z}^{IJ}$ to bring it to block diagonal form

$$\mathcal{Z}^{IJ} = \begin{pmatrix}
0 & \mathcal{Z}_1 & & & & & & \\
-\mathcal{Z}_1 & 0 & & & & & & \\
 & & 0 & \mathcal{Z}_2 & & & & \\
 & & -\mathcal{Z}_2 & 0 & & & & \\
 & & & & \cdots & \cdots & & \\
 & & & & \cdots & \cdots & & \\
 & & & & & & 0 & \mathcal{Z}_{\mathcal{N}/2} \\
 & & & & & & -\mathcal{Z}_{\mathcal{N}/2} & 0
\end{pmatrix} \tag{2.43}$$

Such a transformation is allowed since it preserves the algebra. Now we can define alternate ladder

operators

$$\begin{aligned}
a_{1\alpha} =& \frac{1}{\sqrt{2}}\left(Q^1_\alpha + \epsilon_{\alpha\beta}\bar{Q}_{2\beta}\right)\\
b_{1\alpha} =& \frac{1}{\sqrt{2}}\left(Q^1_\alpha - \epsilon_{\alpha\beta}\bar{Q}_{2\beta}\right)\\
a_{2\alpha} =& \frac{1}{\sqrt{2}}\left(Q^3_\alpha + \epsilon_{\alpha\beta}\bar{Q}_{4\beta}\right)\\
b_{2\alpha} =& \frac{1}{\sqrt{2}}\left(Q^3_\alpha - \epsilon_{\alpha\beta}\bar{Q}_{4\beta}\right)\\
\ldots =& \ldots\\
\ldots =& \ldots\\
a_{\mathcal{N}/2\alpha} =& \frac{1}{\sqrt{2}}\left(Q^{\mathcal{N}-1}_\alpha + \epsilon_{\alpha\beta}\bar{Q}_{\mathcal{N}\beta}\right)\\
b_{\mathcal{N}/2\alpha} =& \frac{1}{\sqrt{2}}\left(Q^{\mathcal{N}-1}_\alpha - \epsilon_{\alpha\beta}\bar{Q}_{\mathcal{N}\beta}\right)
\end{aligned} \tag{2.44}$$

These satisfy the algebra

$$\left\{a_{r\alpha}, a^\dagger_{s\beta}\right\} = \left(2m + \mathcal{Z}_r\right)\delta_{rs}\delta_{\alpha\beta} \tag{2.45}$$

$$\left\{b_{r\alpha}, b^\dagger_{s\beta}\right\} = \left(2m - \mathcal{Z}_r\right)\delta_{rs}\delta_{\alpha\beta} \tag{2.46}$$

$$\left\{a_{r\alpha}, b^\dagger_{s\beta}\right\} = \{a_{r\alpha}, a_{s\beta}\} = \ldots = 0 \tag{2.47}$$

Therefore we again have $2\mathcal{N}$ creation operators. As before, the creation operators with $\alpha = 1$ lower the spin by half whereas those with $\alpha = 2$ increase it by half. From (2.45) and (2.46) we conclude that for positivity of inner products we must have

$$2m \geq |\mathcal{Z}_r| \tag{2.48}$$

This implies that for massless representations the central charges are always trivially realized, i.e. $\mathcal{Z}^{IJ} = 0$, as we already saw before. Therefore massive multiplets are in general larger than massless ones since in the massive case one can have $2^{2\mathcal{N}}$ states (with spins from $j_0 - \frac{\mathcal{N}}{2}$ to $j_0 + \frac{\mathcal{N}}{2}$) whereas in massless case we always have $2^{\mathcal{N}}$ states. In some cases the massive multiplets are shorter than usual. This happens when the bound (134) is saturated for some value of r, because then the corresponding ladder operators are trivially realized. Then the multiplet we get is called a **short supermultiplet** and the states in such a multiplet are called **BPS states**. If the bound is saturated for k values of 'r' then the size of the multiplet reduces to $2^{2(\mathcal{N}-k)}$. If the bound is saturated for all $\mathcal{N}/2$ values of 'r' then the it is called an **ultrashort supermultiplet** and the size reduces to just $2^{\mathcal{N}}$.

Chapter 3

Representations of Superconformal Algebra in Four Dimensions

3.1 Introduction

In this section we will finally introduce the superconformal group in four dimensions. We will start by reviewing the representation theory of conformal group in a manner that will be useful later. We will discuss how to construct irreducible representations of the conformal group using the method of induced representations discussed in the last chapter. We will then introduce the superconformal group in four dimensions and justify the form of the algebra to some extent. Then we will construct representations of the superconformal group for $\mathcal{N} = 1, 2, 3$ & 4 and discuss shortening condtions and BPS states.

3.2 Unitary Representations of the Conformal Group

In chapter 1 we defined a primary field as a field which transforms according to (1.23). We now look into the representation theory of the conformal group SO(d,2) to see how the concept of a primary operator emerges from there. We take the conformal group (1.37) as the starting point and construct its representations. To do this we again follow the method of induced representations (as in the paper by Mack and Salam [4]) by identifying a suitable little group which in this case is the subgroup which keeps the origin fixed. Therefore the little group consists of rotations, dilatations as well as special conformal transformations. The Lie algebra of the little group is generated by $S_{\mu\nu}$, $\tilde{\Delta}$ and κ_μ(corresponding to rotations, dilatations and SCTs), and is isomorphic to the Lie algebra of the Poincare group plus dilatations, with κ_μ acting as translations.

$$\begin{gathered}[\kappa_\mu, \kappa_\nu] = 0;\ \left[\tilde{\Delta}, \kappa_\mu\right] = -i\kappa_\mu \\ [\kappa_\rho, S_{\mu\nu}] = \eta_{\rho\mu}\kappa_\nu - \eta_{\rho\nu}\kappa_\mu;\ [S_{\mu\nu}, S_{\rho\sigma}] = i\left(\eta_{\mu\rho}S_{\nu\sigma} - \eta_{\mu\sigma}S_{\rho\nu} - \eta_{\nu\rho}S_{\mu\sigma} + \eta_{\nu\sigma}S_{\mu\rho}\right)\end{gathered} \tag{3.1}$$

Consider a set of operators $\mathcal{O}^A(x)$ belonging to a representation of the Lorentz group. The action of little group on such an operator at the origin is then given by

$$\begin{aligned}\left[M_{\mu\nu}, \mathcal{O}^A(0)\right] &= (S_{\mu\nu})^A_B \, \mathcal{O}^B(0) \\ \left[D, \mathcal{O}^A(0)\right] &= \tilde{\Delta}^A_B \mathcal{O}^B(0) \\ \left[K_\mu, \mathcal{O}^A(0)\right] &= (\kappa_\mu)^A_B \, \mathcal{O}^B(0)\end{aligned} \tag{3.2}$$

We can then obtain the field representations by translating the commutators (3.2) using $e^{iP.x}$. For example

$$\left[M_{\mu\nu}, \mathcal{O}^A(x)\right] = e^{-iP.x}\left[e^{iP.x} M_{\mu\nu} e^{-iP.x}, \mathcal{O}^A(0)\right] e^{iP.x}$$

Using the Baker Hausdorff identity $e^{-A} B e^A = B + [B, A] + \frac{1}{2!}\left[[B, A], A\right] + ...$, we get

$$e^{iP.x} M_{\mu\nu} e^{-iP.x} = M_{\mu\nu} - x_\mu P_\nu + x_\nu P_\mu$$

$$\begin{aligned}\left[M_{\mu\nu}, \mathcal{O}^A(x)\right] &= e^{-iP.x}\left[M_{\mu\nu} - x_\mu P_\nu + x_\nu P_\mu, \mathcal{O}^A(0)\right] e^{iP.x} \\ &= -i\left(x_\mu \partial_\nu - x_\nu \partial_\mu\right) \mathcal{O}^A(x) + i\left(S_{\mu\nu}\right)^A_B \mathcal{O}^B(x)\end{aligned}$$

where we've used equation (3.2) and the fact that all operators transform as a scalar under translations and therefore - $\left[P_\mu, \mathcal{O}^A(0)\right] = i\partial_\mu \mathcal{O}^A(0)$. Then similarly the field representation for other generators are

$$\left[D, \mathcal{O}^A(x)\right] = i x^\mu \partial_\mu \mathcal{O}^A(x) + \tilde{\Delta}^A_B \mathcal{O}^B(x)$$

$$\left[K_\mu, \mathcal{O}^A(x)\right] = i\left(2x_\mu \left(x.\partial\right) - x^2 \partial_\mu\right) \mathcal{O}^A(x) + i\left[\kappa_\mu - 2i x_\mu \tilde{\Delta} - x^\nu S_{\mu\nu}\right]^A_B \mathcal{O}^B(x)$$

Now if we assume that $\mathcal{O}^A(x)$ belongs to an irreducible representation of the Lorentz subgroup, then since $\left[\tilde{\Delta}, S_{\mu\nu}\right] = 0$ we see that using Schur's lemma, $\tilde{\Delta}^A_B = i\Delta\delta^A_B$. Also, if index A runs over a finite set, we have two cases -

1. $\boldsymbol{\kappa_\mu = 0}$. This case corresponds to primary operators since

   $$\left[K_\mu, \mathcal{O}^A(0)\right] = 0$$

   and

   $$\left[K_\mu, \mathcal{O}^A(x)\right] = i\left(2x_\mu \left(x.\partial\right) - x^2 \partial_\mu\right) \mathcal{O}^A(x) + \left[2i x_\mu \Delta - x^\nu S_{\mu\nu}\right]^A_B \mathcal{O}^B(x)$$

2. $\boldsymbol{\kappa_\mu}$ **is non zero but nilpotent, i.e** $\boldsymbol{(\kappa_\mu)^s = 0}$ **for some finite integer s.** This follows from the fact that for any finite dimensional representations of the Poincare group, the generators of translations are nilpotent. The operators here correspond to the descendant operators. The statement here then says that after a finite number of actions by K_μ on descendant operators at the origin, we must hit a primary and get zero. Therefore for any operator, the dilatation and SCT generators become

   $$\left[D, \mathcal{O}^A(x)\right] = i x^\mu \partial_\mu \mathcal{O}^A(x) + i\Delta \mathcal{O}^A(x)$$

$$\left[K_\mu, \mathcal{O}^A(x)\right] = i\left(2x_\mu\left(x.\partial\right) - x^2\partial_\mu\right)\mathcal{O}^A(x) + \left[\kappa_\mu + 2ix_\mu\Delta - x^\nu S_{\mu\nu}\right]^A_B\mathcal{O}^B(x)$$

For some purposes it is useful to define an alternate set of generators. These have the advantage that the corresponding eigenstates are normalisable. For this, let us recall that the conformal generators can be written in the form,

$$J_{mn} = \begin{pmatrix} M_{\mu\nu} & -\frac{1}{2}\left(P_\mu - K_\mu\right) & -\frac{1}{2}\left(P_\mu + K_\mu\right) \\ \frac{1}{2}\left(P_\mu - K_\mu\right) & 0 & D \\ \frac{1}{2}\left(P_\mu + K_\mu\right) & -D & 0 \end{pmatrix} \tag{3.3}$$

Then we can define alternate generators H,ξ^+_r,ξ^-_s,M_{rs} as -

$$H = J_{0,d+1} = -\frac{1}{2}\left(P_0 + K_0\right) \tag{3.4}$$

$$\xi^\pm_r = J_{d+1,r} \pm iJ_{0r} \tag{3.5}$$

$$M_{rs} = J_{rs} \tag{3.6}$$

where r,s=1,2...d. These satisfy the commutators

$$\begin{aligned} \left[\xi^-_r, \xi^+_s\right] =& 2\delta_{rs}H - 2iM_{rs} \\ \left[H, \xi^\pm_s\right] =& \pm\,\xi^\pm_s \\ \left[\xi^+_r, \xi^+_s\right] =& \left[\xi^-_r, \xi^-_s\right] = 0 \\ \left[M_{rs}, M_{pq}\right] =& i\left(\delta_{rp}M_{sp} - \delta_{sp}M_{rq} - \delta_{rq}M_{sp} + \delta_{sq}M_{rp}\right) \end{aligned} \tag{3.7}$$

From the hermiticity of the generators J_{mn}, we also have

$$\begin{aligned} H^\dagger =& H \\ \xi^{+\dagger}_r =& \xi^-_r \\ M^\dagger_{rs} =& M_{rs} \end{aligned} \tag{3.8}$$

These new generators are related to the old ones by a similarity transformation

$$\begin{aligned} -e^{-\frac{\pi}{2}J_{0d}}iDe^{\frac{\pi}{2}J_{0d}} =& H \\ e^{-\frac{\pi}{2}J_{0d}}\left(J_{ab}, -iJ_{0a}\right)e^{\frac{\pi}{2}J_{0d}} =& \left(M_{ab}, M_{da}\right) \\ e^{-\frac{\pi}{2}J_{0d}}\left(P_a, -iP_0\right)e^{\frac{\pi}{2}J_{0d}} =& \left(\xi^+_a, \xi^+_d\right) \\ e^{-\frac{\pi}{2}J_{0d}}\left(K_a, -iK_0\right)e^{\frac{\pi}{2}J_{0d}} =& \left(\xi^-_a, \xi^-_d\right) \end{aligned} \tag{3.9}$$

where a,b=1,2...d-1. Suppose now we have a set of primary operator $\mathcal{O}^A\left(x\right)$. Then we have the corresponding states as $\left|\mathcal{O}\right\rangle^A$. Since $\mathcal{O}^A(x)$ are local operators, the states $\left|\mathcal{O}\right\rangle^A$ are dilatation eigenstates satisfying

$$D\left|\mathcal{O}\right\rangle^A = i\Delta\left|\mathcal{O}\right\rangle^A$$

Then if we define

$$|\Delta\rangle^A = e^{-\frac{\pi}{2}J_{0d}} |\mathcal{O}\rangle^A \tag{3.10}$$

then

$$H |\Delta\rangle^A = \Delta |\Delta\rangle^A \tag{3.11}$$

Also since $M_{\mu\nu} |\mathcal{O}\rangle^A = (S_{\mu\nu})^A_B |\mathcal{O}\rangle^A$, we have

$$M_{rs} |\mathcal{O}\rangle^A = \left(\hat{S}_{rs}\right)^A_B |\Delta\rangle^A \tag{3.12}$$

where

$$\hat{S}_{ab} = S_{ab};\ \hat{S}_{da} = -iS_{0a} \tag{3.13}$$

Also if the states are primary we have $K_\mu |\mathcal{O}\rangle^A = 0$, therefore

$$\xi_r^- |\Delta\rangle^A = 0 \tag{3.14}$$

The advantage of transforming to this representation is that unlike $|\mathcal{O}\rangle^A$, the states $|\Delta\rangle^A$ are normalisable. We can show that $|\mathcal{O}\rangle^A$ are not normalisable by observing that even though D is a Hermitian operator, it has imaginary eigenvalues, and hence the eigenstates must be non-normalisable.

3.3 The Superconformal Algebra in Four Dimensions

The superconformal algebra is a supersymmetric extension of the conformal algebra. In four dimensions, we have the usual supercharges Q^I_α and $\bar{Q}_{\dot{\alpha}I}$ as in the super Poincare case as well as new superconformal charges S^α_I and $\bar{S}^{\dot{\alpha}I}$. We also have R-symmetry generators R^I_J which play a more important role now. In principle one can construct the superconformal algebra by starting with the conformal and the super Poincare algebra and then try to satisfy the Jacobi identities. Satisfying the Jacobi identities then gives us the required new generators as well as new commutation relations. These relations are simply born out of the necessity that the Jacobi identities must hold. The only other condition being that the resulting algebra must still have the super Poincare algebra and the conformal algebra as subalgebras. Here we will just lay out the superconformal algebra resulting from this procedure and simply indicate how the important commutators can be derived by satisfying the Jacobi identities.

$$
\begin{aligned}
\{Q^I_\alpha, \bar{Q}_{\dot{\alpha}J}\} = 2\delta^I_J P_{\alpha\dot{\alpha}};\ \left\{Q^I_\alpha, Q^J_\beta\right\} = \left\{\bar{Q}_{\dot{\alpha}I}, \bar{Q}_{\dot{\beta}J}\right\} = 0 \ldots (i)\\
\left\{\bar{S}^{\dot{\alpha}I}, S^\alpha_J\right\} = 2\delta^I_J \bar{K}_{\alpha\dot{\alpha}};\ \left\{\bar{S}^{\dot{\alpha}I}, \bar{S}^{\dot{\beta}J}\right\} = \left\{S^\alpha_I, S^\beta_J\right\} = 0 \ldots (ii)\\
\left\{Q^I_\alpha, \bar{S}^{\dot{\alpha}J}\right\} = 0;\ \{S^\alpha_I, \bar{Q}_{\dot{\alpha}J}\} = 0 \ldots (iii)\\
\left\{Q^I_\alpha, S^\beta_J\right\} = 4\left(\delta^I_J\left(M^\beta_\alpha - \frac{i}{2}\delta^\beta_\alpha D\right) - \delta^\beta_\alpha R^I_J\right);\ \left\{\bar{S}^{I\dot{\alpha}}, \bar{Q}_{\dot{\beta}J}\right\} = 4\left(\delta^I_J\left(\bar{M}^{\dot{\alpha}}_{\dot{\beta}} + \frac{i}{2}\delta^{\dot{\alpha}}_{\dot{\beta}} D\right) - \delta^{\dot{\alpha}}_{\dot{\beta}} R^I_J\right) \ldots (iv)\\
\left[M^\beta_\alpha, Q^I_\gamma\right] = \delta^\beta_\gamma Q^I_\alpha - \frac{1}{2}\delta^\beta_\alpha Q^I_\gamma;\ \left[\bar{M}^{\dot{\alpha}}_{\dot{\beta}}, \bar{Q}_{\dot{\gamma}I}\right] = -\delta^{\dot{\alpha}}_{\dot{\gamma}}\bar{Q}_{\dot{\beta}I} + \frac{1}{2}\delta^{\dot{\alpha}}_{\dot{\beta}}\bar{Q}_{\dot{\gamma}I} \ldots (v)\\
\left[M^\beta_\alpha, S^\gamma_I\right] = -\delta^\gamma_\alpha S^\beta_I + \frac{1}{2}\delta^\beta_\alpha S^\gamma_I;\ \left[\bar{M}^{\dot{\alpha}}_{\dot{\beta}}, \bar{S}^{\dot{\gamma}I}\right] = \delta^{\dot{\gamma}}_{\dot{\beta}}\bar{S}^{\dot{\alpha}I} - \frac{1}{2}\delta^{\dot{\alpha}}_{\dot{\beta}}\bar{S}^{\dot{\gamma}I} \ldots (vi)\\
\left[D, Q^I_\alpha\right] = \frac{i}{2}Q^I_\alpha;\ \left[D, \bar{Q}_{\dot{\alpha}I}\right] = \frac{i}{2}\bar{Q}_{\dot{\alpha}I} \ldots (vii)\\
\left[D, S^\alpha_I\right] = -\frac{i}{2}S^\alpha_I;\ \left[D, \bar{S}^{\dot{\alpha}I}\right] = -\frac{i}{2}\bar{S}^{\dot{\alpha}I} \ldots (viii)\\
\left[K_\mu, Q^I_\alpha\right] = -\left(\sigma_\mu\right)_{\alpha\dot{\alpha}}\bar{S}^{\dot{\alpha}I};\ \left[K_\mu, \bar{Q}_{I\dot{\alpha}}\right] = S^\alpha_I\left(\sigma_\mu\right)_{\alpha\dot{\alpha}} \ldots (ix)\\
\left[P_\mu, \bar{S}^{\dot{\alpha}I}\right] = -\left(\bar{\sigma}_\mu\right)^{\dot{\alpha}\alpha}Q^I_\alpha;\ \left[P_\mu, S^\alpha_I\right] = \bar{Q}_{\dot{\alpha}I}\left(\bar{\sigma}_\mu\right)^{\dot{\alpha}\alpha} \ldots (x)\\
\left[R^I_J, R^K_L\right] = \delta^K_J R^I_L - \delta^I_L R^K_J \ldots (xi)\\
\left[R^I_J, Q^K_\alpha\right] = \delta^K_J Q^I_\alpha - \frac{1}{4}\delta^I_J Q^K_\alpha;\ \left[R^I_J, \bar{Q}_{\dot{\alpha}K}\right] = -\delta^I_K\bar{Q}_{\dot{\alpha}J} + \frac{1}{4}\delta^I_J Q_{\dot{\alpha}K} \ldots (xii)\\
\left[R^I_J, S^\alpha_K\right] = -\delta^I_K S^\alpha_J + \frac{1}{4}\delta^I_J S^\alpha_K;\ \left[R^I_J, \bar{S}^{\dot{\alpha}K}\right] = \delta^K_J\bar{S}^{\dot{\alpha}I} - \frac{1}{4}\delta^I_J S^{\dot{\alpha}K} \ldots (xiii)
\end{aligned}
\tag{3.15}
$$

First we must note that the dimension of P_μ and K_μ are 1 and -1 respectively. Also. from (3.15)-(i) it is clear that the 'Q' supercharges have dimension 1/2. This is reflected in the commutators with the dilatation generator in line (3.15)-(vii). Now one can see is that since $[Q, K]$ is a quantity with dimension -1/2. Therefore, we must either have a spinor generator which has dimension -1/2 or the commutator must be zero. The commutator cannot be zero since the Jacobi identity with K, Q and

$\bar{Q}$ will not be satisfied. Therefore we must have additional fermionic generators S and $\bar{S}$ with the commutation relations (3.15)-(ix). These new generators satisfy a similar algebra among themselves given in (3.15)-(ii). Note that in (3.15)-(i) and (ii) there are no central charges. This is a fact that is easily seen by realizing that if a central charge exists it must have dimension 1 and hence a commutator with D would be non-zero, which is not possible since the charge is 'central' (i.e it commutes with all other generators). We also cannot have any non-central charges in that commutator since the Jacobi identities would not close. To see the later, we must first justify (3.15)-(x). The RHS in both those equations must be a spin 1/2 object. It can't be zero because then the Jacobi Identity with P, S and $\bar{S}$ would not close. The index structure then fixes the commutator completely. Then we can use the Jacobi identity for Q, P and $\bar{S}$ to prove that the commutator $\{Q, Q\}$ has to be zero and no central or non-central charges can be present.

$$\left[P_\mu, \left\{\bar{S}^{\dot{\beta}J}, Q^I_\alpha\right\}\right] - \left\{Q^I_\alpha, \left[P_\mu, \bar{S}^{\dot{\beta}J}\right]\right\} + \left\{\bar{S}^{\dot{\beta}J}, \left[Q^I_\alpha, P_\mu\right]\right\} = 0$$

$$\Rightarrow (\bar{\sigma}_\mu)^{\dot{\beta}\beta} \left\{Q^I_\alpha, Q^J_\beta\right\} = 0$$

$$\Rightarrow \mathcal{Z}^{IJ} = 0$$

Let us now jump to (3.15)-(iv). These are anticommutators which are most unfamiliar. At a first glance one doesn't expect R-symmetry generators to pop-up in these commutators. If we just ignore them (put them to zero) and try to build the algebra we will find that the Jacobi identity for Q, Q and S will not be satisfied. Let us explicitly see this for $\mathcal{N} = 1$, so that $R^I_J \to R$. Now if put R=0 in (3.15)-(iv) and try to evaluate the mentioned Jacobi identity we get

$$\left\{Q_\alpha, S^\beta\right\} = 4\left[M^\beta_\alpha - \frac{i}{2}\delta^\beta_\alpha D\right] \tag{3.16}$$

$$[Q_\alpha, \{Q_\beta, S^\gamma\}] + [Q_\beta, \{S^\gamma, Q_\alpha\}] + [S^\gamma, \{Q_\alpha, Q_\beta\}]$$
$$= \left[Q_\alpha, 4M^\gamma_\beta - \frac{4i}{2}\delta^\gamma_\beta D\right] + \left[Q_\beta, 4M^\beta_\alpha - \frac{4i}{2}\delta^\gamma_\alpha D\right]$$
$$= -3\delta^\gamma_\alpha Q_\beta - 3\delta^\gamma_\beta Q_\alpha \neq 0$$

Therefore the Jacobi identity is not satisfied. If we add an additional term $-4\delta^\beta_\alpha R$ in (3.16) satisfying,

$$[R, Q_\alpha] = \frac{3}{4} Q_\alpha$$

the Jacobi identity closes. This is also a nice way to arrive ats the commutators (3.15)-(xii) and (3.15)-(xiii). The other commutators look quite familiar and can be easily justified using various Jacobi identites.

Just as for the conformal case, we can introduce alternate generators for superconformal charges Q and S using a similarity transformation as follows -

$$\begin{aligned}
\mathcal{Q}_{\alpha}^{+I} &= e^{-\frac{\pi}{2}J_{0d}} Q_{\alpha}^{I} e^{\frac{\pi}{2}J_{0d}} = \frac{1}{\sqrt{2}}\left(Q_{\alpha}^{I} + \sigma_{0\alpha\dot{\alpha}}\bar{S}^{\dot{\alpha}I}\right) \\
\mathcal{Q}_{I}^{-\alpha} &= e^{-\frac{\pi}{2}J_{0d}} S_{I}^{\alpha} e^{\frac{\pi}{2}J_{0d}} = \frac{1}{\sqrt{2}}\left(S_{I}^{\alpha} + \bar{Q}_{I\dot{\alpha}}\bar{\sigma}_{0}^{\alpha\dot{\alpha}}\right) \\
-\mathcal{S}^{-\dot{\alpha}I} &= e^{-\frac{\pi}{2}J_{0d}} \bar{S}^{\dot{\alpha}I} e^{\frac{\pi}{2}J_{0d}} = \frac{1}{\sqrt{2}}\left(\bar{S}^{\dot{\alpha}I} - \bar{\sigma}_{0}^{\dot{\alpha}\alpha} Q_{\alpha}^{I}\right) \\
\mathcal{S}_{\dot{\alpha}I}^{+} &= e^{-\frac{\pi}{2}J_{0d}} \bar{Q}_{\dot{\alpha}I} e^{\frac{\pi}{2}J_{0d}} = \frac{1}{\sqrt{2}}\left(\bar{Q}_{\dot{\alpha}I} - S_{I}^{\alpha}\sigma_{0\alpha\dot{\alpha}}\right)
\end{aligned} \tag{3.17}$$

The algebra in (3.15)-(i),(ii),(iii),(iv),(vii) can then be written as

$$\begin{aligned}
\left\{\mathcal{Q}_{\alpha}^{+I}, \mathcal{Q}_{J}^{-\beta}\right\} &= 2\delta_{J}^{I}\delta_{\alpha}^{\beta} H + 4\delta_{j}^{I}\tilde{M}_{\alpha}^{\beta} - 4\delta_{\alpha}^{\beta} R_{J}^{I} \\
\left\{\mathcal{S}^{-\dot{\alpha}I}, \mathcal{S}_{\beta J}^{+}\right\} &= 2\delta_{J}^{I}\delta_{\dot{\beta}}^{\dot{\alpha}} H - 4\delta_{J}^{I}\tilde{\bar{M}}_{\dot{\beta}}^{\dot{\alpha}} + 4\delta_{\dot{\beta}}^{\dot{\alpha}} R_{J}^{I} \\
\left\{\mathcal{Q}_{\alpha}^{+I}, \mathcal{S}_{\dot{\alpha}J}^{+}\right\} &= 2\delta_{J}^{I}\left(e_{r}\right)_{\alpha\dot{\alpha}}\xi_{r}^{+} \\
\left\{\mathcal{S}^{-\dot{\alpha}I}, \mathcal{Q}_{j}^{-\alpha}\right\} &= 2\delta_{J}^{I}\left(\bar{e}_{r}\right)^{\dot{\alpha}\alpha}\xi_{r}^{-} \\
\left[H, \mathcal{Q}^{\pm}\right] &= \pm\frac{1}{2}\mathcal{Q}^{\pm} \\
\left[H, \mathcal{S}^{\pm}\right] &= \pm\frac{1}{2}\mathcal{S}^{\pm}
\end{aligned} \tag{3.18}$$

where $\tilde{M}_{\alpha}^{\beta} = e^{-\frac{\pi}{2}J_{0d}} M_{\alpha}^{\beta} e^{\frac{\pi}{2}J_{0d}}$ and $\tilde{\bar{M}}_{\dot{\beta}}^{\dot{\alpha}} = e^{-\frac{\pi}{2}J_{0d}} \bar{M}_{\dot{\beta}}^{\dot{\alpha}} e^{\frac{\pi}{2}J_{0d}}$, $e_r = (\sigma_a, -i\sigma_0)$ and $\bar{e}_r = (-\bar{\sigma}_a, i\bar{\sigma}_0)$. From the usual hermiticity conditions $Q^{\dagger} = \bar{Q}$ and $S^{\dagger} = \bar{S}$ we have

$$\left(\mathcal{Q}_{\alpha}^{+I}\right)^{\dagger} = \mathcal{Q}_{I}^{-\beta}\sigma_{0\beta\dot{\alpha}}; \left(\mathcal{S}_{\dot{\alpha}I}^{+}\right)^{\dagger} = \sigma_{0\alpha\dot{\beta}}\mathcal{S}^{-\dot{\beta}I} \tag{3.19}$$

We also have $\left(\tilde{M}_{\alpha}^{\beta}\right)^{\dagger} = \bar{\sigma}_{0}^{\dot{\beta}\gamma}\tilde{M}_{\gamma}^{\delta}\sigma_{0\delta\dot{\alpha}}$. Therefore we have two subalgebras with generators $\left\{\mathcal{Q}^{+}, \mathcal{Q}^{-}, \tilde{M}, R, H\right\}$ and $\left\{\mathcal{S}^{+}, \mathcal{S}^{-}, \tilde{\bar{M}}, R, H\right\}$ respectively. Again H will have a positive spectrum and the lowest energy state will in addition also satisfy

$$\mathcal{Q}_{I}^{-\alpha}\left|\Delta\right\rangle = \mathcal{S}^{-\dot{\alpha}I}\left|\Delta\right\rangle = 0 \tag{3.20}$$

3.4 $\mathcal{N} = 1$ and $\mathcal{N} = 2$ Representations

In four dimensional superconformal theories the most important physical cases are $\mathcal{N} = 2$ and $\mathcal{N} = 4$ and hence we will discuss these in considerable detail. We will also discuss the less interesting cases of $\mathcal{N} = 1$ and $\mathcal{N} = 3$ since they have a similar structure to $\mathcal{N} = 2$ and $\mathcal{N} = 4$ cases respectively. Let us then consider the simpler case of $\mathcal{N} = 2$. We will closely follow [2] for this and the subsequent section. We start with the $U(\mathcal{N})$ subalgebra satisfied by the R symmetry generators in (3.15)-(xi). For $\mathcal{N} = 2$, the R-symmetry is U(2) which is locally isomorphic to $SU(2) \times U(1)$. We can show this by writing

the generators R^I_J as -

$$\left[R^I_J\right] = \begin{pmatrix} R_3 & R_+ \\ R_- & -R_3 \end{pmatrix} + \frac{1}{2}\hat{R}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \tag{3.21}$$

where we have essentially just separated the trace from the algebra. Then using (3.15)-(xi) we see that $R_\pm, R_3$ form a standard SU(2) algebra and the trace $\hat{R}$ corresponds to a U(1) subalgebra.

$$\begin{aligned} [R_3, R_\pm] = \pm R_\pm;\ [R_+, R_-] = 2R_3 \\ \left[R_3, \hat{R}\right] = \left[R_\pm, \hat{R}\right] = 0 \end{aligned} \tag{3.22}$$

Now we assume that the lowest dimension states belong to a representation of $U(2) = SU(2) \times U(1)$ with SU(2) charge R and U(1) charge r. Then the highest weight state $|R, r\rangle^{hw}$ in SU(2) is annihilated by R_+ and satisfies

$$\begin{aligned} R_+ |R, r\rangle^{hw} = 0;\ R_3 |R, r\rangle^{hw} = R |R, r\rangle^{hw} \\ \hat{R} |R, r\rangle^{hw} = r |R, r\rangle^{hw} \end{aligned} \tag{3.23}$$

Also from (3.15)-(xii) we have

$$\begin{aligned} \left[R_+, Q^1_\alpha\right] = 0;\ \left[R_+, Q^2_\alpha\right] = Q^1_\alpha \\ \left[R_-, Q^1_\alpha\right] = Q^2_\alpha;\ \left[R_-, Q^2_\alpha\right] = 0 \\ \left[R_3, Q^1_\alpha\right] = \frac{1}{2}Q^1_\alpha;\ \left[R_3, Q^2_\alpha\right] = -\frac{1}{2}Q^2_\alpha \end{aligned} \tag{3.24}$$

Similarly

$$\begin{aligned} \left[R_+, \bar{Q}_{\dot{\alpha}1}\right] = -\bar{Q}_{\dot{\alpha}2};\ \left[R_+, \bar{Q}_{\dot{\alpha}2}\right] = 0 \\ \left[R_-, \bar{Q}_{\dot{\alpha}1}\right] = 0;\ \left[R_-, \bar{Q}_{\dot{\alpha}2}\right] = -\bar{Q}_{\dot{\alpha}1} \\ \left[R_3, \bar{Q}_{\dot{\alpha}1}\right] = -\frac{1}{2}\bar{Q}_{\dot{\alpha}1};\ \left[R_3, \bar{Q}_{\dot{\alpha}2}\right] = \frac{1}{2}\bar{Q}_{\dot{\alpha}2} \end{aligned} \tag{3.25}$$

Therefore, Qs and $\bar{Q}$s belong to $R = \frac{1}{2}$ representations. Also,

$$\left[\hat{R}, Q_{\dot{\alpha}I}\right] = \frac{1}{2}Q_{\dot{\alpha}I};\ \left[\hat{R}, \bar{Q}_{\dot{\alpha}I}\right] = -\frac{1}{2}\bar{Q}_{\dot{\alpha}I} \tag{3.26}$$

Hence for Qs the U(1) charge is $r = \frac{1}{2}$ and for $\bar{Q}$s it is $r = -\frac{1}{2}$.

Let these states also belong to an $(j, \bar{j})$ spin representation of the Lorentz subalgebra SO(3,1). Therefore

$$|R, r\rangle_{\alpha_1 \ldots \alpha_{2j}, \dot{\alpha}_1 \ldots \dot{\alpha}_{2\bar{j}}} = |R, r\rangle_{(\alpha_1 \ldots \alpha_{2j}), (\dot{\alpha}_1 \ldots \dot{\alpha}_{2\bar{j}})}$$

and using $M_{\mu\nu} |\mathcal{O}\rangle_\alpha = (\sigma_{\mu\nu})^\beta_\alpha |\mathcal{O}\rangle_\beta$ and $M_{\mu\nu} |\mathcal{O}\rangle_{\dot{\alpha}} = (\bar{\sigma}_{\mu\nu})^{\dot{\beta}}_{\dot{\alpha}} |\mathcal{O}\rangle_{\dot{\beta}}$ we get in general

$$\begin{aligned} M^\beta_\alpha |R, r\rangle_{\alpha_1 \ldots \alpha_{2j}, \dot{\alpha}_1 \ldots \dot{\alpha}_{2\bar{j}}} = & j \left(2\delta^\beta_{(\alpha_1} |R, r\rangle_{\alpha_2 \ldots \alpha_{2j})\alpha, \dot{\alpha}_1 \ldots \dot{\alpha}_{2\bar{j}}} - \delta^\beta_\alpha |R, r\rangle_{\alpha_1 \ldots \alpha_{2j}, \dot{\alpha}_1 \ldots \dot{\alpha}_{2\bar{j}}}\right) \\ \bar{M}^{\dot{\beta}}_{\dot{\alpha}} |R, r\rangle_{\alpha_1 \ldots \alpha_{2j}, \dot{\alpha}_1 \ldots \dot{\alpha}_{2\bar{j}}} = & -\bar{j} \left(2 |R, r\rangle_{\alpha_1 \ldots \alpha_{2j}, \dot{\alpha}(\dot{\alpha}_1 \ldots \dot{\alpha}_{2\bar{j}-1}} \delta^{\dot{\beta}}_{\dot{\alpha}_{2\bar{j}})} - \delta^{\dot{\beta}}_{\dot{\alpha}} |R, r\rangle_{\alpha_1 \ldots \alpha_{2j}, \dot{\alpha}_1 \ldots \dot{\alpha}_{2\bar{j}}}\right) \end{aligned} \tag{3.27}$$

The representations which satisfy (3.23) and (3.27) are represented as $R_{(j,\bar{j})}$. In addition if the highest

weight state is also a conformal primary it satisfies

$$K_\mu \left|R,r\right\rangle^{hw}_{\alpha_1\ldots\alpha_{2j},\dot\alpha_1\ldots\dot\alpha_{2\bar j}} = 0;\; D\left|R,r\right\rangle^{hw}_{\alpha_1\ldots\alpha_{2j},\dot\alpha_1\ldots\dot\alpha_{2\bar j}} = i\Delta\left|R,r\right\rangle^{hw}_{\alpha_1\ldots\alpha_{2j},\dot\alpha_1\ldots\dot\alpha_{2\bar j}} = 0 \tag{3.28}$$

A superconformal primary is defined as a conformal primary which also satisfies $S^\alpha_I\left|\mathcal{O}\right\rangle = \bar S^{\dot\alpha I}\left|\mathcal{O}\right\rangle = 0$. This is required since the superconformal primary is defined to have the lowest dimension in a multiplet and hence must be annihilated by S since it lowers the dimension by -1. Therefore a superconformal primary belonging to the $R_{(j,\bar j)}$ representation in addition also satisfies

$$\boldsymbol{S^I_\alpha\left|R,r\right\rangle^{hw}_{\alpha_1\ldots\alpha_{2j},\dot\alpha_1\ldots\dot\alpha_{2\bar j}} = 0;\; \bar S^{\dot\alpha I}\left|R,r\right\rangle^{hw}_{\alpha_1\ldots\alpha_{2j},\dot\alpha_1\ldots\dot\alpha_{2\bar j}} = 0} \tag{3.29}$$

We start building the representation with a superconformal primary state $\left|R,r\right\rangle^{hw}_{\alpha_1\ldots\alpha_{2j},\dot\alpha_1\ldots\dot\alpha_{2\bar j}}$ which is also the highest weight state of some $R_{(j,\bar j)}$ representation of R-symmetry and Lorentz subalgebras. The whole representation can then be obtained by acting on this state with R_-, P_μ, Q^I_α and $\bar Q_{\dot\alpha K}$. Here we will supress the conformal descendants and hence will not act by P_μ explicitly anywhere. We define a set of states $\mathcal{S}_{l,\bar l}$ given by

$$\prod_{I,K,\alpha,\dot\alpha}\left(Q^I_\alpha\right)^{n_{\alpha I}}\left(\bar Q_{\dot\alpha K}\right)^{\bar n_{\dot\alpha K}}\left|R,r\right\rangle^{hw}_{\alpha_1\ldots\alpha_{2j},\dot\alpha_1\ldots\dot\alpha_{2\bar j}} \tag{3.30}$$

where $\left|R,r\right\rangle^{hw}_{\alpha_1\ldots\alpha_{2j},\dot\alpha_1\ldots\dot\alpha_{2\bar j}}$ is as described above and $n_{\alpha I},\bar n_{\dot\alpha K} = 0,1; l = \sum_{\alpha,I} n_{\alpha I}; \bar l = \sum_{\dot\alpha K}\bar n_{\dot\alpha K}$. Also, $l,\bar l = 0,1,2,3,4$, since if we have more than four Qs or $\bar Q$s then at least one of them will be repeated twice and hence will give zero. The states in $\mathcal{S}_{l,\bar l}$ along with those obtained by the action of R_- on $\left|R,r\right\rangle^{hw}_{\alpha_1\ldots\alpha_{2j},\dot\alpha_1\ldots\dot\alpha_{2\bar j}}$ span a vector space denoted by $\mathcal{V}_{l,\bar l}$. All states in this vector space have dimension $\Delta_{l,\bar l} = \Delta + \frac{1}{2}\left(l+\bar l\right)$ and U(1) charge, $r_{l,\bar l} = r + \frac{1}{2}\left(l-\bar l\right)$ as is apparent from (3.26). Now let us consider the action of K_μ on states in $\mathcal{V}_{l,\bar l}$. It can be explicitly shown using the previous results that $K_\mu\mathcal{V}_{l,\bar l} \longrightarrow \mathcal{V}_{l-1,\bar l-1}$. Also, $K_\mu\mathcal{V}_{l,0} = K_\mu\mathcal{V}_{0,\bar l} = 0$ and hence all states in $\mathcal{V}_{l,0}$ and $\mathcal{V}_{0,\bar l}$ are conformal primaries. To see this, consider for example the state $Q^I_\alpha Q^J_\beta\left|R,r\right\rangle_{\alpha_1\ldots\alpha_{2j},\dot\alpha_1\ldots\dot\alpha_{2\bar j}}$ belonging to $\mathcal{V}_{2,0}$. Then applying K_μ on this state we get

$$\begin{aligned}
K_\mu Q^I_\alpha Q^J_\beta\left|R,r\right\rangle^{hw}_{\alpha_1\ldots\alpha_{2j},\dot\alpha_1\ldots\dot\alpha_{2\bar j}} &= \left(-\left(\sigma_\mu\right)_{\alpha\dot\alpha}\bar S^{\dot\alpha I} + Q^I_\alpha K_\mu\right)Q^J_\beta\left|R,r\right\rangle^{hw}_{\alpha_1\ldots\alpha_{2j},\dot\alpha_1\ldots\dot\alpha_{2\bar j}} \\
&= Q^I_\alpha K_\mu Q^J_\beta\left|R,r\right\rangle^{hw}_{\alpha_1\ldots\alpha_{2j},\dot\alpha_1\ldots\dot\alpha_{2\bar j}} \\
&= Q^I_\alpha\left(-\left(\sigma_\mu\right)_{\beta\dot\beta}\bar S^{\dot\beta J} + Q^J_\beta K_\mu\right)\left|R,r\right\rangle^{hw}_{\alpha_1\ldots\alpha_{2j},\dot\alpha_1\ldots\dot\alpha_{2\bar j}} \\
&= 0
\end{aligned}$$

where we've used equations (3.28) and (3.29). The states belonging to $\mathcal{V}_{l,\bar l}$ may be decomposed into representations $R'_{(j',\bar j')}$ which are obtained by suitable addition of spin weights R, m and $\bar m$ for the supercharges as long as R',j',$\bar j'$ are positive (see the Racah Speiser Algorithm mentioned in the next section for more details). From equation (3.24), (3.25) and (3.26) we know that the Qs and the $\bar Q$s

have weights

$$\begin{aligned} &Q_1^1 \sim \frac{1}{2}_{(+\frac{1}{2},0)};\ Q_2^1 \sim \frac{1}{2}_{(-\frac{1}{2},0)};\ Q_1^2 \sim -\frac{1}{2}_{(\frac{1}{2},0)};\ Q_2^1 \sim -\frac{1}{2}_{(-\frac{1}{2},0)} \\ &\bar{Q}_{\dot{1}2} \sim \frac{1}{2}_{(0,\frac{1}{2})};\ \bar{Q}_{\dot{2}2} \sim \frac{1}{2}_{(0,-\frac{1}{2})};\ \bar{Q}_{\dot{1}1} \sim -\frac{1}{2}_{(0,\frac{1}{2})};\ \bar{Q}_{\dot{2}1} \sim -\frac{1}{2}_{(0,-\frac{1}{2})} \end{aligned} \tag{3.31}$$

We can then add these weights appropriately in (3.30) using the familiar SU(2) rule to obtain the resulting representations $R'_{(j',\bar{j}')}$, where $R' = R + \frac{1}{2}\sum_\alpha (n_{\alpha 1} - n_{\alpha 2}) - \frac{1}{2}\sum_{\dot{\alpha}} (\bar{n}_{\dot{\alpha}1} - \bar{n}_{\dot{\alpha}2})$. Therefore there are ${}^4C_l \times^4 C_{\bar{l}}$ such representations.

A general long supermultiplet $\mathcal{A}^{\Delta}_{R,r(j,\bar{j})}$ is obtained by application of all 8 supercharges Q^I_α, $\bar{Q}^I_{\dot{\alpha}}$ to a highest weight state $|R,r\rangle^{\ hw}_{\alpha_1\ldots\alpha_{2j},\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}}}$. The representations belonging to $\mathcal{V}_{l,0}$ are given by the action of only the four Qs as follows -

$$R_{(j,\bar{j})} \xrightarrow{Q} \begin{matrix}(R+\frac{1}{2})_{(j\pm\frac{1}{2},\bar{j})} \\ (R-\frac{1}{2})_{(j\pm\frac{1}{2},\bar{j})}\end{matrix} \xrightarrow{Q^2} \begin{matrix}(R\pm 1)_{(j,\bar{j})}, R_{(j,\bar{j})} \\ R_{(j\pm1,\bar{j})}, R_{(j,\bar{j})}\end{matrix} \xrightarrow{Q^3} \begin{matrix}(R+\frac{1}{2})_{(j\pm\frac{1}{2},\bar{j})} \\ (R-\frac{1}{2})_{(j\pm\frac{1}{2},\bar{j})}\end{matrix} \xrightarrow{Q^4} R_{(j,\bar{j})}$$

Then a representation belonging to $\mathcal{V}_{l,\bar{l}}$ may be obtained by applying the conjugate of the above to all representations in $\mathcal{V}_{l,0}$. The dimension of the resulting long multiplet $\mathcal{A}^{\Delta}_{R,r(j,\bar{j})}$ is then given by

$$dim\mathcal{A}^{\Delta}_{R,r(j,\bar{j})} = 2^8\,(2R+1)\,(2j+1)\,(2\bar{j}+1)$$

The unitarity conditions for such a supermultiplet are found similarly as in the conformal case,

$$\Delta \geq 2 + 2j + 2R + r, 2 + 2\bar{j} + 2R - r \tag{3.32}$$

We may get shorter multiplets if one or more **shortening conditions** are satisfied. For example, if j=0, one can impose a shortening condition

$$Q^1_\alpha \big|R,r\big\rangle^{hw}_{\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}}} = 0 \tag{3.33}$$

This results in another constraint on the dimension of the operators in the multiplet. To see this consider the anti-commutator $\left\{Q^1_\alpha, S^\beta_J\right\}$. When this anti-commutator acts on a highest weight state $|R,r\rangle^{\ hw}_{\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}}}$ it gives zero in accordance with (3.29) and (3.33). Therefore

$$\left\{Q^1_\alpha, S^\beta_J\right\}\Big|R,r\Big\rangle^{hw}_{\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}}} = 0$$

$$\Rightarrow -\delta^\beta_\alpha\left(2i\delta^1_J D + 4R^1_J\right)\big|R,r\big\rangle^{hw}_{\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}}} = 0$$

where we've used the fact that j=0 and hence $M^\beta_\alpha\,|R,r\rangle^{\ hw}_{\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}}} = 0$. Therefore for J=1 in the above equation we get

$$\Delta = 2R + r \tag{3.34}$$

whereas the J=2 component simply gives $\delta^\beta_\alpha R^+\,|R,r\rangle^{\ hw}_{\alpha_1\ldots\alpha_{2j},\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}}} = 0$ which is already true. Similarly

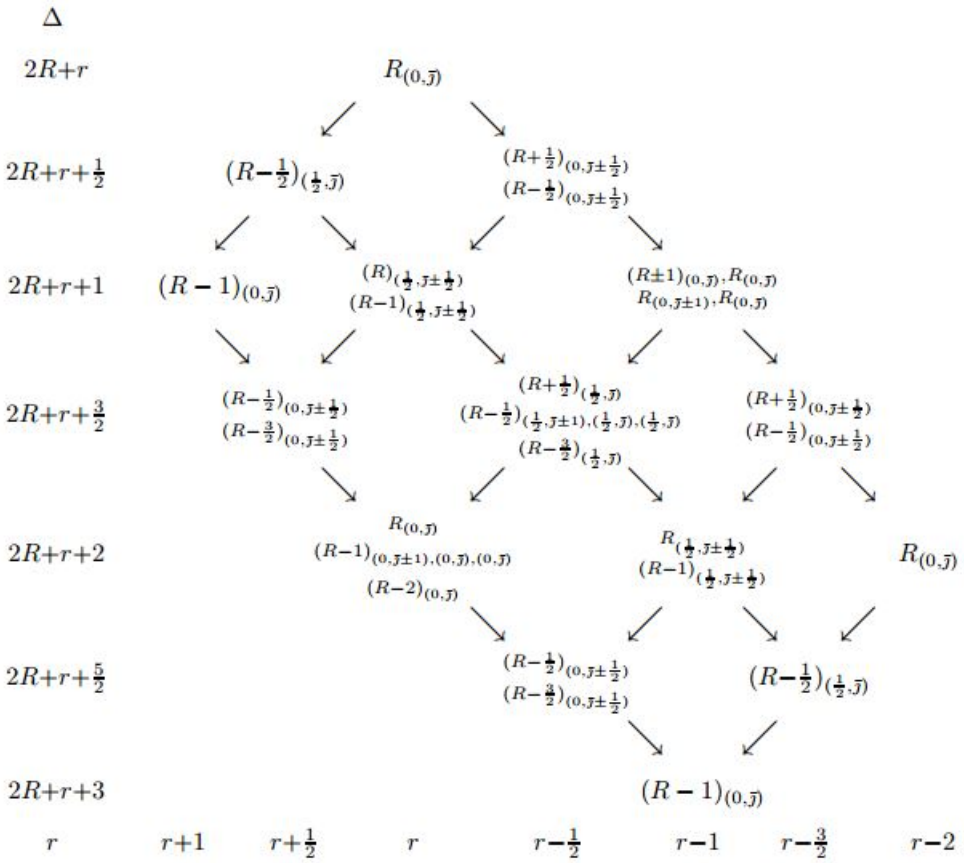
if $\bar{j} = 0$, we can impose the shortening condition

$$\bar{Q}_{\dot{\alpha}2}\big|R, r\big\rangle^{hw}_{\alpha_1 \ldots \alpha_{2j}} = 0 \tag{3.35}$$

which leads to

$$\Delta = 2R - r \tag{3.36}$$

A multiplet satisfying the shortening condition (3.33) is denoted as $\mathcal{B}_{R,r(0,\bar{j})}$ and is given by the diagram



This multiplet has dimension

$$dim\mathcal{B}_{R,r(0,\bar{j})} = 2^7 R\,(2\bar{j} + 1) \tag{3.37}$$

The corresponding unitarity condition is

$$r \geq \bar{j} + 1 \tag{3.38}$$

Similarly, the conjugate supermultiplet $\bar{\mathcal{B}}_{R,r(j,0)}$ which satisfies (3.35) has dimension

$$dim\bar{\mathcal{B}}_{R,r(j,0)} = 2^7 R\,(2j + 1) \tag{3.39}$$

and satisfies the unitarity bound

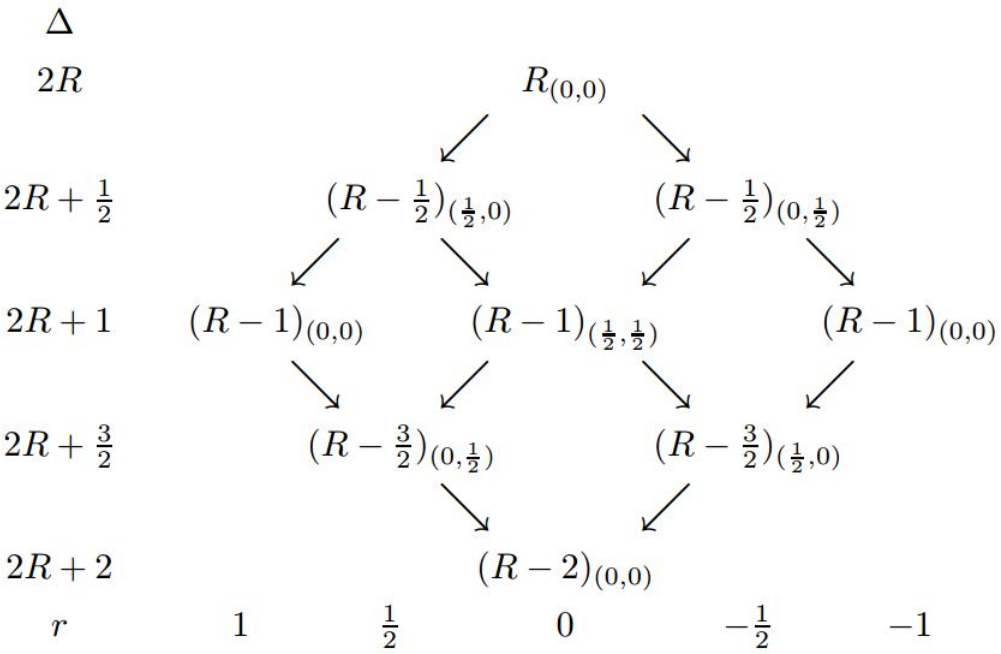
$$-r \geq j + 1 \tag{3.40}$$

If we impose (3.33) and (3.35) simultaneously we must have $j = \bar{j} = r = 0$ and hence $\Delta = 2R$. The

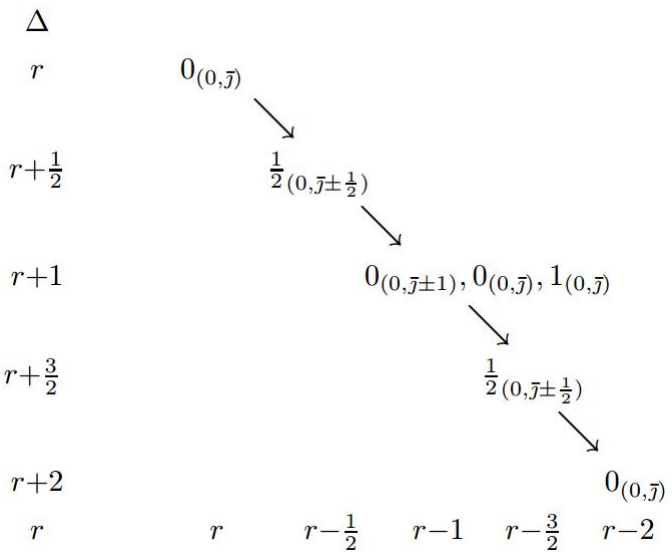
multiplet obtained is denoted by $\hat{\mathcal{B}}_R = \mathcal{B}_{R,r(0,\bar{j})} \cap \bar{\mathcal{B}}_{R,r(j,0)}$ and has dimensions

$$dim\hat{\mathcal{B}}_R = 2^4\,(2R-1) \tag{3.41}$$

This multiplet $\hat{\mathcal{B}}_R$ is given by



If R=0, we can impose a shortening condition where the highest weight state is annihilated by both Q^1_α and Q^2_α. This multiplet denoted by $\mathcal{E}_{r(0,\bar{j})}$ is then generated solely by the action of $\bar{Q}$ charges since acting with Q on a second level state will give only give a conformal descendant of the highest weight primary state (which is suppressed). It is given by



This multiplet has dimension

$$\mathcal{E}_{r(0,\bar{j})} = 2^4 \left(2\bar{j}+1\right) \tag{3.42}$$

A conjugate multiplet $\bar{\mathcal{E}}_{r(j,0)}$ can also be defined similarly.

The shortening conditions (3.33) and (3.35) are quite restrictive. We can impose a weaker **semi-shortening condition** given by

$$\begin{cases} \epsilon^{\alpha\beta} Q^1_\beta \left|R,r\right\rangle^{hw}_{\alpha_1\ldots\alpha_{2j-1}\alpha,\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}}} = 0 & j>0 \\ \left(Q^1\right)^2 \left|R,r\right\rangle^{hw}_{\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}}} = 0 & j=0 \end{cases} \tag{3.43}$$

The application of this condition leads to the absence of the representations $\left(j-\frac{1}{2},\bar{j}\right)$ at the first level. This is because the above condition corresponds to the vanishing of the antisymmetric component of $j\otimes\frac{1}{2} = \left(j+\frac{1}{2}\right)\oplus\left(j-\frac{1}{2}\right)$ which is just the $\left(j-\frac{1}{2}\right)$ representation. Then for $j>0$ we have,

$$\epsilon^{\alpha\beta}\left\{Q^1_\beta, S^\gamma_L\right\}\left|R,r\right\rangle^{hw}_{\alpha_1\ldots\alpha_{2j-1}\alpha,\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}}} = 0$$

$$\Rightarrow \left(\delta^1_L\left(-j-1+\frac{1}{2}\Delta\right) - R^I_L\right)\left|R,r\right\rangle^{hw}_{\alpha_1\ldots\alpha_{2j},\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}}} = 0 \tag{3.44}$$

For L=1 this gives

$$\Delta = 2+2j+2R+r \tag{3.45}$$

whereas the condition for L=2 is again trivial. For j=0, the corresponding condition is obtained by the equation

$$\left\{Q^I_\alpha, S^\gamma_L\right\} Q^I_\beta \left|R,r\right\rangle^{hw}_{\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}}} = Q^I_\alpha \left\{Q^I_\beta, S^\gamma_L\right\}\left|R,r\right\rangle^{hw}_{\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}}}$$

This leads to the condition

$$\left(\delta^1_L\left(-1-\frac{1}{2}\Delta\right) - R^I_L\right)\left|R,r\right\rangle^{hw}_{\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}}} = 0 \tag{3.46}$$

where again for L=1 we have

$$\Delta = 2+2R+r \tag{3.47}$$

The action of Q supercharges on the highest weight state then gives

$$R_{(j,\bar{j})} \xrightarrow{Q} \begin{matrix}(R+\frac{1}{2})_{(j+\frac{1}{2},\bar{j})} \\ (R-\frac{1}{2})_{(j\pm\frac{1}{2},\bar{j})}\end{matrix} \xrightarrow{Q^2} \begin{matrix}R_{(j+1,\bar{j}),(j,\bar{j})} \\ (R-1)_{(j,\bar{j})}\end{matrix} \xrightarrow{Q^3} (R-\tfrac{1}{2})_{(j+\frac{1}{2},\bar{j})}$$

The multiplet satisfying these conditions is denoted as $\mathcal{C}_{R,r(j,\bar{j})}$ and has dimension

$$dim\mathcal{C}_{R,r(j,\bar{j})} = 2^7\left(R\left(4j+3\right)+j+1\right)\left(2\bar{j}+1\right) \tag{3.48}$$

In a similar fashion if we impose the shortening conditions

$$\begin{cases} \epsilon^{\dot{\alpha}\dot{\beta}}\bar{Q}_{\dot{\beta}2}\big|R,r\big\rangle^{hw}_{\alpha_1\ldots\alpha_{2j},\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}-1}\dot{\alpha}} = 0 & \bar{j} > 0 \\ \left(\bar{Q}_2\right)^2\big|R,r\big\rangle^{hw}_{\alpha_1\ldots\alpha_{2j}} = 0 & \bar{j} = 0 \end{cases} \tag{3.49}$$

(corresponding to the absence of $\left(j,\bar{j}-\frac{1}{2}\right)$ at the first level) we get the restriction

$$\Delta = 2 + 2\bar{j} + 2R - r \tag{3.50}$$

The multiplet satisfying this condition is the conjugate multiplet $\bar{\mathcal{C}}_{R,r(j,\bar{j})}$. If R=0, the condition (3.43) implies the same for I=2, i.e (3.43) implies

$$\begin{cases} \epsilon^{\alpha\beta}Q^2_{\beta}\big|R,r\big\rangle^{hw}_{\alpha_1\ldots\alpha_{2j-1}\alpha,\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}}} = 0 & j > 0 \\ \left(Q^2\right)^2\big|R,r\big\rangle^{hw}_{\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}}} = 0 & j = 0 \end{cases} \tag{3.51}$$

Similarly if R=0 the condition (3.49) implies the same for I=1,

$$\begin{cases} \epsilon^{\dot{\alpha}\dot{\beta}}\bar{Q}_{\dot{\beta}1}\big|R,r\big\rangle^{hw}_{\alpha_1\ldots\alpha_{2j},\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}-1}\dot{\alpha}} = 0 & \bar{j} > 0 \\ \left(\bar{Q}_1\right)^2\big|R,r\big\rangle^{hw}_{\alpha_1\ldots\alpha_{2j}} = 0 & \bar{j} = 0 \end{cases} \tag{3.52}$$

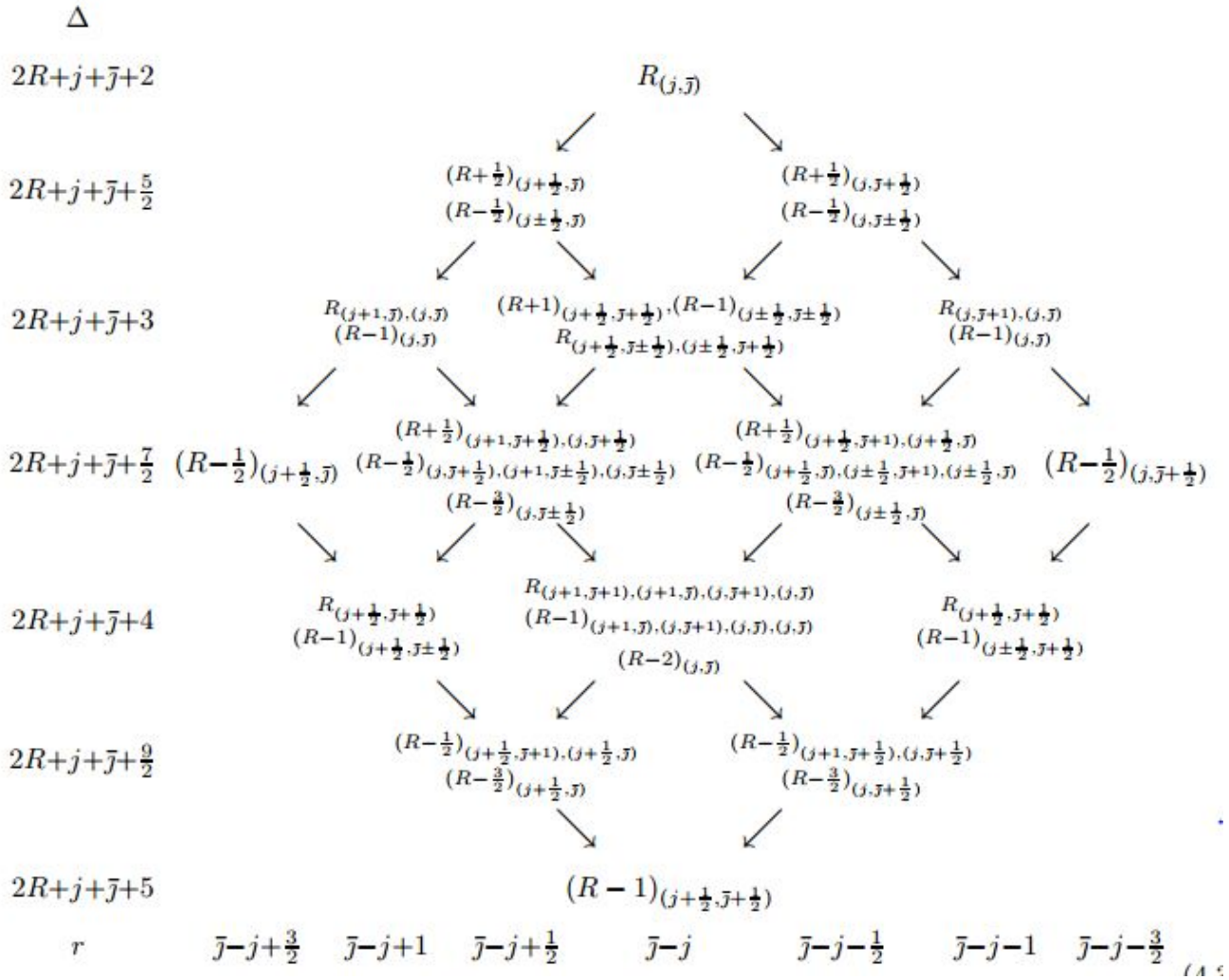
If semi-shortening conditions (3.43) and (3.49) are satisfied simultaneously then for consistency with (3.45) and (3.50) we require $r = j - \bar{j}$ and hence

$$\Delta = 2 + j + \bar{j} \tag{3.53}$$

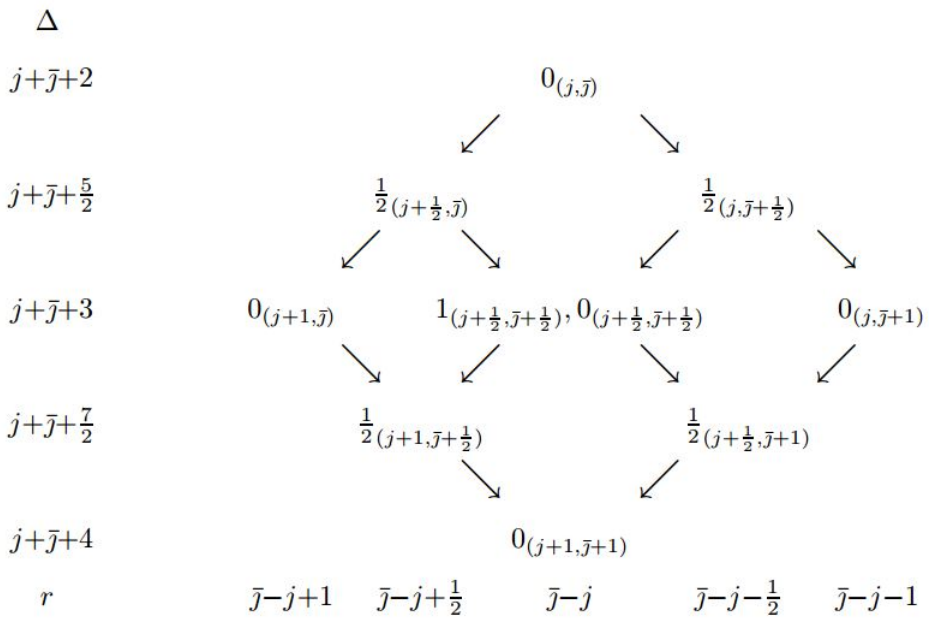
The multiplet satisfying these is denoted by $\hat{\mathcal{C}}_{R(j,\bar{j})} = \mathcal{C}_{R,r(j,\bar{j})} \cap \bar{\mathcal{C}}_{R,r(j,\bar{j})}$ and has dimension

$$dim\hat{\mathcal{C}}_{R(j,\bar{j})} = 2^5 R\,(4j+3)\,(4\bar{j}+3) + 2^5\left(j+\bar{j}+\frac{3}{2}\right) \tag{3.54}$$

It is given by



If R=0, we can also apply (3.51) and (3.52) and get the $\hat{\mathcal{C}}_{0(j,\bar{j})}$ multiplet given by



Then using commutation relation (3.15)-(i) we must have for consistency the conservation equation -

$$\begin{cases} \tilde{P}^{\dot{\alpha}\alpha}\big|0,\bar{j}-j\big\rangle_{\alpha\alpha_1\ldots\alpha_{2j-1},\dot{\alpha}\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}-1}}=0 & j\neq\bar{j}\neq 0 \\ \tilde{P}^{\dot{\alpha}\alpha}Q^I_\alpha\big|0,\bar{j}\big\rangle_{\dot{\alpha}\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}-1}}=0 & j=0,\bar{j}\neq 0 \\ \tilde{P}^{\dot{\alpha}\alpha}\bar{Q}_{\dot{\alpha}I}\big|0,-j\big\rangle_{\alpha\alpha_1\ldots\alpha_{2j-1}}=0 & \bar{j}=0,j\neq 0 \\ \tilde{P}^{\dot{\alpha}\alpha}\left[Q^I_\alpha,\bar{Q}_{\dot{\alpha}J}\right]\big|0,0\big\rangle=0 & j=\bar{j}=0 \end{cases} \tag{3.55}$$

This multiplet is special since it contains states corresponding to conserved currents. These conserved currents include the Noether currents corresponding to the symmetries in the theory. For example, the $\hat{\mathcal{C}}_{0(0,0)}$ multiplet contains the stress tensor and the R-symmetry current. One could guess that the state corresponding to the stress tensor belongs to the four dimensional $0_{(1,1)}$ multiplet (since it is a spin-2 traceless symmetric tensor with R=0) in the last level. Therefore it satisfies the first conservation equation of (3.55)

$$\tilde{P}^{\dot{\alpha}\alpha}\big|0,0\big\rangle_{\alpha\alpha_1,\dot{\alpha}\dot{\alpha}_1}=0\longleftrightarrow\partial^\mu\big|0,0\big\rangle_{\mu\nu}=0$$

The R-symmetry current on the other hand corresponds to the $1_{\left(\frac{1}{2},\frac{1}{2}\right)}$ multiplet at the third level and satisfies

$$\tilde{P}^{\dot{\alpha}\alpha}\big|1,0\big\rangle_{\alpha\dot{\alpha}}=0\longleftrightarrow\partial^\mu\big|1,0\big\rangle_{\mu}=0$$

Now consider again the $\mathcal{B}_{R,r(0,\bar{j})}$ multiplet, At the unitarity threshold (3.38) we can add to it the shortening condition (3.49) to get an even shorter multiplet denoted by $\mathcal{D}_{R(0,\bar{j})}=\mathcal{B}_{R,r(0,\bar{j})}\cap\bar{\mathcal{C}}_{R,r(j,\bar{j})}$ with dimension

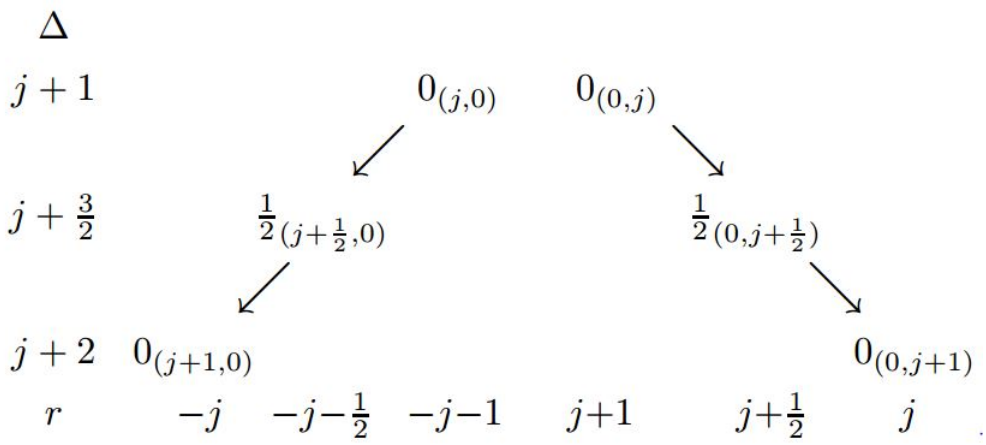
$$dim\mathcal{D}_{R(0,\bar{j})}=2^4\left(4R-1\right)\left(2\bar{j}+1\right)+2^5R \tag{3.56}$$

Similarly at the threshold of (3.40) of $\bar{\mathcal{B}}_{R,r(j,0)}$ we can impose the conditions (3.43) to get the conjugate

multiplet $\bar{\mathcal{D}}_{R(j,0)} = \bar{\mathcal{B}}_{R,r(j,0)} \cap \mathcal{C}_{R,r(j,\bar{j})}$ with dimension

$$dim\bar{\mathcal{D}}_{R(j,0)} = 2^4 (4R-1)(2j+1) + 2^5 R \tag{3.57}$$

If $\Delta = r = 1+\bar{j}$ or $\Delta = -r = 1+j$, we can also impose the semi-shortening conditions (3.49) or (3.43) to multiplets $\mathcal{E}_{r(0,\bar{j})}$ or $\bar{\mathcal{E}}_{r(j,0)}$respectively. For $j = \bar{j}$, the resulting reduced multiplets $\mathcal{F}_{r(0,\bar{j})}$and $\bar{\mathcal{F}}_{r(j,0)}$ are given by



Consistency with (3.15)-(i) then requires that these multiplets satisfy

$$\begin{cases} \tilde{P}^{\dot{\alpha}\alpha} \left|0,-1-j\right\rangle_{\dot{\alpha}\dot{\alpha}_1 \ldots \dot{\alpha}_{2\bar{j}-1}} = 0 \quad for\ \mathcal{E}_{r(0,\bar{j})} \\ \tilde{P}^{\dot{\alpha}\alpha} \left|0,-1-j\right\rangle_{\alpha\alpha_1 \ldots \alpha_{2j-1}} = 0 \quad for\ \bar{\mathcal{E}}_{r(j,0)} \end{cases} \tag{3.58}$$

These are just generalised Dirac equations for any spins since we can see for example that for $j = \frac{1}{2}$ the second equation just means that $(\sigma^\mu \partial_\mu)^{\dot{\alpha}\alpha} \left|0,-1-j\right\rangle_\alpha = 0$, which is just the usual Dirac equation.

The multiplets constructed above can be summarized in the following table -

Multiplet	Shortening Condition	Consistency conditions
$\mathcal{A}^{\Delta}_{R,r(j,\bar{j})}$	None	None
$\mathcal{B}_{R,r\left(0,\bar{j}\right)}$	$Q^1_\alpha \big\vert R,r\big\rangle^{hw}_{\dot\alpha_1\ldots\dot\alpha_{2\bar j}}=0$	$j=0;\ \Delta=2R+r$
$\bar{\mathcal{B}}_{R,r(j,0)}$	$\bar Q_{\dot\alpha 2} \big\vert R,r\big\rangle^{hw}_{\alpha_1\ldots\alpha_{2j}}=0$	$\bar j=0;\ \Delta=2R-r$
$\hat{\mathcal{B}}_R$	$\mathcal{B}\cap\bar{\mathcal{B}}$	$j=\bar j=r=0;\ \Delta=2R$
$\mathcal{C}_{R,r\left(j,\bar{j}\right)}$	$\begin{cases}\epsilon^{\alpha\beta}Q^1_\beta \big\vert R,r\big\rangle^{hw}_{\alpha_1\ldots\alpha_{2j-1}\alpha,\dot\alpha_1\ldots\dot\alpha_{2\bar j}}=0 & j>0\\ \left(Q^1\right)^2\big\vert R,r\big\rangle^{hw}_{\dot\alpha_1\ldots\dot\alpha_{2\bar j}}=0 & j=0\end{cases}$	$\Delta=2+2j+2R+r$
$\bar{\mathcal{C}}_{R,r\left(j,\bar{j}\right)}$	$\begin{cases}\epsilon^{\dot\alpha\dot\beta}\bar Q_{\dot\beta 2} \big\vert R,r\big\rangle^{hw}_{\alpha_1\ldots\alpha_{2j},\dot\alpha_1\ldots\dot\alpha_{2\bar j-1}\dot\alpha}=0 & \bar j>0\\ \left(\bar Q_2\right)^2\big\vert R,r\big\rangle^{hw}_{\alpha_1\ldots\alpha_{2j}}=0 & \bar j=0\end{cases}$	$\Delta=2+2\bar j+2R-r$
$\hat{\mathcal{C}}_{R\left(j,\bar{j}\right)}$	$\mathcal{C}\cap\bar{\mathcal{C}}$	$r=\bar j-j;\ \Delta=2+j+\bar j+2R$
$\mathcal{D}_{R\left(0,\bar{j}\right)}$	$\mathcal{B}\cap\bar{\mathcal{C}}$	$r=\bar j+1$
$\bar{\mathcal{D}}_{R(j,0)}$	$\bar{\mathcal{B}}\cap\mathcal{C}$	$-r=j+1$
$\mathcal{E}_{r\left(0,\bar{j}\right)}$	$Q^I_\alpha \big\vert R,r\big\rangle^{hw}_{\dot\alpha_1\ldots\dot\alpha_{2\bar j}}=0$	$R=0$ and $j=0$
$\bar{\mathcal{E}}_{r(j,0)}$	$\bar Q_{\dot\alpha I} \big\vert R,r\big\rangle^{hw}_{\alpha_1\ldots\alpha_{2j}}=0$	$R=0$ and $\bar j=0$
$\mathcal{F}_{r\left(0,\bar{j}\right)}$	$\mathcal{E}\cap\bar{\mathcal{C}}$	$R=0;\ j=0;\ \Delta=r=1+\bar j;\ \tilde P^{\dot\alpha\alpha}\big\vert 0,1+\bar j\big\rangle_{\dot\alpha\dot\alpha_1\ldots\dot\alpha_{2\bar j-1}}=0$
$\bar{\mathcal{F}}_{r(j,0)}$	$\bar{\mathcal{E}}\cap\mathcal{C}$	$R=0;\ \bar j=0;\ \Delta=-r=1+j;\ \tilde P^{\dot\alpha\alpha}\big\vert 0,-1-j\big\rangle_{\alpha\alpha_1\ldots\alpha_{2j-1}}=0$

At the unitarity threshold of (3.32) with $r=\bar j-j$ we can take the direct sum of certain short and semi-short multiplets to obtain a long multiplet. For example if $j,\bar j>0$ then

$$\boldsymbol{\mathcal{A}^{2R+j+\bar j+2}_{R,\bar j-j\left(j,\bar j\right)}\simeq\hat{\mathcal{C}}_{R\left(j,\bar j\right)}\oplus\hat{\mathcal{C}}_{R+\frac12\left(j-\frac12,\bar j\right)}\oplus\hat{\mathcal{C}}_{R+\frac12\left(j,\bar j-\frac12\right)}\oplus\hat{\mathcal{C}}_{R+1\left(j-\frac12,\bar j-\frac12\right)}} \tag{3.59}$$

If $\bar j=0$ we have

$$\boldsymbol{\mathcal{A}^{2R+j+2}_{R,-j(j,0)}\simeq\hat{\mathcal{C}}_{R(j,0)}\oplus\hat{\mathcal{C}}_{R+\frac12\left(j-\frac12,0\right)}\oplus\bar{\mathcal{D}}_{R+1(j,0)}\oplus\bar{\mathcal{D}}_{R+\frac32\left(j-\frac12,0\right)}} \tag{3.60}$$

If $j=\bar j=0$ then

$$\boldsymbol{\mathcal{A}^{2R+2}_{R,0(0,0)}\simeq\hat{\mathcal{C}}_{R(0,0)}\oplus\mathcal{D}_{R+1(0,0)}\oplus\bar{\mathcal{D}}_{R+1(0,0)}\oplus\hat{\mathcal{B}}_{R+2}} \tag{3.61}$$

We may also get a larger short multiplet as a direct sum of certain shorter multiplets. For example,

$$\boldsymbol{\mathcal{B}_{R,\bar j+1\left(0,\bar j\right)}\simeq\mathcal{D}_{R\left(0,\bar j\right)}\oplus\mathcal{D}_{R+\frac12\left(0,\bar j-\frac12\right)}} \tag{3.62}$$

$$\boldsymbol{\mathcal{B}_{R,1(0,0)}\simeq\mathcal{D}_{R(0,0)}\oplus\hat{\mathcal{B}}_{R+1}} \tag{3.63}$$

We can see that (3.60) and (3.61) are special cases of (3.59) by realizing the equivalence of the following multiplets

$$\boldsymbol{\hat{\mathcal{C}}_{R\left(j,-\frac12\right)}\simeq\bar{\mathcal{D}}_{R+\frac12(j,0)}} \tag{3.64}$$

$$\hat{\mathcal{C}}_{R(-\frac{1}{2},\bar{j})} \simeq \mathcal{D}_{R+\frac{1}{2}(0,\bar{j})} \tag{3.65}$$

$$\bar{\mathcal{D}}_{R(-\frac{1}{2},0)} \cong \mathcal{D}_{R(0,-\frac{1}{2})} \cong \hat{\mathcal{B}}_{R+\frac{1}{2}} \tag{3.66}$$

We can prove these relations by calculating the characters for the multiplets on both sides. Now since all the multiplets are essentially direct products of SU(2), we only need to know the character formula for SU(2). The character for a rotation of angle θ in a spin j representation of SU(2) is given by

$$\chi^j(\theta) = \sum_{m=-j}^{j} e^{im\theta} = \frac{sin\left(j+\frac{1}{2}\right)\theta}{sin\frac{1}{2}\theta} \tag{3.67}$$

Now a representation $R_{(j,\bar{j})}$ is the direct product of a spin-R SU(2), a spin j SU(2) and a spin $\bar{j}$ SU(2). Hence the character is given by

$$\chi^{(R,j,\bar{j})}(\theta,\phi,\alpha) = \sum_{m_R,m_j,m_{\bar{j}}} e^{im_R\theta}e^{im_j\phi}e^{im_{\bar{j}}\alpha} = \frac{sin\left(R+\frac{1}{2}\right)\theta}{sin\frac{1}{2}\theta}\frac{sin\left(j+\frac{1}{2}\right)\phi}{sin\frac{1}{2}\phi}\frac{sin\left(\bar{j}+\frac{1}{2}\right)\alpha}{sin\frac{1}{2}\alpha} \tag{3.68}$$

As in (3.30), a superconformal multiplet may be obtained by applying Qs and $\bar{Q}s$ to the states of $R_{(j,\bar{j})}$ resulting in a vector space $\mathcal{V}_{l,\bar{l}}$ which may be decomposed into the sum of representations $R'_{(j',\bar{j}')}$ by suitable addition of weights for the corresponding Qs nd $\bar{Q}s$.Then the character for the whole supermultiplet is the sum of characters for all $R'_{(j',\bar{j}')}$ in the supermultiplet. Thus we need to add terms like (3.68) corresponding to each $R'_{(j',\bar{j}')}$ in the tree diagram of a supermultiplet.

$$R'_{(j',\bar{j}')} \sim \frac{sin\left(R'+\frac{1}{2}\right)\theta}{sin\frac{1}{2}\theta}\frac{sin\left(j'+\frac{1}{2}\right)\phi}{sin\frac{1}{2}\phi}\frac{sin\left(\bar{j}'+\frac{1}{2}\right)\alpha}{sin\frac{1}{2}\alpha} = \chi^{(R',j',\bar{j}')}$$

For example the character for a long multiplet $\mathcal{A}^{\Delta}_{R(j,\bar{j})}$ is given as

$$\begin{aligned}\chi^{\mathcal{A}^{\Delta}_{R(j,\bar{j})}} =& \chi^{(R,j,\bar{j})} + \chi^{\left(R+\frac{1}{2},j+\frac{1}{2},\bar{j}\right)} + \chi^{\left(R+\frac{1}{2},j-\frac{1}{2},\bar{j}\right)} + \chi^{\left(R-\frac{1}{2},j+\frac{1}{2},\bar{j}\right)} + \chi^{\left(R-\frac{1}{2},j-\frac{1}{2},\bar{j}\right)} \\ &+ \chi^{\left(R+\frac{1}{2},j,\bar{j}+\frac{1}{2}\right)} + \chi^{\left(R+\frac{1}{2},j,\bar{j}-\frac{1}{2}\right)} + \chi^{\left(R-\frac{1}{2},j,\bar{j}+\frac{1}{2}\right)} + \chi^{\left(R-\frac{1}{2},j,\bar{j}-\frac{1}{2}\right)} + \chi^{(R+1,j+1,\bar{j})} \\ &+ \textit{other second and higher level terms } \left(2^8 \textit{ terms}\right)\end{aligned} \tag{3.69}$$

As you can see the first eight terms are the possible terms of applying any supercharge once, then we will have second and higher level terms as well. In total, there are $2^8 = 256$ terms here corresponding to each representation $R'_{(j',\bar{j}')}$ for every possible combination of the eight supercharges in accordance with (3.30) and (3.31). Similarly, the character for the multiplet $\mathcal{B}_{R,r(0,\bar{j})}$ will have all the 48 terms present in the tree diagram on page 55. As one can notice it is hard to keep track of so many terms in this manner. To ease our way, we can define a polynomial corresponding to each supermultiplet which will be equivalent to the character of a supermultiplet. The rule is as follows - corresponding to each $R'_{(j',\bar{j}')}$ in the supermultiplet (and hence each term $\chi^{(R',j',\bar{j}')} = \frac{sin\left(R'+\frac{1}{2}\right)\theta}{sin\frac{1}{2}\theta}\frac{sin\left(j'+\frac{1}{2}\right)\phi}{sin\frac{1}{2}\phi}\frac{sin\left(\bar{j}'+\frac{1}{2}\right)\alpha}{sin\frac{1}{2}\alpha}$ in the character) there is a term $x^{R'}u^{j'}v^{\bar{j}'}$ in the polynomial. The polynomial is then the sum of all such terms. The advantage here though is that we can write this polynomial in a relatively compact form as a product of sum of terms. For example we can write the polynomial $P(x,u,v)$ for the long

multiplet $\mathcal{A}^{\Delta}_{R(j,\bar{j})}$ as

$$\begin{aligned} P^{\mathcal{A}^{\Delta}_{R(j,\bar{j})}}(x,u,v) =& x^R u^j v^{\bar{j}} \\ & \times \left(1+x^{\frac{1}{2}}u^{\frac{1}{2}}v^0\right)\left(1+x^{\frac{1}{2}}u^{-\frac{1}{2}}v^0\right) \\ & \times \left(1+x^{-\frac{1}{2}}u^{\frac{1}{2}}v^0\right)\left(1+x^{-\frac{1}{2}}u^{-\frac{1}{2}}v^0\right) \\ & \times \left(1+x^{\frac{1}{2}}u^0v^{\frac{1}{2}}\right)\left(1+x^{\frac{1}{2}}u^0v^{-\frac{1}{2}}\right) \\ & \times \left(1+x^{-\frac{1}{2}}u^0v^{\frac{1}{2}}\right)\left(1+x^{-\frac{1}{2}}u^0v^{-\frac{1}{2}}\right) \end{aligned} \tag{3.70}$$

Notice that (3.70) contains all the 256 terms equivalent to those in (3.69) in just a few lines. Note that the we have written this term by taking the term $x^R u^j v^{\bar{j}}$ for the base representation $R_{(j,\bar{j})}$ and multiplying it with a term corresponding to the action of each supercharge with appropriate weights as given in (3.31). We can construct the polynomial for the short multiplet $\mathcal{B}_{R,r(0,\bar{j})}$ in a similar fashion. The difference here is that for each of the four $\bar{Q}s$ we will again have a term like $\left(1+x^{q_1}u^0v^{q_2}\right)$ but we will have a single term $\left(1+x^{-\frac{1}{2}}u^{\frac{1}{2}}v^0+x^{-\frac{1}{2}}u^{\frac{1}{2}}v^0x^{-\frac{1}{2}}u^{-\frac{1}{2}}v^0\right)$ for all the Qs since due to the shortening condition (3.33) and the fact that j=0 the charge Q_2^2 can only act on the second level (or else j' will go negative). Therefore

$$\begin{aligned} P^{\mathcal{B}_{R,r(0,\bar{j})}}(x,u,v) =& x^R u^0 v^j \\ & \times \left(1+x^{-\frac{1}{2}}u^{\frac{1}{2}}v^0+x^{-\frac{1}{2}}u^{\frac{1}{2}}v^0x^{-\frac{1}{2}}u^{-\frac{1}{2}}v^0\right) \\ & \times \left(1+x^{\frac{1}{2}}u^0v^{\frac{1}{2}}\right)\left(1+x^{\frac{1}{2}}u^0v^{-\frac{1}{2}}\right) \\ & \times \left(1+x^{-\frac{1}{2}}u^0v^{\frac{1}{2}}\right)\left(1+x^{-\frac{1}{2}}u^0v^{-\frac{1}{2}}\right) \end{aligned} \tag{3.71}$$

Notice again that this has all 48 terms. Similarly, we can also construct the character for $\mathcal{C}_{R,r(j,\bar{j})}$ by simply omitting the action of the charge Q_2^1 on any level.

$$\begin{aligned} P^{\mathcal{C}_{R,r(j,\bar{j})}}(x,u,v) =& x^R u^j v^{\bar{j}} \\ & \times \left(1+x^{\frac{1}{2}}u^{\frac{1}{2}}v^0\right) \\ & \times \left(1+x^{-\frac{1}{2}}u^{\frac{1}{2}}v^0\right)\left(1+x^{-\frac{1}{2}}u^{-\frac{1}{2}}v^0\right) \\ & \times \left(1+x^{\frac{1}{2}}u^0v^{\frac{1}{2}}\right)\left(1+x^{\frac{1}{2}}u^0v^{-\frac{1}{2}}\right) \\ & \times \left(1+x^{-\frac{1}{2}}u^0v^{\frac{1}{2}}\right)\left(1+x^{-\frac{1}{2}}u^0v^{-\frac{1}{2}}\right) \end{aligned} \tag{3.72}$$

Once we obtain the polynomial P(x,u,v) for each multiplet we can obtain their characters by simply expanding the factors and replacing each $x^{R'}u^{j'}v^{\bar{j}'}$ with $\frac{sin\left(R'+\frac{1}{2}\right)\theta}{sin\frac{1}{2}\theta}\frac{sin\left(j'+\frac{1}{2}\right)\phi}{sin\frac{1}{2}\phi}\frac{sin\left(\bar{j}'+\frac{1}{2}\right)\alpha}{sin\frac{1}{2}\alpha}$. Similarly, we can construct the polynomials and hence characters for other multiplets since they are mostly just conjugates or intersections of Bs and Cs. Now, note that if the terms in the polynomials of two multiplets match then so do the terms in the characters of those multiplets and hence those multiplets are equal. Now using these we can check the multiplet recombination formulae (3.59)-(3.63). For

example, let us prove the easiest one - eqn (3.62).

$$\begin{aligned}
P^{\mathcal{D}_{R(0,\bar{j})}}(x,u,v)+P^{\mathcal{D}_{R+\frac{1}{2}\left(0,\bar{j}-\frac{1}{2}\right)}}(x,u,v) =& \left[x^R u^0 v^{\bar{j}}+x^{R+\frac{1}{2}}u^0 v^{\bar{j}-\frac{1}{2}}\right] \\
&\times\left(1+x^{-\frac{1}{2}}u^{\frac{1}{2}}v^0+x^{-\frac{1}{2}}u^{\frac{1}{2}}v^0x^{-\frac{1}{2}}u^{-\frac{1}{2}}v^0\right) \\
&\times\left(1+x^{\frac{1}{2}}u^0v^{\frac{1}{2}}\right)\left(1+x^{-\frac{1}{2}}u^0v^{\frac{1}{2}}\right)\left(1+x^{-\frac{1}{2}}u^0v^{-\frac{1}{2}}\right) \\
=& x^R u^0 v^{\bar{j}}\left(1+x^{\frac{1}{2}}u^0v^{-\frac{1}{2}}\right) \\
&\times\left(1+x^{-\frac{1}{2}}u^{\frac{1}{2}}v^0+x^{-\frac{1}{2}}u^{\frac{1}{2}}v^0x^{-\frac{1}{2}}u^{-\frac{1}{2}}v^0\right) \\
&\times\left(1+x^{\frac{1}{2}}u^0v^{\frac{1}{2}}\right)\left(1+x^{-\frac{1}{2}}u^0v^{\frac{1}{2}}\right)\left(1+x^{-\frac{1}{2}}u^0v^{-\frac{1}{2}}\right) \\
=& P^{\mathcal{B}_{R,\bar{j}+1(0,\bar{j})}}(x,u,v)
\end{aligned}$$

Now since the polynomials match, so do the characters and hence the multiplets are the same on both sides.

$$\chi^{\mathcal{D}_{R(0,\bar{j})}}+\chi^{\mathcal{D}_{R+\frac{1}{2}\left(0,\bar{j}-\frac{1}{2}\right)}}=\chi^{\mathcal{B}_{R,\bar{j}+1(0,\bar{j})}} \tag{3.73}$$

This proves (3.62). Let us now prove the big one, eqn (3.59).

$$\begin{aligned}
P^{\hat{\mathcal{C}}_{R(j,\bar{j})}}+P^{\hat{\mathcal{C}}_{R+\frac{1}{2}\left(j-\frac{1}{2},\bar{j}\right)}}+P^{\hat{\mathcal{C}}_{R+\frac{1}{2}\left(j,\bar{j}-\frac{1}{2}\right)}}+P^{\hat{\mathcal{C}}_{R+1\left(j-\frac{1}{2},\bar{j}-\frac{1}{2}\right)}} =& \Big[x^R u^j v^{\bar{j}}+x^{R+\frac{1}{2}}u^{j-\frac{1}{2}}v^{\bar{j}} \\
&+x^{R+\frac{1}{2}}u^j v^{\bar{j}-\frac{1}{2}}+x^{R+1}u^{j-\frac{1}{2}}v^{\bar{j}-\frac{1}{2}}\Big] \\
&\times\left(1+x^{\frac{1}{2}}u^{\frac{1}{2}}v^0\right)\left(1+x^{-\frac{1}{2}}u^{\frac{1}{2}}v^0\right) \\
&\times\left(1+x^{-\frac{1}{2}}u^{-\frac{1}{2}}v^0\right)\left(1+x^{\frac{1}{2}}u^0v^{\frac{1}{2}}\right) \\
&\times\left(1+x^{-\frac{1}{2}}u^0v^{\frac{1}{2}}\right)\left(1+x^{-\frac{1}{2}}u^0v^{-\frac{1}{2}}\right) \\
=& x^R u^j v^{\bar{j}}\left(1+x^{\frac{1}{2}}u^{-\frac{1}{2}}v^0+x^{\frac{1}{2}}u^0v^{-\frac{1}{2}}+x^1u^{-\frac{1}{2}}v^{-\frac{1}{2}}\right) \\
&\times\left(1+x^{\frac{1}{2}}u^{\frac{1}{2}}v^0\right)\left(1+x^{-\frac{1}{2}}u^{\frac{1}{2}}v^0\right) \\
&\times\left(1+x^{-\frac{1}{2}}u^{-\frac{1}{2}}v^0\right)\left(1+x^{\frac{1}{2}}u^0v^{\frac{1}{2}}\right) \\
&\times\left(1+x^{-\frac{1}{2}}u^0v^{\frac{1}{2}}\right)\left(1+x^{-\frac{1}{2}}u^0v^{-\frac{1}{2}}\right) \\
=& x^R u^j v^{\bar{j}}\left(1+x^{\frac{1}{2}}u^{-\frac{1}{2}}v^0\right)\left(1+x^{\frac{1}{2}}u^0v^{-\frac{1}{2}}\right) \\
&\times\left(1+x^{\frac{1}{2}}u^{\frac{1}{2}}v^0\right)\left(1+x^{-\frac{1}{2}}u^{\frac{1}{2}}v^0\right) \\
&\times\left(1+x^{-\frac{1}{2}}u^{-\frac{1}{2}}v^0\right)\left(1+x^{\frac{1}{2}}u^0v^{\frac{1}{2}}\right) \\
&\times\left(1+x^{-\frac{1}{2}}u^0v^{\frac{1}{2}}\right)\left(1+x^{-\frac{1}{2}}u^0v^{-\frac{1}{2}}\right) \\
=& P^{\mathcal{A}^{2R+j+\bar{j}+2}_{R,\bar{j}-j(j,\bar{j})}}(x,u,v)
\end{aligned}$$

Hence

$$\chi^{\hat{\mathcal{C}}_{R(j,\bar{j})}}+\chi^{\hat{\mathcal{C}}_{R+\frac{1}{2}\left(j-\frac{1}{2},\bar{j}\right)}}+\chi^{\hat{\mathcal{C}}_{R+\frac{1}{2}\left(j,\bar{j}-\frac{1}{2}\right)}}+\chi^{\hat{\mathcal{C}}_{R+1\left(j-\frac{1}{2},\bar{j}-\frac{1}{2}\right)}}=\chi^{\mathcal{A}^{2R+j+\bar{j}+2}_{R,\bar{j}-j(j,\bar{j})}} \tag{3.74}$$

Therefore, we have proven (3.59). Since (3.60) and (3.61) follow from (3.59) (using (3.64)-(3.66)), we don't need to prove them separately.

Now that we have studied the $\mathcal{N}=2$ case we can easily see what's going to happen for the simpler case of $\mathcal{N}=1$. Here the R-symmetry is just U(1). Hence the basis states are labeled by a U(1) charge r and are represented as $|r\rangle_{\alpha_1\ldots\alpha_{2j},\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}}}$. Hence

$$R\,|r\rangle_{\alpha_1\ldots\alpha_{2j},\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}}} = \frac{r}{2}\,|r\rangle_{\alpha_1\ldots\alpha_{2j},\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}}} \tag{3.75}$$

where we've taken the eigenvalue to be $r/2$ for aesthetic purpose. Consquently, the various short and semi-short multiplets are summarised in the following table

Multiplet	Shortening Condition	Consistency Conditions
$\mathcal{A}^{\Delta}_{r(j,\bar{j})}$	None	None
$\mathcal{B}_{r(0,\bar{j})}$	$Q_\alpha\big\vert r\big\rangle^{hw}_{\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}}}=0$	$j=0;\Delta=2r$
$\bar{\mathcal{B}}_{R,r(j,0)}$	$\bar{Q}_{\dot{\alpha}}\big\vert R,r\big\rangle^{hw}_{\alpha_1\ldots\alpha_{2j}}=0$	$\bar{j}=0;\Delta=-2r$
$\hat{\mathcal{B}}_R$	$\mathcal{B}\cap\bar{\mathcal{B}}$	$j=\bar{j}=r=0;\Delta=0$
$\mathcal{C}_{R,r(j,\bar{j})}$	$\begin{cases}\epsilon^{\alpha\beta}Q_\beta\vert R,r\rangle^{hw}_{\alpha_1\ldots\alpha_{2j-1}\alpha,\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}}}=0 & j>0\\ (Q)^2\,\vert R,r\rangle^{hw}_{\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}}}=0 & j=0\end{cases}$	$\Delta=2+2j+2r$
$\bar{\mathcal{C}}_{R,r(j,\bar{j})}$	$\begin{cases}\epsilon^{\dot{\alpha}\dot{\beta}}\bar{Q}_{\dot{\beta}2}\vert R,r\rangle^{hw}_{\alpha_1\ldots\alpha_{2j},\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}-1}\dot{\alpha}}=0 & \bar{j}>0\\ (\bar{Q}_2)^2\,\vert R,r\rangle^{hw}_{\alpha_1\ldots\alpha_{2j}}=0 & \bar{j}=0\end{cases}$	$\Delta=2+2\bar{j}-2r$
$\hat{\mathcal{C}}_{R(j,\bar{j})}$	$\mathcal{C}\cap\bar{\mathcal{C}}$	$r=\frac{\bar{j}-j}{2};\Delta=2+j+\bar{j}$
$\mathcal{D}_{R(0,\bar{j})}$	$\mathcal{B}\cap\bar{\mathcal{C}}$	$r=\frac{\bar{j}+1}{2}$
$\bar{\mathcal{D}}_{R(j,0)}$	$\bar{\mathcal{B}}\cap\mathcal{C}$	$-r=\frac{j+1}{2}$

3.5 $\mathcal{N}=3$ and $\mathcal{N}=4$ Representations

Next, let us consider the more complicated case $\mathcal{N}=4$. For $\mathcal{N}=4$, we see that $R^I_I=0$ and therefore the R-symmetry is just SU(4) instead of U(4). Then the first step is to write the generators R^I_J in a Chevalley basis which we now describe briefly. For SU(4) we have a three dimensional Cartan subalgebra which we denote by H. On the weight space which is dual to the Cartan subspace, we can choose a basis of simple roots $\{\alpha_i\}$. This induces a basis $\{h_i\}$ on the Cartan subspace due to a correspondence between α_i and h_i. Consequently, each simple root α_i has an SU(2) subalgebra $\{e^{\pm}_i, h_i\}$ associated with it. The commutation relations between the generators are then given by

$$\begin{aligned}[h_i,h_j]=0;\left[e^+_i,e^-_j\right]&=\delta_{ij}h_j\\ \left[h_i,e^{\pm}_j\right]&=\pm\,A_{ij}e^{\pm}_j\end{aligned} \tag{3.76}$$

where A_{ij} is the Cartan matrix for SU(4) and is given by

$$[A_{ij}] = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix} \tag{3.77}$$

Here we have written the complete algebra in (3.76). This is a Chevalley basis for the algebra. The relation between these generators and the usual ones is given by

$$\left[R_J^I\right] = \begin{pmatrix} \frac{1}{4}(3h_1+2h_2+h_3) & e_1^+ & \left[e_1^+, e_2^+\right] & \left[e_1^+, \left[e_2^+, e_3^+\right]\right] \\ e_1^- & \frac{1}{4}(-h_1+2h_2+h_3) & e_2^+ & \left[e_2^+, e_3^+\right] \\ -\left[e_1^-, e_2^-\right] & e_2^- & -\frac{1}{4}(h_1+2h_2-h_3) & e_3^+ \\ \left[e_1^-, \left[e_2^-, e_3^-\right]\right] & -\left[e_2^-, e_3^-\right] & e_3^- & -\frac{1}{4}(h_1+2h_2+3h_3) \end{pmatrix} \tag{3.78}$$

An SU(4) representation is characterized by the Dynkin labels $[\lambda_1, \lambda_2, \lambda_3]$ which are eigenvalues of the representation of h_i. Such a representation has dimension given by

$$d(\lambda_1, \lambda_2, \lambda_3) = \frac{1}{12}(\lambda_1+\lambda_2+\lambda_3)(\lambda_1+\lambda_2+2)(\lambda_2+\lambda_3+2)(\lambda_1+1)(\lambda_2+1)(\lambda_3+1) \tag{3.79}$$

The above formula may be derived using the Hook's rule for the Young Tablaeux of SU(4). Now the standard basis states are labeled as $|k,p,q\rangle_{\alpha_1..\alpha_{2j},\dot{\alpha}_1...\dot{\alpha}_{2\bar{j}}}$ where k,p,q are the Dynkin labels (eigenvalues of h_1, h_2 and h_3 respectively). Then along with (3.28) and (3.29), the highest weight states also satisfy

$$e_i^+ |k,p,q\rangle^{hw}_{\alpha_1..\alpha_{2j},\dot{\alpha}_1...\dot{\alpha}_{2\bar{j}}} = 0 \tag{3.80}$$

The representation satisfying these conditions is represented by $[k,p,q]_{(j,\bar{j})}$. Then we can generate a supermultiplet by acting with supercharges Qs and $\bar{Q}$s on the highest weight state. Using (3.15) and (3.78) we see that the Qs and $\bar{Q}s$ satisfy

$$\begin{gathered} \left[h_1, Q_\alpha^1\right] = Q_\alpha^1; \left[h_{2,3}, Q_\alpha^1\right] = 0 \\ \left[h_3, \bar{Q}_{\dot{\alpha}4}\right] = \bar{Q}_{\dot{\alpha}4}; \left[h_{1,2}, \bar{Q}_{\dot{\alpha}4}\right] = 0 \\ \left[e_i^+, Q_\alpha^1\right] = 0; \left[e_i^+, \bar{Q}_{\dot{\alpha}4}\right] = 0 \end{gathered} \tag{3.81}$$

We also see that the other charges are related by action of e_i^- to Q_α^1 or $\bar{Q}_{\dot{\alpha}4}$ (i.e taking the commutator with e_i^-)

$$Q_\alpha^1 \xrightarrow{e_1^-} Q_\alpha^2 \xrightarrow{e_2^-} Q_\alpha^3 \xrightarrow{e_3^-} Q_\alpha^4 \tag{3.82}$$

$$\bar{Q}_{\dot{\alpha}4} \xrightarrow{e_1^-} -\bar{Q}_{\dot{\alpha}3} \xrightarrow{e_2^-} \bar{Q}_{\dot{\alpha}2} \xrightarrow{e_3^-} -\bar{Q}_{\dot{\alpha}1} \tag{3.83}$$

Therefore, the SU(4) part of the Q representation (meaning the representation on Qs) has Dynkin labels $[1,0,0]$ which are weights of the highest state Q_α^1, and that of $\bar{Q}$ representation has Dynkin labels $[0,0,1]$ with $\bar{Q}_{\dot{\alpha}4}$ as the highest weight. Using the Cartan matrix we can write the weights (in

terms of indices corresponding to the standard bases h_i) of the Qs and $\bar{Q}s$ as follows

$$\begin{aligned}
Q_1^1 \sim [1,0,0]_{(\frac{1}{2},0)}\,;\; Q_2^1 \sim [1,0,0]_{(-\frac{1}{2},0)}\,;\; \bar{Q}_{\dot{1}1} \sim [-1,0,0]_{(0,\frac{1}{2})}\,;\; \bar{Q}_{\dot{2}1} \sim [-1,0,0]_{(0,-\frac{1}{2})} \\
Q_1^2 \sim [-1,1,0]_{(\frac{1}{2},0)}\,;\; Q_2^2 \sim [-1,1,0]_{(-\frac{1}{2},0)}\,;\; \bar{Q}_{\dot{1}2} \sim [1,-1,0]_{(0,\frac{1}{2})}\,;\; \bar{Q}_{\dot{2}2} \sim [1,-1,0]_{(0,-\frac{1}{2})} \\
Q_1^3 \sim [0,-1,1]_{(\frac{1}{2},0)}\,;\; Q_2^3 \sim [0,-1,1]_{(-\frac{1}{2},0)}\,;\; \bar{Q}_{\dot{1}3} \sim [0,1,-1]_{(0,\frac{1}{2})}\,;\; \bar{Q}_{\dot{2}3} \sim [0,1,-1]_{(0,-\frac{1}{2})} \\
Q_1^4 \sim [0,0,-1]_{(\frac{1}{2},0)}\,;\; Q_2^4 \sim [0,0,-1]_{(-\frac{1}{2},0)}\,;\; \bar{Q}_{\dot{1}4} \sim [0,0,1]_{(0,\frac{1}{2})}\,;\; \bar{Q}_{\dot{2}4} \sim [0,0,1]_{(0,-\frac{1}{2})}
\end{aligned} \tag{3.84}$$

Then the representations $[k',p',q']_{(j',\bar{j}')}$ formed by acting on $[k,p,q]_{(j,\bar{j})}$ with Qs and $\bar{Q}s$ are found using the Racah Speiser Algorithm explained below.

Racah Speiser Algorithm

The Racah Speiser Algorithm is used to find out the irreducible components of the direct product of two irreducible representations. Suppose we have two representations $\mathcal{R}_\Lambda$ and $\mathcal{R}_{\Lambda'}$ of some semi-simple Lie algebra. Then the irreducible components of the direct product are

$$\mathcal{R}_\Lambda \otimes \mathcal{R}_{\Lambda'} = \sum_{\lambda \in V_{\Lambda'}} \mathcal{R}_{\Lambda+\lambda} \tag{3.85}$$

where $V_{\Lambda'}$ is the set of all weights in the representation $\mathcal{R}_{\Lambda'}$. We also require

$$\mathcal{R}_\Lambda = sign\,(S)\,\mathcal{R}_{\lambda^s};\; \lambda^s = S\,(\lambda+\delta) - \delta \tag{3.86}$$

where S is an element of the Weyl group $\mathcal{W}$ and $\delta = \frac{1}{2}\sum_{\alpha>0}\alpha$ is half the sum of all positive roots. Let us see how this algorithm translates for SU(2) and see if we get the familiar SU(2) addition rule back. We have two roots α and $-\alpha$. Also, $h = J_3$, $e_{\pm\alpha} = J_\pm$ and $\delta = \frac{1}{2}\alpha$. The set of weights is - $V_{j'} = \{-j', -j'+1, ..., j'-1, j'\}$. Therefore we get

$$\mathcal{R}_j \otimes \mathcal{R}_{j'} = \sum_{m=-j'}^{j'} \mathcal{R}_{j+m} \tag{3.87}$$

on the condition that

$$\mathcal{R}_m = -\mathcal{R}_{-m-1} \tag{3.88}$$

In particular, this implies that

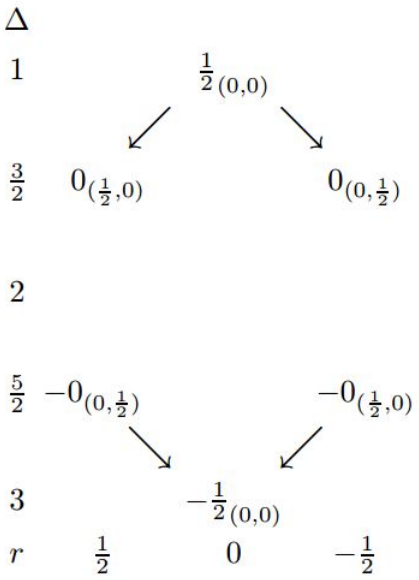
$$\mathcal{R}_{-\frac{1}{2}} = -\mathcal{R}_{-\frac{1}{2}} = 0 \tag{3.89}$$

You can check that (3.87) along with (3.88) gives us back the usual SU(2) rule. The catch is that whenever j+m<0 in a term on the RHS of (3.87) we use (3.88) to convert it back to a representation with positive weight but with an overall '-' sign. This negative term will always cancel with one of the positive terms in the sum, leaving only a sum of positive representations which match with the SU(2) addition rule. When you use the Racah Speiser algorithm for $\mathcal{N}=2$ supermultiplets, you may encounter multiplets with negative values of R', j' and $\bar{j}'$. Then using (3.88), we can convert them into positive representations which are subtracted in the supermultiplets. For this case, the condition

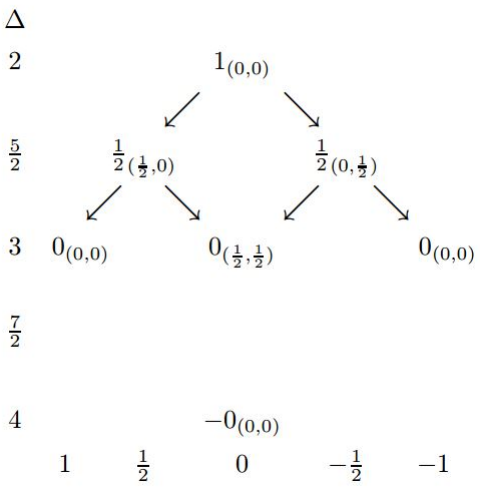
(3.85) translates to

$$(-R)_{(j,\bar{j})} = -\left(R-1\right)_{(j,\bar{j})}; R_{(-j,\bar{j})} = -R_{(j-1,\bar{j})}; R_{(j,-\bar{j})} = -R_{(j,\bar{j}-1)} \tag{3.90}$$

Let us see this for the $\hat{\mathcal{B}}_{\frac{1}{2}}$ multiplet. The actual multiplet with the negative contributions from eqn (3.87) can be written as



We can use the equations of motion to put the contributions of the negative terms to zero. For example if we denote the $\frac{1}{2}_{(0,0)}$ representation by a scalar Φ and the $0_{(\frac{1}{2},0)}$ and $0_{(0,\frac{1}{2})}$ by two spinors ψ_α and $\chi^{\dot{\alpha}}$ we can see that $\partial^2\Phi$ matches the $\frac{1}{2}_{(0,0)}$ in both spin and dimensions. Similarly the $0_{(0,\frac{1}{2})}$ and $0_{(\frac{1}{2},0)}$ terms match with $\partial^{\dot{\alpha}\alpha}\psi_\alpha$ and $\partial_{\alpha\dot{\alpha}}\chi^{\dot{\alpha}}$ (here we've used the usual correspondence to write $(\sigma^\mu\partial_\mu)^{\dot{\alpha}\alpha} = \partial^{\dot{\alpha}\alpha}$). The equations of motion which put these contributions to zero are precisely the Klein Gordon equation $\partial^2\Phi = 0$ and the Dirac equations $\partial^{\dot{\alpha}\alpha}\psi_\alpha = 0$ and $\partial_{\alpha\dot{\alpha}}\chi^{\dot{\alpha}} = 0$. We previously neglected these contributions while writing the multiplets. Sometimes the negative contributions are also eliminated by conservation equations. To see this, let us examine what happens in the case for $\hat{\mathcal{B}}_1$. We have



Here we have the current $J_{\alpha\dot{\alpha}}$ corresponding to the $0_{\left(\frac{1}{2},\frac{1}{2}\right)}$ representation. Then $\partial^{\dot{\alpha}\alpha}J_{\alpha\dot{\alpha}}$ corresponds to the negative term $0_{(0,0)}$ and hence can be put to zero by demanding that the current $J_{\alpha\dot{\alpha}}$ be conserved.

For SU(4) we simply calculate the weight vectors of one representation and add them to the highest weight of the other to get the highest weights of the irreducible components. A more visual approach is provided by the Young's Tablaeux which can be applied to any SU($\mathcal{N}$). After a little bit of analysis one can see that for SU(4) the condition (3.86) for a representation with Dynkin labels $[k,p,q]$ translates to

$$[k,p,q] = -[-k-2, p+k+1, , q] = -[k+p+1, -p-2, q+p+1] = -[k, p+q+1, -q-2] \quad (3.91)$$

These identities generate a group of order 24 so there are 23 non trivial relations which allow $[k,p,q] = \pm[k\prime,p\prime,q\prime]$ with $k\prime,p\prime,q\prime \geq 0$ or $[k,p,q]=0$, as when any of k, p, q are equal to -1. These rules allow the construction of supermultiplets to be extended to cases when the lowest state representation has low k, p, q.

Now let us continue with our discussion of $\mathcal{N}=4$ representations. A general long multiplet is denoted by $\mathcal{A}^{\Delta}_{[k,p,q](j,\bar{j})}$. It has dimension

$$dim\mathcal{A}^{\Delta}_{[k,p,q](j,\bar{j})} = 2^{16}d\,(k,p,q)\,(2j+1)\,(2\bar{j}+1) \quad (3.92)$$

where d(k,p,q) is given by eqn (3.79). The unitarity condition gives

$$\Delta \geq 2+2j+\frac{1}{2}\,(3k+2p+q)\,,\ 2+2\bar{j}+\frac{1}{2}\,(k+2p+3q) \quad (3.93)$$

As before, if j=0 we can apply a shortening condition as follows

$$Q^I_\alpha\,|k,p,q\rangle^{hw}_{\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}}} = 0 \quad (3.94)$$

Then using (3.4a) we can see that

$$\left\{Q^I_\alpha, S^\beta_J\right\}|k,p,q\rangle^{hw}_{\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}}} \Longrightarrow -\delta^\beta_\alpha\left(2i\delta^I_J D + 4R^I_J\right)|k,p,q\rangle^{hw}_{\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}}} = 0$$

The values of R^I_J in terms of Cartan basis can be taken from eqn (3.78) and substituted to obtain the solution to the above equation. The shortening condition (3.94) can only hold for I=1, I=1,2, I=1,2,3 or I=1,2,3,4. Corresponding to this, let s denote the fraction of supercharges for which (3.94) is satisfied. Then we get the following solutions for each case

$$\begin{cases} s=\frac{1}{4};\ I=1 & \Delta=\frac{1}{2}\,(3k+2p+q) \\ s=\frac{1}{2};\ I=1,2 & \Delta=\frac{1}{2}\,(2p+q)\,;\ k=0 \\ s=\frac{3}{4};\ I=1,2,3 & \Delta=\frac{1}{2}q;\ k=p=0 \\ s=1;\ I=1,2,3,4 & \Delta=0;\ k=p=q=0 \end{cases} \quad (3.95)$$

Similarly if $\bar{j}=0$ we can apply a shortening condition of the form

$$\bar{Q}_{\dot{\alpha}I}\big|k,p,q\big\rangle^{hw}_{\alpha_1\ldots\alpha_{2j}}=0 \tag{3.96}$$

For this we have the solutions

$$\begin{cases} \bar{s}=\frac{1}{4};\ I=4 & \Delta=\frac{1}{2}\left(k+2p+3q\right) \\ \bar{s}=\frac{1}{2};\ I=3,4 & \Delta=\frac{1}{2}\left(k+2p\right);\ q=0 \\ \bar{s}=\frac{3}{4};\ I=2,3,4 & \Delta=\frac{1}{2}k;\ p=q=0 \\ \bar{s}=1;\ I=1,2,3,4 & \Delta=0;\ k=p=q=0 \end{cases} \tag{3.97}$$

A multiplet satisfying both (3.94) and (3.96) with non zero s and $\bar{s}$ (also $j=\bar{j}=0$) is denoted by $\mathcal{B}^{s,\bar{s}}_{[k,p,q](0,0)}$. We can see that for the two solutions to be consistent with each other we must have $s=\bar{s}$ if they are non zero. For $s=\bar{s}=1$ we have k=p=q=0 and $P_\mu\big|0,0,0\big\rangle=0$ and hence this corresponds to the vacuum representation. Same is true for $s=\bar{s}=\frac{3}{4}$. Therefore there are only two non trivial representations.The first one is $\mathcal{B}^{\frac{1}{4},\frac{1}{4}}_{[q,p,q]}$ for which we must have k=q and $\Delta=p+2q$ for consistency. The other one is $\mathcal{B}^{\frac{1}{2},\frac{1}{2}}_{[0,p,0]}$ for which k=q=0 and $\Delta=p$. The dimensions of these two multiplets are

$$dim\mathcal{B}^{\frac{1}{2},\frac{1}{2}}_{[0,p,0]}=2^8 d\left(0,p-2,0\right) \tag{3.98}$$

$$dim\mathcal{B}^{\frac{1}{4},\frac{1}{4}}_{[q,p,q]}=2^{12}d\left(q-2,p,q\right)+\frac{256}{3}\left(p+1\right)\left(2p+3\right)\left(2p+1\right)\left(2q+p+1\right) \tag{3.99}$$

Just as before we can also apply certain semi-shortening conditions given by

$$\begin{cases} \epsilon^{\alpha\beta}Q^I_\beta\big|k,p,q\big\rangle^{hw}_{\alpha_1\ldots\alpha_{2j-1}\alpha,\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}}}=0 & j>0 \\ \left(Q^I\right)^2\big|k,p,q\big\rangle^{hw}_{\alpha_1\ldots\alpha_{2j},\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}}}=0 & j=0 \end{cases} \tag{3.100}$$

Now using (3.15) we can see that

$$\epsilon^{\alpha\beta}\left\{Q^I_\beta,S^\gamma_L\right\}\big|k,p,q\big\rangle^{hw}_{\alpha_1\ldots\alpha_{2j-1}\alpha,\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}}}=0$$

$$\Rightarrow\left(\delta^I_L\left(-j-1+\frac{1}{2}\Delta\right)-R^I_L\right)\big|k,p,q\big\rangle^{hw}_{\alpha_1\ldots\alpha_{2j},\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}}}=0$$

Again the values of R^I_J in terms of Cartan basis can be taken and substituted to obtain the solution. Just as before, the semi-shortening condition (3.100) can only hold for I=1, I=1,2, I=1,2,3 or I=1,2,3,4. Corresponding to this, we denote the fraction of supercharges for which (3.100) is satisfied by t. Then

we get the following solutions for each case

$$\begin{cases} t=\frac{1}{4};\ I=1 & \Delta=2+2j+\frac{1}{2}\left(3k+2p+q\right) \\ t=\frac{1}{2};\ I=1,2 & \Delta=2+2j+\frac{1}{2}\left(2p+q\right);\ k=0 \\ t=\frac{3}{4};\ I=1,2,3 & \Delta=2+2j+\frac{1}{2}q;\ k=p=0 \\ t=1;\ I=1,2,3,4 & \Delta=2+2j;\ k=p=q=0 \end{cases} \tag{3.101}$$

Similarly we can have the conjugate condition

$$\begin{cases} \epsilon^{\dot{\alpha}\dot{\beta}}\bar{Q}_{\dot{\beta}I}\left|k,p,q\right\rangle^{hw}_{\alpha_1\ldots\alpha_{2j},\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}-1}\dot{\alpha}}=0 & \bar{j}>0 \\ \left(\bar{Q}_I\right)^2\left|k,p,q\right\rangle^{hw}_{\alpha_1\ldots\alpha_{2j}}=0 & \bar{j}=0 \end{cases} \tag{3.102}$$

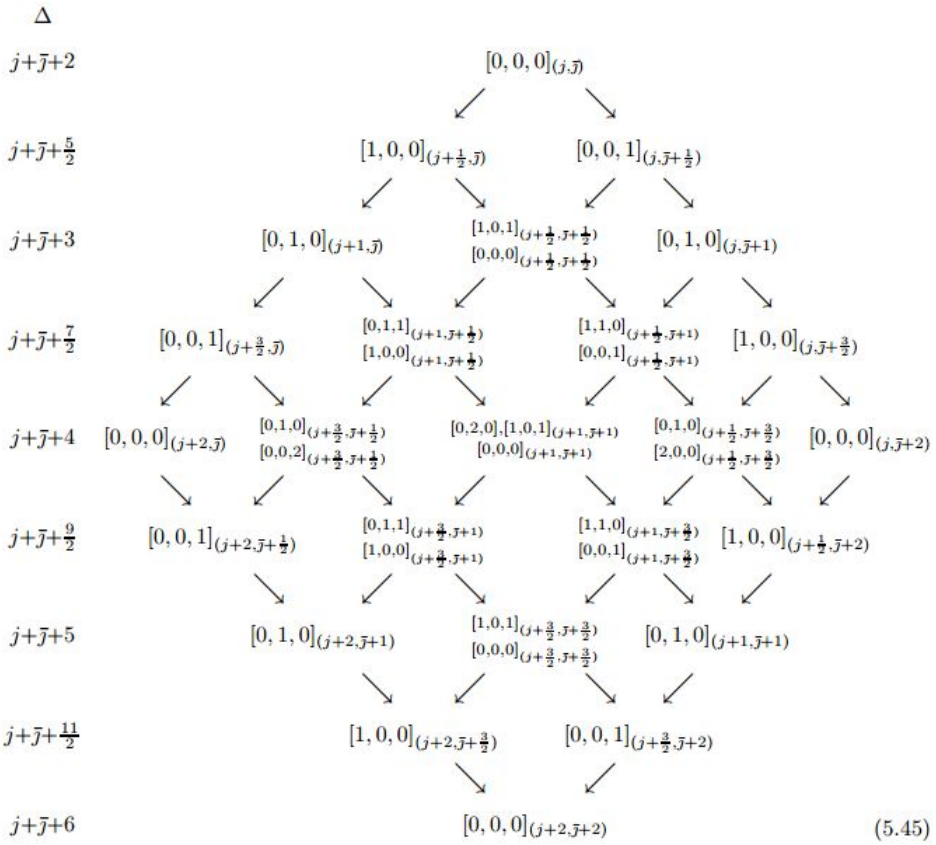
and the corresponding solutions

$$\begin{cases} \bar{t}=\frac{1}{4};\ I=4 & \Delta=2+2\bar{j}+\frac{1}{2}\left(k+2p+3q\right) \\ \bar{t}=\frac{1}{2};\ I=3,4 & \Delta=2+2\bar{j}+\frac{1}{2}\left(k+2p\right);\ q=0 \\ \bar{t}=\frac{3}{4};\ I=2,3,4 & \Delta=2+2\bar{j}+\frac{1}{2}k;\ p=q=0 \\ \bar{t}=1;\ I=1,2,3,4 & \Delta=2+2\bar{j};\ k=p=q=0 \end{cases} \tag{3.103}$$

If both the semi-shortening conditions (3.100) and (3.102) are applied for non zero t and $\bar{t}$ we obtain a semi-short multiplet denoted by $\mathcal{C}^{t,\bar{t}}_{[k,p,q](j,\bar{j})}$. A special case is $\mathcal{C}^{1,1}_{[0,0,0](j,\bar{j})}$ since for $j,\bar{j}>0$ according to (3.15) it leads to a conservation equation of the form

$$\tilde{P}^{\dot{\alpha}\alpha}\left|0,0,0\right\rangle^{hw}_{\alpha\alpha_1\ldots\alpha_{2j-1},\dot{\alpha}\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}-1}}=0 \tag{3.104}$$

Results similar to $\mathcal{N}=2$ also hold here if j or $\bar{j}$ is zero. This multiplet is given by



(5.45)

Other important examples are as follows

$$\mathcal{C}^{\frac{1}{4},\frac{1}{4}}_{[k,p,q](j,\bar{j})},\ k-q=2\left(\bar{j}-j\right),\ \Delta=2+j+\bar{j}+k+p+q$$

$$\mathcal{C}^{\frac{1}{2},\frac{1}{2}}_{[0,p,0](j,j)},\quad \Delta=2+2j+p$$

$$\mathcal{C}^{\frac{1}{4},\frac{1}{2}}_{[k,p,0](j,\bar{j})},\ k=2\left(\bar{j}-j\right),\ \Delta=2+j+\bar{j}+k+p$$

$$\mathcal{C}^{\frac{1}{4},\frac{3}{4}}_{[k,0,0](j,\bar{j})},\ k=2\left(\bar{j}-j\right),\ \Delta=2+j+\bar{j}+k$$

$$\mathcal{C}^{1,1}_{[0,0,0](j,j)},\quad \Delta=2+2j$$

If j=0 we can apply the shortening condition (3.94) together with the semi-shortening condition (3.102) to obtain a multiplet which is denoted by $\mathcal{D}^{s,\bar{t}}_{[k,p,q](0,\bar{j})}$. Similarly applying (3.96) and (3.100) for $\bar{j}=0$ gives the conjugate multiplet $\bar{\mathcal{D}}^{\bar{s},t}_{[k,p,q](j,0)}$. For $\mathcal{D}^{s,\bar{t}}_{[k,p,q](0,\bar{j})}$ only $s=\frac{1}{4}$ and $\bar{t}=\frac{1}{4},\frac{1}{2},\frac{3}{4}$ are possible (with appropriate consistency constraints on k,p and q). For $\bar{\mathcal{D}}^{\bar{s},t}_{[k,p,q](j,0)}$ we again only have $\bar{s}=\frac{1}{4}$ and $t=\frac{1}{4},\frac{1}{2},\frac{3}{4}$. Just as before we have the following multiplet recombinations

$$\boldsymbol{\mathcal{A}^{2-j+3\bar{j}+p+2q}_{[k,p,q]}\Big|_{k-q=2(\bar{j}-j)} \simeq \mathcal{C}^{\frac{1}{4},\frac{1}{4}}_{[k,p,q](j,\bar{j})} \oplus \mathcal{C}^{\frac{1}{4},\frac{1}{4}}_{[k+1,p,q](j-\frac{1}{2},\bar{j})} \oplus \mathcal{C}^{\frac{1}{4},\frac{1}{4}}_{[k,p,q+1](j,\bar{j}-\frac{1}{2})} \oplus \mathcal{C}^{\frac{1}{4},\frac{1}{4}}_{[k+1,p,q+1](j-\frac{1}{2},\bar{j}-\frac{1}{2})}} \tag{3.105}$$

$$\boldsymbol{\mathcal{A}^{2+2j+p}_{[0,p,0](j,\bar{j})} \simeq \mathcal{C}^{\frac{1}{2},\frac{1}{2}}_{[0,p,0](j,,j)} \oplus \mathcal{C}^{\frac{1}{4},\frac{1}{2}}_{[1,p,0](j-\frac{1}{2},,j)} \oplus \mathcal{C}^{\frac{1}{2},\frac{1}{4}}_{[0,p,1](j,j-\frac{1}{2})} \oplus \mathcal{C}^{\frac{1}{4},\frac{1}{4}}_{[1,p,1](j-\frac{1}{2},j-\frac{1}{2})}} \tag{3.106}$$

$$\boldsymbol{\mathcal{A}^{2+2j}_{[0,0,0](j,j)} \simeq \mathcal{C}^{1,1}_{[0,0,0](j,j)} \oplus \mathcal{C}^{\frac{1}{4},\frac{3}{4}}_{[1,0,0]\left(j-\frac{1}{2},j\right)} \oplus \mathcal{C}^{\frac{3}{4},\frac{1}{4}}_{[0,0,1]\left(j,j-\frac{1}{2}\right)} \oplus \mathcal{C}^{\frac{1}{4},\frac{1}{4}}_{[1,0,1]\left(j-\frac{1}{2},j-\frac{1}{2}\right)}} \tag{3.107}$$

$$\boldsymbol{\mathcal{A}^{2+p+2q}_{[q,p,q](0,0)} \simeq \mathcal{C}^{\frac{1}{4},\frac{1}{4}}_{[q,p,q](0,0)} \oplus \mathcal{D}^{\frac{1}{4},\frac{1}{4}}_{[q+2,p,q](0,0)} \oplus \bar{\mathcal{D}}^{\frac{1}{4},\frac{1}{4}}_{[q,p,q+2](0,0)} \oplus \mathcal{B}^{\frac{1}{4},\frac{1}{4}}_{[q+2,p,q+2](0,0)}} \tag{3.108}$$

$$\boldsymbol{\mathcal{A}^{2+p}_{[0,p,0](0,0)} \simeq \mathcal{C}^{\frac{1}{2},\frac{1}{2}}_{[0,p,0](0,0)} \oplus \mathcal{D}^{\frac{1}{4},\frac{1}{2}}_{[2,p,0](0,0)} \oplus \bar{\mathcal{D}}^{\frac{1}{2},\frac{1}{4}}_{[0,p,2](0,0)} \oplus \mathcal{B}^{\frac{1}{4},\frac{1}{4}}_{[2,p,2](0,0)}} \tag{3.109}$$

$$\boldsymbol{\mathcal{A}^{2}_{[0,0,0](0,0)} \simeq \mathcal{C}^{1,1}_{[0,0,0](0,0)} \oplus \mathcal{D}^{\frac{1}{4},\frac{3}{4}}_{[2,0,0](0,0)} \oplus \bar{\mathcal{D}}^{\frac{3}{4},\frac{1}{4}}_{[0,0,2](0,0)} \oplus \mathcal{B}^{\frac{1}{4},\frac{1}{4}}_{[2,0,2](0,0)}} \tag{3.110}$$

where we may identify the following equivalence relations

$$\boldsymbol{\mathcal{C}^{\frac{1}{4},\frac{1}{4}}_{[k,p,q]\left(j,-\frac{1}{2}\right)} \simeq \bar{\mathcal{D}}^{\frac{1}{4},\frac{1}{4}}_{[k,p,q+1](j,0)};\ \mathcal{C}^{\frac{1}{4},\frac{1}{4}}_{[k,p,q]\left(-\frac{1}{2},-\frac{1}{2}\right)} \simeq \mathcal{B}^{\frac{1}{4},\frac{1}{4}}_{[k+1,p,q+1]}} \tag{3.111}$$

Now let us now discuss the $\mathcal{N}=3$ case. The R-symmetry here is U(3). We know that we can separate the algebra as $U(3) \simeq SU(3) \times U(1)$ by removing the trace. We choose a Chevalley basis for the SU(3) subalgebra - $\left\{h_1, h_2, e_1^{\pm}, e_2^{\pm}\right\}$ for which the commutation relations (3.76) stay the same only with the Cartan matrix taking the form

$$[A_{ij}] = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \tag{3.112}$$

The R-symmetry generators then can be written in terms of this basis as follows

$$\left[R^I_J\right] = \begin{pmatrix} \frac{1}{3}\left(2h_1+h_2\right) & e_1^+ & \left[e_1^+, e_2^+\right] \\ e_1^- & \frac{1}{3}\left(-h_1+h_2\right) & e_2^+ \\ -\left[e_1^-, e_2^-\right] & e_2^- & -\frac{1}{3}\left(h_1+2h_2\right) \end{pmatrix} + \frac{1}{3}\hat{R}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{3.113}$$

An SU(3) representation characterized by the Dynkin labels $[\lambda_1, \lambda_2]$ has dimensions given by

$$d\left(\lambda_1, \lambda_2\right) = \frac{1}{2}\left(\lambda_1+1\right)\left(\lambda_2+1\right)\left(\lambda_1+\lambda_2+2\right) \tag{3.114}$$

The standard basis states are labeled as $|k,p,r\rangle_{\alpha_1..\alpha_{2j},\dot{\alpha}_1...\dot{\alpha}_{2\bar{j}}}$. Again along with (3.28) and (3.29), the highest weight states also satisfy

$$e_i^+ \left|k,p,r\right\rangle^{hw}_{\alpha_1..\alpha_{2j},\dot{\alpha}_1...\dot{\alpha}_{2\bar{j}}} = 0;\ \hat{R}\left|k,p,r\right\rangle^{hw}_{\alpha_1..\alpha_{2j},\dot{\alpha}_1...\dot{\alpha}_{2\bar{j}}} = r\left|k,p,r\right\rangle^{hw}_{\alpha_1..\alpha_{2j},\dot{\alpha}_1...\dot{\alpha}_{2\bar{j}}} \tag{3.115}$$

The representation satisfying these conditions is denoted by $[k,p]_{(j,\bar{j})}$. Then we can generate a super-multiplet by acting with supercharges Qs and $\bar{Q}$s on the highest weight state. Using (3.15) and (3.78) we see that the Qs and $\bar{Q}s$ satisfy

$$\begin{gathered} \left[h_1, Q^1_\alpha\right] = Q^1_\alpha; \left[h_{2,3}, Q^1_\alpha\right] = 0 \\ \left[h_3, \bar{Q}_{\dot{\alpha}3}\right] = \bar{Q}_{\dot{\alpha}3}; \left[h_{1,2}, \bar{Q}_{\dot{\alpha}3}\right] = 0 \\ \left[e_i^+, Q^1_\alpha\right] = 0; \left[e_i^+, \bar{Q}_{\dot{\alpha}3}\right] = 0 \end{gathered} \tag{3.116}$$

We also see that the other charges are related by action of e_i^- to Q^1_α or $\bar{Q}_{\dot{\alpha}3}$ (i.e taking the commutator with e_i^-)

$$Q^1_\alpha \xrightarrow{e_1^-} Q^2_\alpha \xrightarrow{e_2^-} Q^3_\alpha \tag{3.117}$$

$$\bar{Q}_{\dot{\alpha}3} \xrightarrow{e_1^-} -\bar{Q}_{\dot{\alpha}2} \xrightarrow{e_2^-} \bar{Q}_{\dot{\alpha}1} \tag{3.118}$$

Therefore, the SU(3) part of the Q representation has Dynkin labels $[1,0,0]$ which are weights of the highest state Q^1_α, and that of $\bar{Q}$ representation has Dynkin labels $[0,0,1]$ with $Q_{3\dot{\alpha}}$ as the highest weight. From (3.15) and (3.113) we also have

$$\left[\hat{R}, Q^I_\alpha\right] = \frac{1}{4} Q^I_\alpha;\ \left[\hat{R}, \bar{Q}_{\dot{\alpha}I}\right] = -\frac{1}{4}\bar{Q}_{\dot{\alpha}I} \tag{3.119}$$

Therefore the Qs and the $\bar{Q}s$ have U(1) charge $r = \frac{1}{4}$ and $r = -\frac{1}{4}$ respectively. Using the Cartan matrix we can write the weights (in terms of indices corresponding to the standard bases h_i) of the Qs and $\bar{Q}s$ as follows

$$\begin{gathered} Q^1_1 \sim [1,0]_{(\frac{1}{2},0)}\,;\ Q^1_2 \sim [1,0]_{(-\frac{1}{2},0)}\,;\ \bar{Q}_{\dot{1}3} \sim [0,1]_{(0,\frac{1}{2})}\,;\ \bar{Q}_{\dot{2}3} \sim [0,1]_{(0,-\frac{1}{2})} \\ Q^2_1 \sim [-1,1]_{(\frac{1}{2},0)}\,;\ Q^2_2 \sim [1,-1]_{(-\frac{1}{2},0)}\,;\ \bar{Q}_{\dot{1}2} \sim [-1,1]_{(0,\frac{1}{2})}\,;\ \bar{Q}_{\dot{2}2} \sim [-1,1]_{(0,-\frac{1}{2})} \\ Q^3_1 \sim [0,-1]_{(\frac{1}{2},0)}\,;\ Q^3_2 \sim [0,-1]_{(-\frac{1}{2},0)}\,;\ \bar{Q}_{\dot{1}3} \sim [-1,0]_{(0,\frac{1}{2})}\,;\ \bar{Q}_{\dot{2}3} \sim [-1,0]_{(0,-\frac{1}{2})} \end{gathered} \tag{3.120}$$

Also action of each Q adds a $\frac{1}{4}$ factor to the state and $\bar{Q}$ adds a $-\frac{1}{4}$ factor. Then the representations $[k',p']_{(j',\bar{j}')}$ formed by acting on $[k,p]_{(j,\bar{j})}$ with Qs and $\bar{Q}s$ are found using the Racah Speiser Algorithm. The negative indices can be converted by using (3.86) for SU(3) as follows

$$[k,p] = -\left[-k-2, k+p+1\right] = -\left[k+p+1, -p-2\right] \tag{3.121}$$

A general long multiplet is denoted by $\mathcal{A}^{\Delta}_{[k,p]r(j,\bar{j})}$. It has dimension

$$dim\mathcal{A}^{\Delta}_{[k,p]r(j,\bar{j})} = 2^{12} d\,(k,p)\,(2j+1)\,(2\bar{j}+1) \tag{3.122}$$

where d(k,p) is given by eqn (3.114). The unitarity condition gives

$$\Delta \geq 2 + 2j + \frac{2}{3}\,(2k+p+r)\,,\ 2 + 2\bar{j} + \frac{2}{3}\,(k+2p-r) \tag{3.123}$$

As before, if j=0 we can apply a shortening condition as follows

$$Q^I_\alpha \left|k,p,r\right\rangle^{hw}_{\dot{\alpha}_1 \ldots \dot{\alpha}_{2\bar{j}}} = 0 \tag{3.124}$$

Then using (3.4a) we can see that

$$\left\{Q^I_\alpha, S^\beta_J\right\} \left|k,p,r\right\rangle^{hw}_{\dot{\alpha}_1 \ldots \dot{\alpha}_{2\bar{j}}} = 0 \Rightarrow -\delta^\beta_\alpha \left(2i\delta^I_J D + 4R^I_J\right) \left|k,p,q\right\rangle^{hw}_{\dot{\alpha}_1 \ldots \dot{\alpha}_{2\bar{j}}} = 0$$

The values of R^I_J in terms of Cartan basis can be taken from eqn (3.113) and substituted to obtain the solution to the above equation. The shortening condition (3.124) can only hold for I=1, I=1,2 or I=1,2,3. Corresponding to this, let s denote the fraction of supercharges for which (3.124) is satisfied.

Then we get the following solutions for each case

$$\begin{cases} s=\frac{1}{3};\ I=1 & \Delta=\frac{2}{3}\left(2k+p+r\right) \\ s=\frac{2}{3};\ I=1,2 & \Delta=\frac{2}{3}\left(p+r\right);\ k=0 \\ s=1;\ I=1,2,3 & \Delta=\frac{2}{3}r;\ k=p=0 \end{cases} \tag{3.125}$$

Similarly, applying the conjugate condition for $\bar{j}=0$ gives

$$\bar{Q}_{\dot{\alpha}I}\left|k,p,r\right\rangle^{hw}_{\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}}}=0 \tag{3.126}$$

$$\begin{cases} \bar{s}=\frac{1}{3};\ I=3 & \Delta=\frac{2}{3}\left(k+2p-r\right) \\ \bar{s}=\frac{2}{3};\ I=3,2 & \Delta=\frac{2}{3}\left(k-r\right);\ p=0 \\ \bar{s}=1;\ I=3,2,1 & \Delta=-\frac{2}{3}r;\ k=p=0 \end{cases} \tag{3.127}$$

If a multiplet satisfies both (3.124) and (3.126) with non zero s and $\bar{s}$ (also $j=\bar{j}=0$) then the multiplet is denoted by $\mathcal{B}^{s,\bar{s}}_{[k,p]r(0,0)}$. For consistency we must have $p-k=2r$.

If $j\geq 0$, we can also apply the semi-shortening conditions given by

$$\begin{cases} \epsilon^{\alpha\beta}Q^I_\beta\left|k,p,r\right\rangle^{hw}_{\alpha_1\ldots\alpha_{2j-1}\alpha,\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}}}=0 & j>0 \\ \left(Q^I\right)^2\left|k,p,r\right\rangle^{hw}_{\alpha_1\ldots\alpha_{2j},\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}}}=0 & j=0 \end{cases} \tag{3.128}$$

Now using (3.15) we can see that

$$\epsilon^{\alpha\beta}\left\{Q^I_\beta,S^\gamma_L\right\}\left|k,p,r\right\rangle^{hw}_{\alpha_1\ldots\alpha_{2j-1}\alpha,\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}}}=0$$

$$\Rightarrow\left(\delta^I_L\left(-j-1+\frac{1}{2}\Delta\right)-R^I_L\right)\left|k,p,r\right\rangle^{hw}_{\alpha_1\ldots\alpha_{2j},\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}}}=0$$

Again the values of R^I_J in terms of Cartan basis can be taken and substituted to obtain the solution. Just as before, the semi-shortening condition (3.128) can only hold for I=1, I=1,2 or I=1,2,3. Corresponding to this, we denote the fraction of supercharges for which (3.128) is satisfied by t. Then we get the following solutions for each case

$$\begin{cases} t=\frac{1}{3};\ I=1 & \Delta=2+2j+\frac{2}{3}\left(2k+p+r\right) \\ t=\frac{2}{3};\ I=1,2 & \Delta=2+2j+\frac{2}{3}\left(p+r\right);\ k=0 \\ t=1;\ I=1,2,3 & \Delta=2+2j+\frac{2}{3}r;\ k=p=0 \end{cases} \tag{3.129}$$

Similarly we can have the conjugate condition

$$\begin{cases} \epsilon^{\dot{\alpha}\dot{\beta}}\bar{Q}_{\dot{\beta}I}\left|k,p,r\right\rangle^{hw}_{\alpha_1\ldots\alpha_{2j},\dot{\alpha}_1\ldots\dot{\alpha}_{2\bar{j}-1}\dot{\alpha}}=0 & \bar{j}>0 \\ \left(\bar{Q}_I\right)^2\left|k,p,r\right\rangle^{hw}_{\alpha_1\ldots\alpha_{2j}}=0 & \bar{j}=0 \end{cases} \tag{3.130}$$

and the corresponding solutions

$$\begin{cases} \bar{t} = \frac{1}{3};\ I = 3 & \Delta = 2 + 2\bar{j} + \frac{2}{3}(k + 2p - r) \\ \bar{t} = \frac{2}{3};\ I = 3, 2 & \Delta = 2 + 2\bar{j} + \frac{2}{3}(k - r);\ p = 0 \\ \bar{t} = 1;\ I = 3, 2, 1 & \Delta = 2 + 2\bar{j} - \frac{2}{3}r;\ k = p = 0 \end{cases} \tag{3.131}$$

If both the semi-shortening conditions (3.128) and (3.130) are applied for non zero t and $\bar{t}$ we obtain a semi-short multiplet denoted by $\mathcal{C}^{t,\bar{t}}_{[k,p]r(j,\bar{j})}$. Again a special case is $\mathcal{C}^{1,1}_{[0,0]r(j,\bar{j})}$ (for which we must have $r = \frac{3}{2}(\bar{j} - j)$) since for $j, \bar{j} > 0$ according to (3.15) it leads to a conservation equation of the form

$$\tilde{P}^{\dot{\alpha}\alpha} \left| 0, 0, \frac{3}{2}(\bar{j} - j) \right\rangle^{hw}_{\alpha\alpha_1 \ldots \alpha_{2j-1}, \dot{\alpha}\dot{\alpha}_1 \ldots \dot{\alpha}_{2\bar{j}-1}} = 0 \tag{3.132}$$

If j=0 we can apply the shortening condition (3.124) together with the semi-shortening condition (3.130) to obtain a multiplet which is denoted by $\mathcal{D}^{s,\bar{t}}_{[k,p]r(0,\bar{j})}$. For consistency we need $p - k = 2r - \frac{3}{2}(1 + \bar{j})$. Similarly applying (3.126) and (3.128) for $\bar{j} = 0$ gives the conjugate multiplet $\bar{\mathcal{D}}^{\bar{s},t}_{[k,p,q](j,0)}$, for which we must have $p - k = 2r + \frac{3}{2}(1 + j)$.

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